

ACTUARIAL NON LIFE

1. THE GENERAL FRAMEWORK

[12/09]

$$[1] \text{PL}_t = B_t - X'_t - V_t^S + V_{t-1}^S - E_t$$

$$[2] \text{PL}_t = (B'_t - V_t^P + V_{t-1}^P) - X_t^{\text{CY}} - V_t^S(\text{CY}) - E_t + V_{t-1}^S - X_t^{\text{PY}} - V_t^S(\text{PY})$$

Our initial aim is to measure the **technical profit/loss** for the company, which excludes all the other possible P/L (eg. financial).

Throughout the course we will not take into account any type of reinsurance treaty that the company may underwrite, therefore all the results will be **gross of reinsurance**.

Definitions of formula [1]:

- B_t : **earned premiums** of year t
- X'_t : amount **paid** in the calendar year t (disregarding the year of the claim/expense to be paid)
- V_t^S : **reserve** at the end of the year t
- V_{t-1}^S : **reserve** at the end of the year t-1
- E_t : **expenses** regarding **acquisition, collection**, distribution channels, and fixed costs for the business (**management**).

The **settlement expenses**, which are the expenses to evaluate the magnitude of the claims, are inside the X'_t or V_t^S depending on whether they are afforded in the year or not: there are **not included** in the term E_t .

Why do we split the settlement expenses from the rest of the expenses? Because this specific type of expense is covered by the fair premium (or risk premium), so they are included in the expected value of cost for the contract, whereas the rest of the expenses are covered by the expenses loading.

Both for [1] and for [2] we can have **two** different **perspectives**: ex-ante and ex-post.

Ex post: we consider all the values in the formulas as **deterministic results**, since we are at the end of the year t and we can have all the data to know with certainty the value of each of the terms. In this type of analysis, the aim is to evaluate the **performance** of the company, through the computation of some key parameters, such as the **loss ratio, combined ratio** ecc...

Ex-ante: not all the values are known since we are at the beginning of the year t, therefore we consider some of the terms as **random variables**. This type of analysis is crucial for the computation of the **premium**, the assumption underlying the **reserving** process, and also the **capital requirements computation**, therefore in all the processes that require some degree of estimation.

Another possible division that we can make is regarding the year of the claims both for payments and for reserving ([2]).

Why is it crucial to differentiate according to the years? Because the claims of the current year are covered by the premiums whereas the ones of the previous years are covered by the reserve.

We will distinguish the year t, written as CY (current year), from the previous years, PY.

To pass from [1] to [2] we must split both X'_t and V_t^S according to the year of occurrence of the claims:

- $X'_t = X_t^{\text{CY}} + X_t^{\text{PY}} \rightarrow$ Paid in the same year but **covered** by **different** amounts.
- $V_t^S = V_t^S(\text{CY}) + V_t^S(\text{PY})$

It has no sense to split also V_{t-1}^S since the value for the current year would be always null since we are referring to the previous year.

The last splitting in [2] is out of our interest, but it is regarding the possibility to rewrite the earned premiums as written premiums plus the difference between the premium reserve in the previous year and in the current year, namely: $B_t = B'_t - V_t^P + V_{t-1}^P$.

Now we are gonna decompose the P/L from [2] into two components:

- 1) $(B'_t - V_t^P + V_{t-1}^P) - X_t^{\text{CY}} - V_t^S(\text{CY}) - E_t \rightarrow$ This quantity is related only to the **possible profit** coming from **pricing**, which occurs when the premiums are able to cover the claims of

the year. Of course, when we made the ex-ante analysis, it is more likely to have a positive quantity, since the gross premium is composed of the fair premium (what I expect in terms of claims amount), expenses loading (when I expect in terms of expenses) and safety loading, which represents the expected profit for the company and usually it is positive.

- 2) $V_{t-1}^S - X_t^{PY} - V_t^{S(PY)} \rightarrow$ This quantity refers to the difference between the claims reserve at the end of the previous year and the payments and the reserve, for claims which occurred in the previous years, at the end of the current year (so what I have to cover with the claims reserve in the year): the value represents the **sufficiency/insufficiency** of the **claims reserve** of the previous year. Here we are referring only to **P/L** related to **claims reserving** (while before we were referring to premiums). V_{t-1}^S refers to claims occurred and paid in the future so I should be able to cover both X_t^{PY} and $V_t^{S(PY)}$.

We are splitting the two sources of P/L also because it is required for the computation of capital requirements such as CaR and SCR for risk reserve (important for this course).

The **formulas** [1] and [2] are **independent** of the type of **balance sheet** (Local GAAP, Solvency II, IFRS19,...): the difference between these approaches lies in the evaluation and computation of the claims reserve V_t^S and V_{t-1}^S (eg. **Prudence VS Market Consistent Valuation**).

We will see that some model is good for every type of balance sheet whereas some other type is feasible only for a specific balance sheet.

If we consider the expected value of $V_{t-1}^S - X_t^{PY} - V_t^{S(PY)}$, it strictly depends on the balance sheet to which we are referring: in local GAAP we cannot discount, therefore we must compare the current year t-1 for the term V_{t-1}^S and the current year t for the term X_t^{PY} .

For example: given $E[X_t^{PY}] = 1000$, what is the value of V_{t-1}^S ?

- $V_{t-1}^S = 1000$ if there is no discount
- $V_{t-1}^S > 1000$ if we apply some degree of prudence in the reserve
- $V_{t-1}^S < 1000$ if we are discounting with a positive rate the reserve

The expected value of $V_{t-1}^S - X_t^{PY} - V_t^{S(PY)}$ is not univocally positive (as it happens with the expected value of $(B_t' - V_t^P + V_{t-1}^P) - X_t^{CY} - V_t^{S(CY)} - E_t$), because it depends on how the claims reserve has been evaluated: it is equal to 0 when there is no discounting and the only driven factor is the expectation.

The amount of the **reserve** is important because it **affects** both the **capital requirements** and the **OF**: in fact, if we think about the Solvency Ratio, this is the ratio between what I have in the company wallet and what I must keep in order to be solvent in the future, also in bad scenarios

$(SR = \frac{OF}{SCR})$; when I modify the reserve, there are changes both for the OF and for the SCR,

namely on the same direction but with different magnitude.

For example, if I decide to keep a high amount of reserve for a small amount of claims, certainly the SCR will be low but the liabilities will be high and for this reason, the OF will be low (since, disregarding the tiering, the OF can be defined as the difference between assets and liabilities).

Now we will focus on the local GAAP, namely on the technical reserve for the non-life insurance sector.

Premium reserve = UPR + URR

UPR: Unearned Premium Reserve is the part of the premium that must be **delayed** in order to be considered throughout the entire duration of the contract. Typically it is computed with the so-called **pro rata temporis** approach, which consists of the evaluation of the portion of the premium to be postponed with respect to and proportional to the passage of time.

URR: Unexpired Risk Reserve is an additional amount (that can be only positive) present in the local GAAP when the UPR is **not sufficient** to cover the claims.

Claims reserve

- **Not discounted**

- **Gross of recoveries:** in some LoB I can have some **credits to the policyholders**. For example in MTPL it is possible to underwrite contracts with some **retention amount**: in this case, if the amount of the claim is higher than the retention, the company covers the whole amount of the claim and then it has a credit with respect to the policyholder; this is done in order to protect the victim of the damage, who will receive the entire amount immediately.

Regarding the insurance companies, these recoveries are not considered in the claims reserve, in which is reported the whole amounts of the claims, but they are **considered** on the **assets** side. This treatment of the recoveries is different in **SII**: since the BEL is the difference between cash in and cash out, in the SII balance sheet is reported only the **difference** between the whole **claim amount** and the value of the **retention**; this could lead to problems with negative values in the triangle.

- **Case reserve:** reserve evaluated claim by claim by the **claim adjuster**, without any type of correlation in the aggregation process. The **advantage** is that the evaluation is based on the condition of the damaged item and the historical experience; the **drawbacks** are the **subjectivity** of the estimation, the **time-consumingness** of the process, and the **absence of the IBNR**.

Usually, the actuaries have at disposal the computation of the case reserve and it is used as a comparison for the results obtained with the statistical methods.

- **Provision for future development on reported claims (RBNS):** differently from the case reserve, the reserve is computed as an **aggregation** of the different claims, usually compared to the case reserve in order to catch the differences between the two approaches. The **aim** of the aggregation is to exploit some **regularities** in the part of the triangle composed of observed data (upper triangle), such as the value of λ for the C-L method. As said before some method uses the case reserve as input.

- **Claim adjustment expenses** (settlement expenses): these are the expenses for the evaluation of the claims occurred; we've seen that the part of these expenses not yet afforded is considered in the reserve.

We have to differentiate between **two types** of settlement expenses:

ALAE (Allocated Loss Adjustment Expenses) —> it is possible to allocate the expenses to the specific claims, for example, the cost of the claim adjuster.

ULAE (Unallocated Loss Adjustment Expenses) —> it is not possible to allocate the expenses directly to the claims, for example, the salary of a worker outside of the company

It is important to know which are the component included in the triangle (eg. Only payments or payments + expenses) since all the values not considered in the triangle must be estimated separately with other methods.

- **Provision for IBNR:** this type of claim is not included in the case reserve (for obvious reasons) and it is most likely to estimate it **separately** with respect to the triangle.
- **Adjustments:** all the previous parts are strictly related to data (which usually are not selected by the actuaries) so it is not possible to observe the changes in the claims environment (such as the table for bodily injuries, or the inflation). These adjustments can be useful, for example, not to change the assumption in the inflation rate: for the C-L method we know that embedded in the process there is the projection of the inflation rate, but if the prediction is not accurate (eg. 2022) it is possible to adjust for the real inflation after the computation of the results with C-L.

Another possible usage of the adjustments is to control the changes in the pattern of regularities, fundamental for the reliability of the estimations.

2. DETERMINISTIC METHODS

[14/09]

The evaluation of the reserve is crucial, since in the insurance sector we can deal with long time horizon and big amount of money: usually the estimation is a combination of the case reserve and the reserve obtained with the actuarial methods (but we will focus only on the actuarial methods).

The only aim of the **deterministic methods** is the **point estimation** of claims **reserve**: simply a number which doesn't take into consideration either the variability of the result or its skewness, it is relevant only the expected mean of future payments.

Until ten years ago the deterministic methods were the most used, while after the introduction of SII the attention was moved to the computation of the risk, therefore to a measure of the volatility of the results.

One of the first question that we can answer is: which kind of data we have at disposal?

The starting point is the triangle: the claims are grouped according to some specific criteria (eg. the accidental year or the reported year), and this is the main difference between the case reserve and the actuarial one.

The data as the number of claim or the severity of the claims are internal to the company; on the other hand many external factors (eg. Inflation) can affect the estimation procedure of the claims reserve, therefore we must include also these aspects; another example is given by the extreme events, which tend to behave in a profound different way with respect to the normal situation.

(15) **Row: $i \in [0, n]$** is related either to the **accident year (AY)**, most common and provided by the balance sheet) or to the **reported year (RY)**, for internal purposes it is possible to use this classification) (**EXAMPLE Q**)

The **main problem** of the **RY** classification is related to the fact that in the same cell it is possible to **mix** both IBNR and not IBNR, but in the two cases there are different lags between the occurrence and the payment.

On the other hand, the **main advantage** of the **RY** classification is that you know in **advance** the **total number of claims** for the row, differently from the **AY** in which the total number of claims per row is updated year by year.

There is a third possibility for reinsurance: underwriting year (UY), but this is not in the scope of the course (approfondimento nel quaderno).

If we observe the entire row, we can appreciate the development of the claims of a specific accident/reported year.

Column: $j \in [0, n]$ is related to the **development year (DY)**.

If we observe the entire column, we can appreciate the behaviour of the payments with the same lag between AY and payment year.

If we observe the triangle by **diagonal** we can see the payments made in the same **calendar year**: in fact $i + j$ is the CY. If we sum all the values on the same diagonal we can obtain the total amount of payments for the same CY.

If $i > j$ then the triangle is **trapezoid**: this is useful when I have few information regarding the tail (not in the scope of the course).

If $i < j$ then we are including also the **tail**, in order to observe the future development: usually this is useful when in the first AY I don't have paid all the claims; however it is possible to include the tail also when all the claims of the first row are paid within n years.

The upper triangle is useful also for the stochastic model: in fact we are going to compute the variance conditionally to the information included in the upper triangle.

Notation: $D^{(n)} = \{X_{i,j} : i + j \leq n\}$ —> This is the notation to summarise all the information contained in the upper triangle, strictly related to the idea of filtration.

Then to compute the variance: $Var(R | D^{(n)})$ with $D^{(n)}$ updated every year.

Notation for the triangle

- **Incremental paid claims** —> $P_{i,j}$

- **Cumulative paid claims** —> $C_{i,j} = \sum_{h=0}^j P_{i,h}$

- **Incremental amount at reserve** (either case, booked or best estimate) —> $R_{i,j}$

- **Number of paid claims** $\rightarrow n_{i,j}$
 - **Number of open claims** $\rightarrow r_{i,j}$
 - **Reopened claim counts** $\rightarrow re_{i,j}$
 - **Number of closed claims without payments** $\rightarrow s_{i,j}$
 - **Number of reported claims** $\rightarrow$ (AY) $d_{i,j}$ since we don't know in advance the whole number of claims occurred in an accident year; (DY) d_i since you already know the total number of claims reported in each year. (MISSING EXAMPLE Q)
- Control always if the data are expressed in RY or in AY.

(18) Given the triangles with $P_{i,j}$ and $n_{i,j}$ we can compute the **average cost** (useful for the Fisher-

Lange method) as follows: $m_{i,j} = \frac{P_{i,j}}{n_{i,j}}$.

In this case, due to the regulation, the **partial payments** are considered in $P_{i,j}$ but not in $n_{i,j}$ therefore $m_{i,j}$ is **biased**.

(19) We can compute something similar to $m_{i,j}$ also for the **reserving**: in this case we want to

calculate the **average reserve cost** as $\frac{R_{i,j}}{r_{i,j}}$. Even in this case the ratio is biased because the

partial payments (and so the partial reserving) is considered only at the numerator.

The **former triangle** is an **estimation**, therefore it depends on the method used.

The **latter triangle** is **not** an **estimation**, it comes out of a closed formula:

$$- r_{i,0} = d_i - n_{i,0} - s_{i,0} \text{ (DY)} \iff r_{i,0} = d_{i,0} - n_{i,0} - s_{i,0} \text{ (AY)}$$

$$- r_{i,j} = r_{i,j-1} - n_{i,j} - s_{i,j} + re_{i,j} \text{ (DY)} \iff r_{i,j} = r_{i,j-1} + d_{i,j} - n_{i,j} - s_{i,j} + re_{i,j} \text{ (AY)}$$

Why we don't have reopened cases in 0? Closed and reopened cases in the same year are not reported as reopened in the triangle, but only as open.

From these triangles we can see if after n years from the first DY considered we have some payments still to be done: if the last value of the upper triangle is different from 0 then some claims have to be paid in the future.

If I close the triangle after n years then it is possible that I'm missing some payments.

Usually I have higher rating of closure at the beginning, while in the end we tend to have higher rating of reopened claims.

Important indicator for the company (cell by cell) $\rightarrow s_{i,j} - re_{i,j}$: typically (for **MTPL**) **positive** in the **first columns** while in the **central** part it is usually **negative**; anyway the specific behaviour depends on the LoB (eg. GTPL high incidence of $s_{i,j}$ also for higher columns, because the procedure for the assessment of closure can be long) and the company. This will be important for some estimator in the Fisher-Lange.

The triangles can be used also to compute some **diagnostic parameters**, such as the **settlement speed**: according to the settlement speed we can know how fast the company is in the **process of liquidation** of the claims; usually the faster a company is the lower will be per amount to be paid, since the impact of inflation is irrelevant.

We can split the settlement speed according to four different parameters, considering a couple of parameters at the time:

- **Amount** or **Numbers**
- **Cumulative** or **Incremental**

The typical ones for **lvass** are the **cumulative** ones, either based on amount or based on numbers.

Cumulative settlement speed (amount) $\rightarrow V_{i,j}^A = \frac{C_{i,j}}{C_{i,j} + R_{i,j}}$ in which $C_{i,j} = C_{i,j-1} + P_{i,j}$

and the ratio represent how much I have paid until j over the total amount of expected payments for the row (paid + reserved), that is the estimation of the ultimate cost for the row made in the year $i + j$.

(EXAMPLE Q)

(GRAPHIC Q)

In general we can observe that **lower** is the **reserve**, **higher** is the **settlement speed**, for which the maximum value is 1.

If we consider $V_{0,0}^A$ and $V_{0,1}^A$, they have the **same denominator** only when $R_{i,0} = R_{i,1} + P_{i,1}$ and this happens only when the **run-off** is **null**, which means that the reserve is not changed and the expected amount of payments (of the previous year) is equal to the real one.

When $V_{i,j}^A = 1$, then I have **paid all the claims**.

(GRAPHIC Q)

Changing the LoB (and also company) it is possible to observe a change both in the starting point and in the rapidity of convergence: the **longer** is the **tail** of a business, the **lower** is the **starting point** in the figure of the settlement speed.

The behaviour of the settlement speed depends also on the size of the triangle: for some LoB the usual 12x12 triangle can be limiting; the decision on the size of the triangle comes also from the regulation.

$V_{i,j}^A$ depends also on i : it is important to assess if there is a degree of stability through the columns, changing the rows; this stability is fundamental for the reliability of the prediction.

Cumulative settlement speed (numbers) $\rightarrow V_{i,j}^N = \frac{n_{i,j}^{CUM}}{n_{i,j}^{CUM} + r_{i,j}}$ the behaviour of this

settlement speed is similar to the previous one, even if typically $V_{i,j}^N > V_{i,j}^A$ since the payments in the first years is usually regarding the lower (in amount) claims (this is true especially for MTPL and GTPL)

If we want to compute the incremental settlement speed, we have simply to make the difference between the column j and the column $j - 1$

PAID CHAIN-LADDER METHOD

The **main assumption** in the PC-L method is that there is a possible way with which relate the **cumulative payments of two adjacent cells**; more specifically the ratio of two cumulative payments present in two adjacent cells is (more or less) constant.

$$\frac{C_{i,j+1}}{C_{i,j}} = \text{const/invariant to } i \approx \lambda_j$$

If this assumption is satisfied, then, it is easily possible to predict the missing values.

An important point for this approach is the **volatility of the ratio**: namely, **lower** is the **volatility**, **higher** is the **goodness** of the predicted values.

When this assumption is **not satisfied**?

- **Changes** in the **settlement speed** for the columns; sometimes it is possible to solve this problem with some indications to the claim adjuster.
- **Inflation** (specifically **claim inflation**): the cells are not adjusted to the nowadays inflation, therefore the different ratio of different RY/AY are affected by different inflation. In the classical PC-L the assumption is to predict the future inflation on the basis of the past inflation, but this is not always satisfied (like today).
- **Changes** in the **portfolio**: these changes could affect the stability through the modification of the settlement speed (eg. M&A)

[15/09]

There can be different version of the C-L (not only the one with payments): for example, we will see the incurred one.

Let's assume that the assumption are satisfied, then through the computation of λ_j we are able to estimate every value of the lower triangle.

Given the triangle with $C_{i,j}$, it is possible to build another triangle, namely the triangle made of the so called **individual loss development factors** (the term individual refers to the specific row i):

$$f_{i,j} = \frac{C_{i,j+1}}{C_{i,j}} \text{ (we make the computation for every couple of adjacent cells).}$$

Why is it useful to create this new triangle? In this way it is easier to see how close the data are to the C-L assumptions of similarity in the ratio: lower is the difference between the ratios, closer the data are to the C-L assumption.

(28)

The box-plot should follow a decreasing behaviour, since the ILDFs tend to 1.

From the box-plot we can also observe if there is the presence of outliers, that are values outside of the box-plot.

We can plot also only for the $DY > 0$ in order to reduce the scaled factor: in fact in the first year the value is quite different from the values in the following years (as we can see from the triangle).

It is unlikely that in real life the C-L is applied without any judgement before, in order to select the better model to describe the behaviour of the data. As a consequence we don't describe λ_j exactly as the ratio of the cumulative payments, whereas as a function of the ratio, meaning that $\lambda_j = f(f_{i,j}) \rightarrow$ In this way it is possible to compute several kind of C-L with different function of $f_{i,j}$.

The final result, whatsoever is the function applied, is a vector of λ_j 's, with at least n components:

$\lambda = [\lambda_0, \dots, \lambda_{n-1}]$, while if we consider the tail is $\lambda = [\lambda_1, \dots, \lambda_\infty]$.

$$\text{Classical C-L} \rightarrow \hat{\lambda}_j = \frac{\sum_{i=0}^{n-j-1} C_{i,j+1}}{\sum_{i=0}^{n-j-1} C_{i,j}} = \frac{\sum_{i=0}^{n-j-1} f_{i,j} C_{i,j}}{\sum_{i=0}^{n-j-1} C_{i,j}} \text{ for } j = 0, \dots, n-1$$

We obtain a row vector $\hat{\lambda} = [\hat{\lambda}_0, \dots, \hat{\lambda}_{n-1}]$

We deduct j from n-1 since when j grows we have less and less terms, meaning less information.

From the formula is clear that $\hat{\lambda}_j$ is a **weighted average** of $C_{i,j}$, in which the **weights** are the cumulative payments of the row: where there are higher payments, we weight more $f_{i,j}$, since I have more information (and this is an assumption).

It is possible also not to considerate all the data in the computation of the settlement speed: for example, if there is a change in settlement speed and the older data are not anymore representative of the new situation, then it is possible to consider some of the last 12 calendar years.

Therefore it is possible to generalise the formula before as $\hat{\lambda}_j = \frac{\sum_{i=0}^{n-j-1} f_{i,j} w_{i,j}}{\sum_{i=0}^{n-j-1} w_{i,j}}$, with $w_{i,j}$ that is

a generic weight.

(EXAMPLE EXCEL)

The usage of different weight is important in the first DY's because we have at disposal more data, while with the passage of time it is less interesting to change the weights since the data with which we can compute the average are few.

Changes in the settlement speed correspond with changes in the behaviour of the CY, therefore we have to control the diagonal to see if there is a modification of the settlement speed.

After we have decided which vector $\hat{\lambda} = [\hat{\lambda}_0, \dots, \hat{\lambda}_{n-1}]$ to select we can proceed with the estimation of the **lower triangle**, in **two** analogous **ways**:

$$1) \hat{C}_{n,1} = C_{n,0} \cdot \hat{\lambda}_0; \hat{C}_{n,2} = \hat{C}_{n,1} \cdot \hat{\lambda}_1; \dots$$

2) The idea is to link the estimation to the last information available, not to an estimated value.

$$\hat{C}_{i,j} = C_{i,n-i} \cdot \prod_{h=n-i}^{j-1} \hat{\lambda}_h \text{ for } i = 1, \dots, n; j = n - i + 1, \dots, n$$

We start from the column that refers to the diagonal (n-i) and we arrive to the previous DY with respect to the cell that we want to estimate. $C_{i,n-i}$ is the last information since $i+n-i=n$, therefore we are in the last diagonal.

This **second approach** is useful for the **stochastic evaluation**: $Var(\hat{C}_{n,2}) = Var(\hat{C}_{n,1} \cdot \hat{\lambda}_1)$ in which we have two random variables; $Var(\hat{C}_{n,2}) = Var(C_{n,0} \cdot \hat{\lambda}_0 \cdot \hat{\lambda}_1) = C_{n,0}^2 Var(\hat{\lambda}_0 \cdot \hat{\lambda}_1)$ in which the variance of $\hat{\lambda}_j$ is easier to be computed.

Without the tail we can **stop here**: the column n is the estimation of the ultimate cost for each row at evaluation date (year n).

If I want to compute the **total estimated reserve** I have simply to deduct the value in the diagonal

from the ultimate cost, namely: $\sum_{i=1}^n \hat{R}_i = \sum_{i=1}^n (\hat{C}_{i,n} - C_{i,n-i})$ and $\hat{R}_0 = 0$ since we pay all within

n years, therefore for the first AY/Ry I don't have the reserve anymore.

It is clear that the reserve computed before is the local GAAP one: with SII we have the concept of BE, therefore we have to include the discounting, which is not considered in this approach.

Now we **introduce the tail**, to include the case in which there are some payments after n years: it is not necessary that $r_{0,n} \neq 0$ to consider the tail but if this is true we have to include it.

Tail factor: we have to add one or more columns in the triangle, namely one column for each year in which I expect to pay something after the first n years; however it is pretty common to add only one column which represents a summary of all the years after n.

The main advantage of this practice is that I have to estimate only one additional $\lambda = \lambda_\infty$, while in the other case I have to compute one λ for each additional column.

The relationship between the 'compact' tail factor and the 'complete' tail factor is $\hat{\lambda}_\infty = \prod_{i=n}^{n+k-1} \hat{\lambda}_i$

with k as number of additional column. This formula is helpful because in some case it is not possible to compute directly $\hat{\lambda}_\infty$ but we have the different values for $\hat{\lambda}_i$.

The **first possible way** in which compute $\hat{\lambda}_\infty$ is when we have the value of the reserve at

evaluation date: $\hat{\lambda}_\infty = \frac{C_{0,n} + R_{0,n}}{C_{0,n}}$. In this case we are using the reserve for computing $\hat{\lambda}_\infty$ that

is again used to estimate the final reserve: if I am an internal actuary I don't have the value of $R_{0,n}$ at disposal, since it is what I want to estimate, therefore it is possible to consider $R_{0,n}$ as the case reserve (with all the limitations); on the other hand if I am an external actuary (eg. Consulting firm actuary) I have at disposal the balance sheet reserve, but I would check the validity of the tail factor computed by the company with the reserve of the company → Conflict of interest: it can be solve with the knowledge of the case reserve or the usage of other approaches to compute an input for PC-L).

$$\hat{C}_{i,\infty} = \hat{C}_{i,n} \cdot \hat{\lambda}_\infty$$

$$R^{PCL} = \sum_{i=1}^n (\hat{C}_{i,\infty} - C_{i,n-i})$$

If I use $\hat{\lambda}_\infty = \frac{C_{0,n} + R_{0,n}}{C_{0,n}}$ then $\hat{R}_0 = R_{0,n} \neq 0$

Remark: it is possible to apply some extrapolation method, in order to adjust some value; for example, we can try to fit the development factors with respect to j , that is trying to fit a curve to explain the behaviour of λ (or some value on which λ depends).

Why we want to do this?

1) Correct some anomalous point

2) Estimate the values for $j > n$ (eg. λ_∞) with several points ($\hat{\lambda}_\infty = \prod_{i=n}^{n+k-1} \hat{\lambda}_i$ is useful at this point).

(RECUPERA EXCEL)

[19/09]

One of the best way to solve the problem of **change** in the **settlement speed** is the computation of the estimators only on the **last k years**: in this way all the information strictly related to the years before the change in the speed, they are not considered in the future.

How the **C-L** deals with **inflation**?

In the **plain C-L** approach we have the **assumption** that **future inflation** must be **in line** with the **past**; obviously this requirement is not always respected (like it happens today), therefore we must find a new solution.

We can consider the so called **inflation adjusted C-L**: the formulas are analogous to the one of the simple C-L, with the only difference regarding the old values (the ones before the year n). In fact all the **values** of the previous years are **adjusted** to the most **recent inflation rate**.

Since the changes are made considering each of the single years, we must capitalise all the diagonals (until the first value of DY) with the exception of the main one; the capitalisation is made according to the last claim inflation rate.

Problem 1: in this way, I **don't consider** the **future inflation** in the lower triangle ($\hat{C}$).

Solution 1: I **correct** the **lower triangle** of cumulative payments with specific **assumption** regarding the **future inflation**.

Problem 2: I **cannot use** $\hat{C}$, since in the cumulative values there are **embedded different years**, therefore we have different inflation rate.

Solution 2: I have to **pass from** $\hat{C}$ to $\hat{P}$ and then I capitalise considering the future inflation.

The remaining problem is **how to estimate the claim inflation rate**: in fact we cannot consider only the value present in the market (economic inflation) but in the claim inflation we have some additional values that can differ LoB by LoB: the companies have internal models to predict the behaviour of the claim inflation.

How do C-L deals with **large claims**? In which part of the triangle are they located usually? How much they are stable?

These are some other of the **problem of C-L**: usually the **large claims** appear for **large** value of **DY**, since the settlement process is longer very often.

We have to assess if the large claims will affect the individual loss (link) ratio, that is analogous to assess if the large claims have to be included in the estimation of the lower triangle: if they depend on an extreme event (**outliers**) then it makes no sense to include them in the estimation, if they depend on some replicable event, then it is necessary to consider them for the lower triangle. In reality, it happens very often that these large claims are cleaned out of the triangle (eg. 17 mln is one of the biggest claim in MTPL for the Italian insurance sector): surely this happens for earthquakes and natural disasters.

The **advantage** of this approach is that the triangle is much more **stable**; the **drawback** is that we have to make some **external evaluation** for **large claims**.

We can have some issues with the evaluation of the threshold (which is usually fixed to 100.000/200.000): which value we have to consider? The estimation made when the claim is reported (case reserve) or the one with the actuarial methods? Since the evaluation can change throughout the time, there is a problem regarding the variability of the approach.

We can now introduce a fundamental factor, the **cumulative development factor** (CDF):

$$CDF_{n-i} = \prod_{h=n-i}^{n-1/\infty} \hat{\lambda}_h$$
, it is the product of the development factor.

This factor is helpful to move from the diagonal to the end of the triangle, namely

$\hat{C}_{i,\infty} = C_{i,n-i} \cdot CDF_{n-i}$. So we can compute the ultimate cost as this product, since the λ 's present in the CDF allow the estimation of the future $\hat{C}_i$.

Remember: $R_i = \hat{C}_{i,\infty} - C_{i,n-i} \rightarrow$ Instead of estimating the whole lower triangle, we can compute the reserve only having two components (CDF and the last cumulative payment, $C_{i,n-i}$).

There is an **important relationship** between **CDF** and the **settlement speed**: $SS^A \approx \frac{1}{CDF_{n-i}}$

This is a **sort of cumulative settlement speed**: the **difference** is that the settlement speed considered in the previous lectures is based on the triangle and it depends only on payments and reserve, while $\frac{1}{CDF_{n-i}}$ is based on the C-L methodology (λ), so there must be respected the C-L assumptions.

Example

$$\frac{1}{CDF_{11}} = \frac{1}{\lambda_{\infty}} = \frac{C_{0,11}}{C_{0,11} + R_{0,11}} \rightarrow \text{This is exactly the settlement speed}$$

$$\frac{1}{CDF_{10}} = \frac{1}{\lambda_{10} \cdot \lambda_{\infty}} = \frac{1}{\frac{C_{0,n}}{C_{0,10}} \cdot \frac{C_{0,n} + R_{0,n}}{C_{0,n}}} = \frac{C_{0,10}}{C_{0,n} + R_{0,n}} \rightarrow \text{This is not exactly equal to the}$$

settlement speed and the difference between the two measures increases when we go back in time (with the decreasing of $n - i$ for the CDF).

$$\frac{1}{CDF_{11}} = 98.77\% \rightarrow \text{with respect to the ultimate cost, the 98.77\% has been paid already.}$$

This value is strictly related to the tail factor: depending on the LoB, if λ_{∞} increases then $\frac{1}{\lambda_{\infty}}$

decreases (as it happens in GTPL), then the reserve is higher.

Different LoB:

- MTPL [34.61%, ..., 98.77%]
- GTPL [7.27%, ..., 73.71%]
- MOD [70.75%, ..., 99.36%]

With this data we can say that if there is an error in the computation of λ_0 , then this is less relevant for GTPL than for MTPL and MOD; on the other hand, after one year, in GTPL there is a low portion of claims that has been paid $\rightarrow$ Maybe even 12 years are not sufficient to capture in a good way the ultimate cost of the claims, since the tail has an high influence on the final result. In MOD the incidence of the tail is very low.

It is possible to use actuarial models also for **checking/validating** the **reserve** computed by someone else. When this is the case, we have to consider the concept of **adequacy**: is the booked reserve sufficient according to my computation?

The **insurer reserve** (reported in slide 34) is computed $R = \sum_{i=0}^n (\hat{C}_{i,+\infty} - C_{i,n-i})$; R^{PCL} is the estimation made **externally** $\rightarrow$ Adequacy = $R_{Company} - R^{PCL}$.

When we apply statistical methodology to the data, we aim to make prediction regarding the future on the basis of the data (back-fitting); but how can we evaluate the goodness of fit? We don't have training and test dataset, since we don't have the lower triangle; then it is not possible either to use old data (not reliable) or to split the triangle (we don't have enough data). A possible solution is given by the residuals (not important the formula for now): in this way it is possible to compare the predictions to the observations, since we apply the statistical approaches to the upper triangle, for which we have the true values.

$$\text{The formula for the residuals is } y_{i,j} = \frac{C_{i,j} - C_{i,j-1} \hat{\lambda}_{j-1}}{C_{i,j-1}} = f_{i,j-1} - \hat{\lambda}_{j-1}$$

If we take for example $i = n - 1$ and $j = 1 \rightarrow y_{n-1,1} = \frac{C_{n-1,1} - C_{n-1,0}\hat{\lambda}_0}{C_{n-1,0}}$ in which $\frac{C_{n-1,1}}{C_{n-1,0}}$ is

the individual development factor while $C_{n-1,0}\hat{\lambda}_0$ is how I would estimate the cell considered if I estimated the upper triangle.

In the end, we are **measuring** how **distant** is the **individual development factor from λ** , in order to assess the **stability** of $f_{i,j}$ ($y_{i,j} = f_{i,j-1} - \hat{\lambda}_{j-1}$).

We have seen how the residuals can be used as a sort of diagnostic for the model: their volatility is an important estimator of the goodness of the model; moreover they are important to capture some possible trend to be included in the estimation process.

INCURRED C-L METHOD

Differently from the PC-L, in this case we consider a triangle made of the **incurred claims**.

If we fix the row and we follow the DY path, we can see how the estimation of the ultimate cost evolves throughout the years, for the same claims (attention, this is true for RY, slightly different for AY, due to the presence of IBNR).

Incurred losses: $I_{i,j} = C_{i,j} + R_{i,j}$, in which $C_{i,j}$ can represent different reserves, having different impacts; we consider $C_{i,j}$ as the **booked reserve** (therefore **un-discounted**).

$I_{i,j}$ is an **estimation** of the **ultimate cost** made in the year **i + j**.

$I_{0,0} - I_{0,1}$ represents the effect of the **CDR** (run-off) only for a specific row.

Proof: $I_{0,0} - I_{0,1} = C_{0,0} + R_{0,0} - C_{0,1} - R_{0,1} = R_{0,0} - P_{0,1} - R_{0,1}$ that is exactly the run-off.

If $I_{0,0} - I_{0,1} > 0$, then I have **reduced** the **estimation** of the **ultimate cost** from the year 1993 to the year 1994 (in the example).

Since until now we have worked with ratio, we can introduce the **individual loss development**

factor $f_{i,j}^I = \frac{I_{i,j+1}}{I_{i,j}}$.

$$\begin{cases} run - off = 0 \rightarrow f_{i,j}^I = 1 \\ run - off > 0 \rightarrow f_{i,j}^I < 1 \\ run - off < 0 \rightarrow f_{i,j}^I > 1 \end{cases}$$

The development factor depends on the run-off, namely on how the estimation of the ultimate cost evolves.

Important: This result is true only when the reserve considered is the booked one

[21/09]

Why do we use the booked reserve? Only in this way it is possible to consider $I_{i,j} - I_{i,j+1}$ as the run-off of the reserve.

From the triangles with $R_{i,j}$ and $C_{i,j}$ we can build the triangle of $I_{i,j}$ (slide 39).

As we've already seen, following a specific row, each difference between two adjacent columns gives indication of the run-off of all the claims occurred in that specific year (again, attention to the difference between RY and AY: with RY I cannot have IBNR).

There are four main factors that can change the run-off:

- 1) **Payments:** when I pay a different amount with respect the one that I reserved the year before.
- 2) **Reserve:** when I reserve a different amount with respect the one that I reserved the year before (revaluation of the claims reserve).
- 3) **Reopened claims** (minor impact)
- 4) **Claims without any payments** (minor impact)

When there are no claims to be paid anymore, then the UC remains constant throughout the DY.

The idea of PC-L is to look at the columns (eg. Ratio between couple of cells stables throughout the years): analogously, for the IC-L we want to assess if $\frac{I_{i,j+1}}{I_{i,j}}$ is stable with respect to $i \rightarrow$ in actuarial terms this assumption can be checked in differed ways, namely through the study of **settlement speed, inflation, portfolio composition**.

If $\frac{I_{i,j+1}}{I_{i,j}} < 1$, for each year I expect an under estimation of the ultimate cost.

The **crucial point** is the **volatility** of $f_{i,j}^I$: lower the volatility, higher the probability of goodness of the approach.

The values in the triangle are the empirical representation of the distribution, but in order to be reliable we need an high amount of data: this is the main problem for some LoB for the claims reserve, since often we don't have sufficiently big triangles.

As we can see from the box-plot in the slide 40, with the booked reserve we have a fluctuation of values, of the **individual loss development factor, around 1**: this behaviour is very different from the trend of PC-L's individual loss development factor, which is monotonically decreasing.

As for the PC-L, we can obtain the development factor with the formula

$$\hat{\lambda}_j^I = \frac{\sum_{i=0}^{n-j-1} I_{i,j+1}}{\sum_{i=0}^{n-j-1} I_{i,j}} = \frac{\sum_{i=0}^{n-j-1} f_{i,j}^I I_{i,j}}{\sum_{i=0}^{n-j-1} I_{i,j}} \text{ for } j = 0, \dots, n-1 \rightarrow \lambda \text{ as weighted average of the}$$

individual development factors, with I 's as weights.

Also in this case it is possible to change the weights with some generic ones: this is interesting for the initial DY's, while for the final ones it is not relevant.

The difference with the PC-L is in the intrinsic value of λ : in fact for the **IC-L**, we have the value of the **tail already embedded** in $\hat{\lambda}_{n-1}^I$. Why? $\hat{\lambda}_{n-1}^I = \frac{I_{0,n}}{I_{0,n-1}} = \frac{C_{0,n} + R_{0,n}}{I_{0,n-1}}$ with $R_{0,n}$ that is the last

reserve at disposal, estimated with the last development factor, which includes the payments of claims after n years.

The underlying assumption is that the last reserve is at the base of $\hat{\lambda}_{n-1}^I$, which means that the tail factor gives credibility to the reservation of the company.

If I have no tail, then $R_{0,n} = 0$; at the same time, $R_{0,n} \neq 0$ can be an indicator of the presence of the tail.

If I am an external actuary, usually I don't want to trust $R_{0,n}$, I would like to find another indicator.

In this case I cannot reproduce the fitting procedure (seen in the previous lectures), since I don't have either an exponential behavior or the monotonic decreasing behaviour (more relevant): the values fluctuate around the value 1, it is almost a straight line.

A possible solution is to find the value of $R_{0,n}$ with other methodologies.

Now we want to estimate the lower triangle: $\hat{I}_{i,n}$ is the final estimation of the ultimate cost, which is the corresponding of the IC-L of $\hat{C}_{i,\infty}$ (ultimate cost with paid amount).

How's the relationship between $\hat{I}_{i,n}$ and $\hat{C}_{i,\infty}$? If they are similar, this is a sign of good reliability of the result; on the other hand, if they are different, we have to choose which is the best model to fit the data.

$$\hat{I}_{i,j} = I_{i,n-i} \prod_{h=n-i}^{j-1} \hat{\lambda}_h^I \text{ for } i = 1, \dots, n ; j = n - i + 1, \dots, n$$

$$\mathbf{R}^{\text{ICL}} = \sum_{i=0}^n (\hat{I}_{i,n} - C_{i,n-i}), \text{ where } \rightarrow \text{ We can see that we are not computing as difference of } I\text{'s}$$

(as it happens in PC-L for C's).

Until now, we are computing un-discounting claims reserve: in SII framework (and IFRS17) we have market consistent evaluation, so we have to consider the discount.

Discounting procedure: from $\hat{C} \rightarrow \hat{P}$: we always have to discount the incremental amounts, because in the cumulative amounts we have different years of the claims. (DISEGNINO SPOT RATE). (0,1) is applied only on the first diagonal of the lower triangle, since they are cash flows expected for the next year.

I have no problem to recover $\hat{P}$ from the PC-L, I can have some problems with the IC-L.

In IC-L I have C and R but I cannot obtain $\hat{P}$ since we don't have $\hat{C} \rightarrow$ I cannot apply IC-L in the market consistent framework.

One possible solution is to estimate $\hat{P}$ with some external model, and then apply IC-L.

EXAMPLE

100€ is the un-discounted estimation of the ultimate cost for the last row (sum of incremental payments): ($\hat{C} \rightarrow \hat{P}$) we have to split 100€ in a set of incremental payments from 1 to n, like

$$100 = \hat{P}_{n,1} + \dots + \hat{P}_{n,n}.$$

We can pass to the incremental payments looking at the settlement speed, taken from some other information (PC-L, from the market...): $SS_{Incremental}^{Amount} = [30\%, 20\%, 15\%, \dots, 10\%]$ which sum up to 100%.

Now we have to rescale, since we don't have to column 0 (in fact the lower triangle starts from column 1) $\rightarrow$ The new sum must be again 100% but between 1 and n:

$$SS_1 = \frac{20\%}{1 - 30\%}; SS_2 = \frac{15\%}{1 - 30\%}; \dots \text{ In this way we have the incremental payments only for one row.}$$

$$50 = \hat{P}_{n-1,2} + \dots + \hat{P}_{n-1,n}, \text{ now I have to rescale between 2 and n.}$$

This is a general approach for whatsoever methodology which is not able to estimate $\hat{P}$.

We have to bear in mind that all these reasoning are made under the assumption that the reserve used is the booked one: this is restrictive both for internal actuary (who wants to estimate the booked reserve) and for external actuary (who doesn't want to rely on the booked reserve, since you want to control its accuracy).

We have to take into consideration also the presence of the **management adjustments**: these create a sort of irregularity in the individual link ratio, often violating the assumption of the methodologies.

It is possible to replace the booked reserve with the case one: the advantage is that this is less afflicted by volatility, the first drawback is that in the case reserve are included the IBNR.

EXAMPLE

$I_{i,j} = C_{i,j} + R_{i,j}$, with $R_{i,j}$ as **case reserve** $\rightarrow f_{i,j}^I$ are **effected** by the **presence of IBNR** when they are **reported**.

In column 0 we don't have IBNR; in column 1 we can have IBNR (reported after 1 year) either in $C_{i,1}$ or in $R_{i,1}$.

The **second drawback** is that $f_{i,j}^I$ is **not** anymore the **run-off**, since it is not related to the booked reserve. We cannot say anymore that $f_{i,j}^I$ is more or less centred in 1: $f_{0,1}^I > 1$ even if the run-off is null, because of the presence of IBNR (the first box-plot on the left may be shifted up).

Something similar happens in other columns, but the effect is lower and lower: for high DY we have less IBNR, especially for rapid LoB. An example that proves the contrary is the 'Asbestos', with large IBNR due to the health condition problems after several years.

With the case reserve, the reporting delay should be stable (eg. Always reported after q lags) otherwise I'm inserting high volatility to my model: in fact, differently from the booked reserve, in Example (referring to slide 33)

We two companies: company A with $C_{n,0} = 60.361$; company B with $C_{n,0} = 120.000$, but with the same triangle for the rest.

We can have **two** different **possibilities**, as sort of diagnostic:

- 1) The **increment** is given by an **increase in the portfolio** dimension, therefore we have more claims: in this case we are consistent in terms of stability, so we can **apply C-L**.
- 2) The **increment** is given by a **change** in the **settlement speed**, therefore I pay differently the claims: in this case we are violating the assumption of stability, so we **cannot apply C-L**.

Of course, it is not possible to base the reasoning only on the absolute values of the payments, I need some other indicators.

Even though the $C_{n,0}$ is not involved in the deviation factor computation, its value affects the whole row for the reserving, therefore is extremely important to know the explanation of this different behaviour.

Less data we have, more the results are uncertain.

BORNHUETTER-FERGUSON

The idea of this approach is to combine two methodologies for the computation of the ultimate cost:

- 1) **Development method** (based on the triangle): in this case we consider either the PC-L or the IC-L (not totally correct), and we give full **credibility** to the **claim experience** through the data.
- 2) **Expected claims method**: considering the **expected loss ratio**, we don't give credibility to the claims experience.

The general rule is that we should give more weight to the PC-L for the older AY/Ry, since we have more collected data, while for the most recent years it assumes more relevance the ELR.

$$U_i^{BF} = C_{i,\infty}^{PCL} \cdot Z_i + \hat{\mu}_i \cdot (1 - Z_i) \text{ with:}$$

- U_i as the **ultimate cost** estimated for the row i
- $C_{i,\infty}$ as the **ultimate cost** estimated with the **PC-L**
- $\hat{\mu}_i$ as the **ultimate cost** estimated with the **ELR**

$$Z_i \text{ as } \textbf{credibility weight}, Z_i \in [0,1] \forall i = 0, \dots, n \text{ and } \sum_{i=1}^n Z_i = 1$$

$\hat{\mu}_i = \widehat{ULR}_i \cdot B_i \rightarrow$ In ELR there is not the usage of the triangle; the assumption is to estimate the ultimate cost as a product of earned premiums of the row and the estimation of the loss ratio of the row (eg. $\widehat{ULR}_0 = \frac{UC_0}{B_0}$).

I can obtain $\widehat{ULR}_i$ from other information, for example coming from the **pricing**, since I estimate the **expected rate of claims** to pay for the following years (eg. $\frac{\text{Risk premium}}{\text{Gross Premium}}$).

Obtaining $\widehat{ULR}_i$ in this way, it is possible to state that it is reliable for the recent AY, since the pricing process has been made recently.

ULR is very important for the start-up companies in which is not possible to compose the triangle (few data).

How do we calibrate Z_i ? $\hat{Z}_i = \frac{1}{\widehat{CDF}_{n-i}}$, so it is a sort of estimation of the cumulative

settlement speed, with $CDF_j = \prod_{h=j}^{n-1} \hat{\lambda}_h$ (assuming no tail) = $\prod_{h=j}^n \hat{\lambda}_h$ (assuming tail)

The relationship between i of the Z and $n - i$ of the CDF is explain by the (DISEGNINO)

In this way, $\hat{Z}_0 = \frac{1}{\widehat{CDF}_n} > \hat{Z}_n = \frac{1}{\widehat{CDF}_0}$, which means that the **weights** are **increasing**

disregarding the type of company and the LoB: only the magnitude of the increase depends on these two features (eg. MOD is a fast LoB, so the weights are higher for PC-L and this is consistent because there are required less data to give credibility to the method).

[22/09]

MY QUESTION?

Having defined the weights we can rewrite the ultimate cost as:

$$U_i^{BF} = C_{i,\infty}^{PCL} \cdot \frac{1}{\widehat{CDF}_{n-i}} + \hat{\mu}_i \cdot \left(1 - \frac{1}{\widehat{CDF}_{n-i}}\right) = C_{i,n-i} + \hat{\mu}_i \cdot \left(1 - \frac{1}{\widehat{CDF}_{n-i}}\right) \text{ because}$$

$$\hat{C}_{i,\infty} = C_{i,n-i} \cdot \widehat{CDF}_{n-i}$$

From ultimate cost to **reserve**: $R_i^{BF} = \hat{\mu}_i \cdot \left(1 - \frac{1}{\widehat{CDF}_{n-i}}\right) \rightarrow$ We can see that it is not obtained

as weighted average of the two methodologies; it is like for the PC-L, with $\hat{\mu}_i$ instead of $C_{i,\infty}^{PCL}$

EXAMPLINO

I can create other methodologies by replacing $\hat{\mu}_i$ or by taking the settlement speed from the market (or from another triangle).

Even if R_i^{BF} doesn't estimate the lower triangle, the methodology is based on the PC-L for the CDF, therefore I can obtain the $\hat{P} \rightarrow$ I can discount.

Example (53)

First three columns are application of PCL:

- 1) $C_{i,n-i}$, the diagonal
- 2) CDF_i (in 0 is 2014); last CDF is the tail factor of PC-L
- 3) U^{CL} is the ultimate cost of PC-L

Z_i are the weights, with monotonic decreasing/increasing (depending on the direction considered, from 1993 to 2004 decreasing) behaviour.

Reserve PC-L: $174421 - 60361 = 174421 \cdot (1 - 34.61\%)$

Ultimate loss ratio based on PC-L $\rightarrow \frac{U^{CL}}{B}$: this ratio is useful to compare different methods, since there is a better graphical comparison than the compensation between reserves, there's a better rescaling.

LR^{EC} is the input for BF (taken from pricing), the a priori expected ultimate loss ratio.

$LR_{1993}^{EC} = 85.82\%$ and $LR_{1993}^{CL} = 89.65\%$, so the BF shall yields a result between 85.82% and 89.65%. In particular I expect something more similar to 89.65% in the initial years and something nearer to 85.82% in the final years.

$LR > 1$ means that I expect a loss (since I am not even considering the expenses)

$$U^{EC} = \mu_i$$

It is interesting to see how the LR for 2004 will be in 2005 estimation.

Why does the **Incurred BF not make sense**? This comes from the definition of Z . In fact

$$\hat{Z}_i = \frac{1}{\widehat{CDF}_{n-i}^I} = \frac{1}{\prod_{h=n-i}^{n-1} \lambda_h^I} \text{ with } \lambda_h^I \text{ that fluctuates around 1, so there is an high probability to}$$

have value of Z near 1, giving all the credibility to the IC-L.

Moreover it is possible to encounter some $\lambda_h^I < 0$: in this case I would have some weights higher than 1, which does not make sense (eg. $\hat{Z}_i = 110\%$; $(1 - \hat{Z}_i) = -10\%$).

A possible solution is given by the replacing of $\hat{\mu}_i$ with $C_{i,n-i}^I$, creating a method that compares PC-L and IC-L.

Some **variants of the BF**

Cape Cod: here the ultimate cost is estimated as $\hat{\mu}_i = B_i \cdot \frac{\sum_{i=0}^n C_{i,n-i}}{\sum_{i=0}^n B_i Z_i}$

with the ratio that represents an estimation of the ultimate loss ratio (in fact loss ratio multiplied by earned premiums gives the ultimate cost).

In this framework, for the **ULR** we consider only the **payments**, not the part reserved; this is valid also for the premium at the denominator, for which I consider only the portion of payments.

If we consider the value of the Z's on the example (53), then $Z_n = 34\% \rightarrow 34\%$ of the premiums are used to cover the total payments.

The **advantage** of this approach is that I **don't** need **external information**, but this is a **disadvantage** at the same time, since with the BF we'd **like** to use **outside info**.

$$U_i^{CC} = C_{i,n-i} + B_i \cdot \frac{\sum_{i=0}^n \frac{C_{i,n-i}}{B_i Z_i} \cdot B_i Z_i}{\sum_{i=0}^n B_i Z_i} \cdot (1 - Z_i) = \left[\frac{C_{i,n-i}}{Z_i} = UC; \frac{UC}{B_i} = ULR \right] = C_{i,n-i} + B_i \cdot \frac{\sum_{i=0}^n ULR_i^{PC-L} \cdot B_i Z_i}{\sum_{i=0}^n B_i Z_i} \cdot (1 - Z_i)$$

Comparison between BF and CC:

- $U_i \rightarrow$ Both from PCL
- $\hat{\mu}_i \rightarrow$ From external information (BF); from PCL (CC)
- $Z_i = \frac{1}{\widehat{CDF}_{n-i}} \rightarrow$ Both from PCL
- LR^{EC} **varies** throughout the time; LR^{CC} is **constantly** the **weighted average of the ULR**.

It is possible to combine two or more deterministic methods, such as BF and CL, using for the older part the CL and for the most recent the BF: the problem regards the variance, namely the dependency between the two methodologies.

[26/09]

FISHER-LANGE

The idea is to **split** the part regarding the **numbers** and the one regarding the **cost of claims**.

$P_{i,j} = n_{i,j} \cdot m_{i,j}$ (same as the rate premium in Introduction) with $n_{i,j}$ as **incremental number** of claims and $m_{i,j}$ as **average paid cost** of claims.

Advantage: it is possible to focus separately on the estimation of number of claims and average cost ($\hat{n}$, $\hat{m}$).

Disadvantage: the number of parameters to be estimated is higher, therefore there is an higher volatility given by the uncertainty of the results on the parameters.

When I have $\hat{n}$ and $\hat{m}$ estimated, then I simply multiply (cell by cell) the two terms to obtain $\hat{p}$.

We start with the estimation of the **number of claims**; there are **two steps** for $n_{i,j}$:

- 1) **Percentage of open claims:** which is the proportion of open claims that are expected to be paid in the future.
- 2) **Settlement speed (based on numbers)**, representing the timing of the claims to be paid in the future

$$aliq_{i,j} = \frac{\sum_{h=j+1}^{n-i} n_{i,h} + r_{i,n-i}}{r_{i,j}} \quad (i = 0, \dots, n-1; j = 0, \dots, n-i-1) \rightarrow \text{Proportion of the}$$

claims having a payment in the future (either reopened, IBNR or nil).

Remember:

- $n_{i,j}$: number of closed claims with payments
- $r_{i,j}$: number of open claims

The summation gives us the total number of claims paid until the CY n (regarding the RY i).

$$\text{Example: } aliq_{0,0} = \frac{\sum_{h=1}^n n_{0,h} + r_{0,n}}{r_{0,0}} \rightarrow r_{0,0} \text{ is the number of open claims at the end of 1993}$$

and reported in 1993; $r_{0,n}$ is the tail factor.

The ratio $\frac{\sum_{h=j+1}^{n-i} n_{i,h}}{r_{i,j}}$ depends on the number of **claims** closed **without payments** and the

claims reopened: $\begin{cases} < 1 \text{ Incidence of CCW/O} \\ > 1 \text{ Incidence of Reopened} \end{cases}$

For the sake of the method, this **ratio** is assumed to be **constant** through the **column**.

$r_{i,n-i}$: claims open at the CY n (the present), therefore it represents the number of claims in the tail, what I expect in terms of numbers for the future. Also the ratio $\frac{r_{i,n-i}}{r_{i,j}}$ depends on the claims

closed without payments and the ones reopened.

In order to use AY there are some adjustment to be made.

The formula is **not used for the last row**: in this case $\frac{\sum_{h=j+1}^{n-i} n_{i,h}}{r_{i,j}} = 0$ and $\frac{r_{i,n-i}}{r_{i,j}} = 1$, but this is

not an indicator that the 100% of the claims will be paid in the future.

Usually in MTPL in the first columns we expect a ratio lower than 1: in fact the number of claims closed without payments should be higher than the one of the reopened claims, since in order to reopen a claim I have to close it before. Of course the magnitude of the ratio depends on the policies of the companies.

Example: $aliq_{i,0} = 90\%$ $\rightarrow$ 90% of the claims reserved in 0 are expected to be paid in future, while the other 10% is closed without payments.

In the central part of the triangle usually we have the opposite behaviours, that is the ratio higher than 1.

Then, for higher DY, we expect to have a ratio around 100%, meaning that there are no more reopened or closed without payments claims. A function of these ratios (usually the arithmetic mean for each development year) allows to obtain a vector $\rightarrow \alpha = [\alpha_h]_{j=0,\dots,n-1}$ with

$\alpha_j = E(aliq_{i,j})$; if we include also the **tail** we have to compute also α_n , with the usual

assumption $\alpha_n = 1$; other possibility: enlarge the triangle, using older data.

Differently from the PC-L, the **F-L** gives **more credibility** to the ratio for the **older years**, since

there are more data available (eg. $aliq_{0,0} = \frac{\sum_{h=1}^n n_{i,h} + r_{0,n}}{r_{0,0}}$ while $aliq_{n-1,0} = \frac{n_{n-1,1} + r_{n-1,1}}{r_{n-1,0}}$).

Nevertheless, the older data are representative of older portfolios, so it is possible to introduce some weights, in order to recalibrate the credibility.

(62)

$\alpha_0 = 91.52 \rightarrow$ On average this percentage of claims opened in 0 will be paid in the future.

There is consistency between $r_{i,n-i}$ and α_{n-i} only if they are related to the same DY.

For each row I can compute the **final number of claims as** $r_{i,n-i} \cdot \alpha_{n-i}$.

Now we want to split this numbers for the different columns. Why? Because in the different columns we have different average cost for the claims. Remember: we split the number of claims only in the lower triangle!

In order to divide them we can use the **settlement speed**: for the first RYs we have less columns to be estimated, so we have to consider the settlement speed for the columns, with the sum equal to 100% (vector normalised in different way for different rows).

Last row: $v_j^{(n)} = \frac{n_{n-j,j} \cdot \frac{d_n}{d_{n-j}}}{\sum_{j=1}^n n_{n-j,j} \cdot \frac{d_n}{d_{n-j}}}$ with d_i as number of reported claims, $j = 1, \dots, n$

$\frac{n_{n-j,j}}{\sum_{j=1}^n n_{n-j,j}} \rightarrow$ **Ratio** between the **number of payments** (at the last information available, so in the **last diagonal**) in the same column of the settlement speed I am computing and the **sum of payments** on the **diagonal**.

This ratio has a value between 0 and 1 and it is already a **settlements speed: incremental** and based on **numbers**.

So why we add $\frac{d_n}{d_{n-j}}$? This is add in order to corrected the cases in which the number of payments could lead to biased information.

Example: $n_{m-1,1} = 100$; $n_{m-2,1} = 200 \rightarrow V_1^n < V_2^n$ but can I say $SS_1 < SS_2$? No, in fact, considering also the number of reported claims $d_{m-1} = 500$; $d_{m-2} = 2000$ we see that the ratio number of claims paid over number of claims reported is $\frac{1}{5}$ for the row m-1 and $\frac{1}{10}$ for the row m-2.

The presence of the reported claims is fundamental for the definition of a correct settlement speed.

$$v_1^{(n)} = \frac{\frac{n_{n-1,1}}{d_{n-1}}}{\frac{n_{n-1,1}}{d_{n-1}} + \dots + \frac{n_{0,n}}{d_0}}$$
 : we consider homogeneous values since in this way we have more consistency between columns.

d_n is used for the diagnostic part.

The sum of the vector V_j^n shall be **equal to 1** between **columns 1 and n**.

If we want introduce the **tail** (with the common assumption of including all the years in the tail in one factor $v_{n+}^{(n)}$) we shall make some **adjustments**:

- $\sum_{j=1}^n n_{n-j,j} \cdot \frac{d_n}{d_{n-j}} + \frac{r_{0,n}}{d_0} \cdot d_n$ at the denominator

-
$$v_{n+}^{(n)} = \frac{\frac{r_{0,n}}{d_0} \cdot d_n}{\sum_{j=1}^n n_{n-j,j} \cdot \frac{d_n}{d_{n-j}} + \frac{r_{0,n}}{d_0} \cdot d_n}$$
 we approximate the number of paid claims after 11 years

with the numbers of the open claims in the DY 11

(64)

The part in white is the computation of the numerator of the formula.

The part in light blue is the settlement speed, computed as value in white over sum of the white values.

31 is for the tail.

Why number are huge w.r.t the market data? SS based on number is around 60%, why here is 78.30%? The **SS** are **normalised** in order to be 100% between 1 and n $\rightarrow$ 78% it doesn't mean that 78% of the claims are paid in 1 but that between 1 and 12 are paid of that percentage in 1 (we are excluding 0).

In general: $\hat{n}_{i,j} = r_{i,n-i} \cdot \alpha_{n-i} \cdot v_j^{(i)}$, for which the sum for the row is exactly the total number,

since $\sum_j v_j^{(i)} = 100\%$.

Focusing on the CY n : one of the **problem** is that $v_j^{(n)}$ **depends** only on the **values** on the

diagonal: if there is an **outlier**, the settlement speed for the whole column is biased.

In order to mitigate this problem is possible not only to consider the last diagonal, but also the last three ones, checking for the stability and computing then the average.

In order to compute $\hat{n}$, so for all the rows beside the last one, we have to rescale $v_j^{(i)}$.

Generic formula, with tail:
$$V_j^{(i)} = \frac{n_{i-j,j} \cdot \frac{d_i}{d_{i-j}}}{\sum_{j=i+1-n}^n n_{i-j,j} \cdot \frac{d_i}{d_{i-j}} + \frac{r_{0,n}}{d_0} \cdot d_n}$$

[28/09]

MISSING START

Average paid cost

Underlying assumption: claims of the same DY are assumed to be similar (meaning similar kind of claims), not equal since they refer to different CY so they are exposed to different inflation rates.

$$\hat{m}_{i,j} = m_{n-j,j} \cdot \prod_{h=n+1}^{i+j} (1 + ir_h) \text{ with } ir_h \text{ as claim inflation rate; I stop the estimation at the}$$

calendar year in which I am (eg. $\hat{m}_{n,1} = m_{n-1,1} \cdot \prod_{h=n+1}^{n+1} (1 + ir_h) = m_{n-1,1} \cdot (1 + ir_{n+1})$); h moves

according to the CY.

If $ir = 0$ all the **values** in the **same column** are **equal**; for MTPL we have lower amount for the first DYs and higher amount for the last ones (due to bodily injuries).

Differently from C-L, with F-L we deal directly with inflation, we have to estimate directly the claim inflation in order to apply the methodology.

Claim inflation = Economic inflation (exogenous inflation) + spread (endogenous inflation).

Usually we cannot apply directly the economic inflation, since there is not perfect correlation with the claim inflation. In fact, the economic one refers to a different basket of goods with respect to insurance companies (eg. Damages to the car, bodily injuries).

Therefore, we introduce the **spread** in order to be **consistent** also with the **different LoBs** (eg. MOD and MTPL requires different inflation for the claims).

The **problem** with the **endogenous inflation** is that I don't have a **market estimation** (each companies do their own one) and the projection requires a **time horizon** of 11-13 years, so there is also a problem related at the volatile markets (just like today).

As for the settlement speed we use only the last diagonal, since is the most recent, but we can introduce some diagnostic (eg. Inflate the past data to compare values in the same column).

It is important not only to look at the numbers but also at the claims from which they come.

In order to consider also the **tail factor** there are **two** principal **possibilities**:

- 1) **Assume** that the **average cost of claims** in $n - 1$ is the same of n^+ , following the diagonal (considering the CY) $\rightarrow m_{0,12^+} = m_{1,11}$ **for construction since** $\hat{m}_{1,11} = m_{0,11} \cdot (1 + ir_{2005})$. This **assumption** is **realistic** since we have very large claims amount.

Drawback: if in the tail I consider different years (12^+), I should consider the inflation for the average duration of the tail.

- 2) Approximation specifically for 12^+ , like the estimation of the average paid cost as average reserve cost $\hat{m}_{0,12^+} = \frac{R_{0,11}}{r_{0,11}}$; by inflating we can estimate the whole column for the tail.

Cell by cell we can obtain the **incremental payments** $\rightarrow \hat{P}_{i,j} = \hat{n}_{i,j} \cdot \hat{m}_{i,j}$

Summing the lower triangle we can have the reserve: $\hat{R}_n = \sum_{j=1}^{12} \hat{P}_{n,j}$

This is not to be wrongly considered as the ultimate cost estimation, for which we have to include

also the element of the column 0: $UC_n = \sum_{j=0}^{12} \hat{P}_{n,j}$

Partial payments: following the balance sheet, the partial payments affect the total payments but not the number of payments (so it is present in P but not in n) $\rightarrow$ **The triangle of m is biased.**

Example: we have only two claims reported in 1993; C1 is totally paid in the 1993 with a cost of 100€; C2 is partially paid in 1993 (50€) and partially paid in 1994 (70€).

Disegnini Triangolo

$m_{1,1} = \frac{150}{1} \text{€}; m_{1,2} = \frac{70}{1} \text{€}$ are both distorted: in the first case the average paid cost is higher than the real one while in the second case it is lower.

This could lead to a problem in the reserving?

$P = m \cdot n \rightarrow \begin{cases} 150 \\ 70 \end{cases}$ so if we limit our analysis on the upper triangle, we see that the triangle of partial payments is not biased.

What happens in the lower triangle? Do we have perfect compensation as in the upper one?

We can state that the degree of the compensation depends on the stability of the partial payments with respect to the column: more stability, more perfectly compensated.

The stability in terms of partial payments is reflected both on the amount and on the numbers.

Why don't we use decimal values in the triangle? It would introduce too much uncertainty and volatility in the number of claims, which we have seen to be deterministically computable, with a closed and precise formula.

FREQUENCY SEVERITY

Instead of splitting the incremental payments, we focus on the cumulative ones: $C_{i,j} = n_{i,j}^c \cdot m_{i,j}^c$

We estimate separately the two amounts:

- $n_{i,j}^c = \begin{cases} n_{i,j} & \text{if } j = 0 \\ n_{i,j} + n_{i,j-1}^c & \text{otherwise} \end{cases}$ similar to the C-L but referring to the **numbers**.

- $m_{i,j}^c = \frac{C_{i,j}}{n_{i,j}^c}$ so it is not the sum of the average paid cost (**common mistake!!**).

For example: $m_{i,0}^c = m_{i,0}$; $m_{i,1}^c = \frac{P_{i,0} + P_{i,1}}{n_{i,0} + n_{i,1}}$ (if I was able to split the first two terms of numerator

and denominator from the second ones, then I would obtain $m_{i,0} + m_{i,1}$ but this is not possible for

obvious reasons) $\rightarrow m_{i,1}^c = \frac{m_{i,0} \cdot n_{i,0} + m_{i,1} \cdot n_{i,1}}{n_{i,0} + n_{i,1}}$ so it is a weighted average of the incremental

average paid cost, with numbers as weights.

If there is an increase in $m_{i,0}$, this is lower in $m_{i,1}^c$ since it is a weighted average

I can compute the **cumulative development factors**: instead of computing the **settlement speed** per amount we will compute it per **numbers** and we have to check for the stability of the cumulative average cost through the columns.

The behaviour of the development factors is similar to the C-L ones, that is a decreasing trend.

Higher the settlement speed, lower are the values of the λ 's; usually $SS^N > SS^A \rightarrow \lambda^N < \lambda^A$

Tail factor: $\lambda_{\infty}^n = \frac{n_{0,n}^c + r_{0,n}}{n_{0,n}^c}$; $\lambda_{\infty}^m = \frac{m_{0,\infty}^c}{m_{0,n}^c}$, with $m_{0,\infty}^c = \frac{C_{0,n} + R_{0,n}}{n_{0,n}^c + r_{0,n}}$

Even in this case if I don't want to use the reserve I can interpolate from the curve.

It is possible also to combine different methods.

SUMMARY OF THE RESULTS

We can compute $ULR = \frac{UC}{\text{Earned Premiums}}$ for each one of the methodologies to compare

them. It would be sufficient to use directly UC, but ULR is more suited since it is not affected by the volume of the premiums.

(75) the column for the insurer reserve is for the external actuaries: the balance sheet values are compared to the result of the external entity.

3. MARKET CONSISTENT VALUATION of technical liabilities (SII)

Market consistent liabilities: value of amount for which liabilities could be transferred between willing parties (**current exit value**).

The MCV is based on the concept of BE and RM.

Hedgeable liabilities: it is possible to find in the market some financial instruments that fully replicates the payoff of the liabilities (eg. Unit/Index linked w/o guaranteed). For this type of liabilities we don't have to use BE + RM, the market value of assets is exactly equal to the value of the liabilities. There is no need of RM since it is intrinsically considered in the market value of assets, as risk premium.

There are some problems with the longevity bonds: they don't hedge perfectly the liabilities, since the longevity is calibrated on the world's values.

There are no liabilities with this characteristic in the non life sector.

Non-Hedgeable liabilities:

- **BE: expected present value of cash out flows** minus cash in flows (used to cover liabilities)

- **RM:** remuneration for risk taking, as sort of **risk premium**; it is a **profit** when **released**.

Another interpretation: assume that the undertaking A sell all the claims reserve to B; A shall pay to B all the expected value for the claims + an additional remuneration for the risk. This is computed as the part of the capital that B must be put aside for the capital requirements (with all the consequent constraints): this is the **cost of opportunity** —> how much capital I'm losing not investing in the market. **CoC (cost of capital) = RM.**

BEST ESTIMATE

As said before, the BE is the expected present value of the cash flow - cash in flow: this is more common in life than in non life. In fact in life the concept of mathematical reserve is quite similar to the one of best estimate, with **4 main differences: technical basis** (locked, **first order** vs updated, **second order**), **revaluation rates** (present vs future ones, so the need for a model to forecast the future dynamic of the assets), **lapses** and **discounting (technical rate vs risk free rate + adjustment)**.

In **non life** the main **difference** is given by the **discounting**, which is not present in local GAAP. Even though the most important item of non life is the claim reserve, the premium reserve is the part that is more affected by the market consistent evaluation required by SII.

Premium reserve

In **local GAAP**: **PR=UPR** (unearned premium reserve) + **URR** (unexpired risk reserve); this last component cannot be negative and it is present when the UPR is not sufficient.

In **SII**: **PR** is obtained from a **projection of cash out and cash in**, for the **contracts already underwritten and open at the end of the year** of evaluation; for this type of contracts I have to consider the **claims** that could happen in the **future**, not the ones that are already occurred (those are in the claims reserve if not yet paid—> Green blocks in the figure 85).

We have also to **estimate** when these future claims will be **paid**, therefore it is crucial to have the computation of the **settlement speed**; why we have to consider when the future claims will be paid? Because we have to **discount** these values.

Finally, the histogram will stop when the last claims, regarding future periods, will be paid.

We can now appreciate that, as we expected, the main **difference** in the premium reserve between **local GAAP** and **SII** is the **perspective**: in the former case we look at the present, assessing the sufficiency (or not) of the UPR (obtained through the pro-rata approach), while in the latter case we are interested in the future, with quite relevance to the discounting process.

It is possible to have **multi-annual contracts**: in this case we have to consider more years for the possible occurrence of claims, so there is an higher probability to have to cover the claims; moreover here it is more relevant the component of **cash in**, since we have the annual part of the **premiums** paid by the policyholders.

Since the interval of time is wider than one year, we have to take into account also the possibility of **lapses** (considered as the stoppage of the coverage): this is consider in the **premiums** of multi-annual policies, that is **net of the probability to receive** the payments.

Inside of Capital Requirements for non life underwriting risk there is a part of capital for the lapse risk: this capital is 0 for the annual contracts, not negligible for the multi-annual ones.

Expenses: management (for all the contracts), settlement (for the claims) and acquisition (for the multi-annual).

With this definition of BE for premium reserve, it could happen that **BE < UPR** (eg. High safety loadings in pricing, high prudence), as the URR was negative.

[29/09]

EXAMPLE

Simplification of BE for premium reserve: **BE = CR · VM + (CR – 1)PVFP + AER · PVFP**
The idea is to use some indicator of the past (such as the Combined Ratio), in order to estimate the future behaviour of claims and expenses.

CR = $\frac{\text{Incurred claims} + \text{Claims expenses} + \text{Expenses}}{\text{Earned premium gross of acquisition cost}}$; if CR > 1, claims + expenses are

higher than premiums, so the reserve is not sufficient and we need URR. If CR < 1 → BE < UPP since the local GAAP is prudent.

VM: volume measure for **unearned premium**, that is the complementary value of the earned premium (UPP of the local GAAP).

Ideally: **VM ⇔ UPR** and **(CR – 1)VM ⇔ URR**

In **SII** we use **CR** while in **local GAAP** we use **modified LR** ($\frac{X_t^{(CY)} + V_t^{S(CY)}}{\text{Earned premiums}}$), without the run-off):

- In **both cases** we consider only the **claims** occurred in the **current year**.
- In **both cases** we **don't** have the **run-off**, why? Because we don't want to affect the computation of the premium reserve with the behaviour of the claims reserve related to claims not linked to the same contracts of the premium reserve.
- In **CR** we include also the **expenses**, so it is more consistent: in fact LR=1 is already a sign of insufficiency since we have to add also the expenses, while $CR = 1 \Leftrightarrow P \& L = 0$.
- In **CR** we include the **acquisition expenses** (both in numerator and in denominator) because this indicator is used by the companies for other purposes.

The following part of the formula for BE is important to consider also the possible existence of future premiums: in fact, also in local GAAP, when there is a correction of the UPR (so the presence of URR), this updating is linked also to the future premiums; analogously here we have the term $(CR - 1) \cdot PVFP$ (PVFP of existing contracts, more important for multi annual premiums) → PVFP is important only for the URR, not for UPR; If CR > 1, we are anticipating today the (CR-1) percent of future premiums because we are estimating that the future premiums are not sufficient. If CR < 1 we can reduce the reserve (of [CR-1]), since we expect future profit.

Embedded in PVFP there is also the **lapse risk**: I have to consider the probability of lapse, otherwise I would realised today a future non-existent profit. So the risk of lapse is present because in SII I can anticipate profit.

AER: acquisition expenses ratio, $\frac{\text{Acq exp}}{\text{Written premiums}}$ is anticipated to the present, even if the

acquisition expenses are in the future for multi-annual contracts; this is made in order to have agents that are interested in maintaining the contracts, so it is a sort of protection strategy for the insurance companies.

The advantage of this formula is that we use indicators and we don't have to forecast (except for the PVFP).

Claims reserve

For the claims reserve the most crucial point is regarding the **discounting** (since here we have to discount, not like in local GAAP): as already seen for some of the actuarial methods, in order to discount we have to take into consideration the incremental payments; only in this way we can split the payments for their corresponding CY and we can apply the right discounting rate (different for each diagonal).

Even if it may seem strange, also in claims reserve we can have some cash-in: this is regarding the **recoverables**, that are some amounts that must be paid to the company by the policyholders (eg. MTPL with retention quota, if the damage is higher than the retention, initially the whole claim is paid by the company, then the policyholder must repay the insurer). This mechanism was created in order to protect the victims of the incidents, who should receive the given amount notwithstanding the financial possibility of whom commits the incidents.

Given the presence of the recoverables, we have to take into account the possibility of default of the policyholders in the capital requirements: it is much more difficult to estimate the probability of insolvency of a policyholder than the one for the financial/reinsurance companies.

Cash out: claims and future settlement expenses.

Cash in: recoverables.

Problem: do we have/should we have to include the recoverables in the triangle?

The answer is not univocal: If we do include them, then we can observe also negative values in the triangle, since there is a time gap between when I pay the whole claim and when I obtain the refund (recoverable).

One possible drawback is that we require stability in the upper triangle to apply many of the actuarial deterministic methodologies, and if there are negative values this stability is difficult to be obtained; the problem is also with the stochastic approaches: for example if we suppose to have log-normal distribution as representation of the claim amount, then it is not possible to insert negative values.

In general is more common to estimate separately the recoverables, with some sort of scoring models for the policyholders.

DISCOUNTING RATES (APPUNTI)

[03/10]

With the introduction of SII, the estimation process for the claims reserve changes for two main reasons: the discounting and the recoverables (with the drawbacks that we have highlighted).

(Figure 94): the first two columns represent the local GAAP reserve.

In reality, it is more practical to group the claims by column, in order to take into account the CY for the reserve, especially if we want to discount these amounts.

The last column is the estimation of the incremental payments of the lower triangle: this is a different way in which we aggregate the amounts, therefore the final results of the two approaches are equal ($\sum \text{reserve AY/DY} = \sum \text{reserve in CY}$).

Reminder: the discounting is possible only for the methodologies that can give $\hat{P}$.

In reality the rates are not flat, therefore for each row in the triangle we have a different discounting rate.

$$BE = \sum_{t=1}^N \frac{E(CF_{n+t})}{(1 + r(0,t))^t} : N \text{ is the last year in which I expect to have the payments (it depends on}$$

the size of the triangle, the LoB and the internal policy of the company); $r(0,t)$ is the spot rate.

We should also decide when the claims are paid throughout the year (eg. At the end, at the beginning, in uniform way): here it is assumed that all the payments are made at the end of the year, so we can use the spot rates. If we use the uniform assumption we have to change the rates (eg. 1.5 years, we have to compute the 1st and the 2nd forward).

For the tail, the usual assumption is to group all the years after the last one in a single factor (12^+ is our examples): this is not a problem for the un-discounted reserve, whereas it is a real problem for the discounted one. We should estimate when the claims are paid, dividing for CY.

$$BE = \sum_{i=0}^n \sum_{j=n-i+1}^N \frac{\hat{P}_{i,j}}{(1 + r(0,j - n + i))^{j-n+i}} : n-i+1 \text{ is the next diagonal (n+1); } j-n+i \text{ is a sort of}$$

estimation of the lag between the current CY and when the claim is paid. Each cell is discounted at the evaluation date.

important: the underlying assumption for the application of the PC-L to obtain the BE is that the data have not any prudence embedded.

Example (95)

Here we have displayed the BE of the claims reserve with different discounting curves: the difference is given by the CY of the calibration.

The aim is to understand the effect of different discounting rates over the time, to analyse the impact of the financial condition of the market on the BE.

QIS2 → 225 mln and BE/UC=91.60%, something around 96%/97% if we consider the RM ($RM \approx 4\% / 5\%$)

June 2018 → 244 mln and BE/UC=99.24%, which is certainly over the 100% if we consider also the RM.

The introduction of MCV could lead to volatility in the reserve due to market condition, with a consequent increase of the volatility of the liabilities.

The reduction/increase of liabilities from local GAAP to SII depends on the market: for this reason, in this framework is fundamental to have A&L that move together (ALM).

In 2018, the rates were negative, therefore the BE was capitalised, not discounted.

Since the liabilities in non-life have low duration, they are affected more by rates of low time horizon.

Moreover, the short tailed LoB are less affected by rates with respect to long tailed LoB, since there is even shorter the time horizon.

The **duration of the liabilities** is a useful indicator also for non-life, since it is related to the reaction of the liabilities to different curves.

In general: $D = \sum_t \frac{\frac{t \cdot CF_t}{(1+i)^t}}{\frac{CF_t}{(1+i)^t}}$ the idea is to consider how the payoffs are distributed according to

the time.

$\frac{\Delta P}{P} = -D \cdot \frac{\Delta i}{i}$ in which P is the price of the asset; usually P and i are known and I want to

know how the price is effected by the change of the rate. As we can see, the price is proportional to the duration and it has a negative relationship with the discounting rate.

With flat rates, P=BE; without flat rate, we can use this formula only with a shift in the discounting curve.

$D_i = \frac{\sum_{j=n-i+1}^N (j-n+i) \hat{P}_{i,j} (1+r(0,j-n+i))^{-(j-n+i)}}{\sum_{j=n-i+1}^N \hat{P}_{i,j} (1+r(0,j-n+i))^{-(j-n+i)}}$ this is an indicator for the single AY/

RY of how much the liabilities are exposed to the variation of the rate curves.

Example (97)

$D_0 \approx 1$ (I expect to conclude the payments in one year time horizon)

D_1 between 1 and 2 (IF POSITIVE RATES), and so on.

Total duration for the time horizon → $D = \sum_{i=0}^n D_i$

Discounted cash flow → Denominator of the formula

Weighted cash flow → Numerator of the formula

Life insurance has always higher value for the duration than non-life, also for MTPL (one of the longest LoB): this happens because of the short maturity of the liabilities. Lower duration means less effect of rates and lower horizon for the asset to cover liabilities.

3.3 RISK MARGIN

The SII approach to quantify the RM is the CoC (cost of capital): cost for managing the liabilities until run-off.

$RM = \sum_{t=0}^T CoC \cdot v(0, t+1) \cdot SCR_t$ (t=0 is the evaluation date) → Capital that I need for the

present liabilities.

CoC is fixed to 6% in SII, while for the VIF depends on the company. Why the regulation fixes CoC? Because it should be fixed by the entity that buys the liabilities but there is no entity in reality, so it is given by the regulation.

6% should reflect the spread between the theoretical rate that can be obtained investing in the market and the real rate that is obtained keeping the capital within the company, given the solvency constraints (SCR). In fact, to assure the liquidity to cover the liabilities, there are some constraints, both in terms of time and in terms of quantity/quality.

Why we are discounting the SCR in t for the year $t+1$? Because the SCR computed in t is the amount that the company must keep at disposal until the end of the year $t+1$: this choice was made by EIOPA, and it is a more favourable scenario if the rates are positive.

Remember: in the discounting rates we don't have VA, there are only the risk free rates: the correction of the volatility is applicable only to the BE.

Which SCR I have to include in the RM computation? We consider only a subset of the risks used in the capital requirements calculation.

- **Underwriting risks** (non-life in this course)
- **Operational risks**
- **Default risks**: only the technical part (to cover the insurance aspects), such as failure of the reinsurance companies, failure of the policyholders, NOT failure of the financial companies (of the assets).
- **Material market risk**: part of risk not able to hedge particular liabilities (important for life)

We are neglecting all the part related to the financial part (market, intangible and part of default). Why? The RM is a sort of financial risk premium, which is already present in the financial instruments.

Diversification of RM, separate between LoB or all together? For sure the two approaches will yield different, since the SCR is different: only consider all LoB together, I can favourite of diversification.

(I) $MTPL \rightarrow RM_{MTPL} + GTPL \rightarrow RM_{GTPL} \geq$ (II) $RM_{MTPL+GTPL}$ (equal iff correlation=1)

(I) is related to the concept of current exit value;

(II) Is related to the concept that the companies in the market can take advantage of diversification.

Usually, in reality, it is preferred to compute the RM as a whole (with all the LoB together), with the only constraint that is related to the division between Life and Non Life business.

On important point that is not yet solved is the possible correlation in time, namely if there is correlation between SCR_t and SCR_{t-1} .

For the computation of the future SCR there are some possible simplification, since it is quite complex to project it for long time periods (both computational complexity and complexity in assumptions).

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The simplifications for the SCR projection are mostly used by the small companies, since for them it is proportionally more costly and harder to perform this kind of analysis.

Possible proxies (from the less to the most simple):

- 1) Approximate the individual risks or sub-risks, within some or all modules and submodules in the future SCR computation.
- 2) Approximate the whole SCR for each future year, using some **proportional approach** (widely

used in Non Life) $\rightarrow SCR_t = SCR_0 \cdot \frac{BE_t^{NET}}{BE_0^{NET}}$ (in which net indicates net of reinsurance).

Obviously the SCRs considered are only the ones used for the RM computation.

The idea is that the ratio $\frac{SCR_t}{SCR_0}$ can be fairly approximated by $\frac{BE_t^{NET}}{BE_0^{NET}}$; it is important to notice

that the BE are computed in a closed portfolio, so without considering any new claims in the following years. Eg. given BE_0 , then BE_1 is the best estimate at the end of the next year, considering only the claims at evaluation date (year 0). We use the same triangle of BE_0 but considering only the claims below the main diagonal (**DISEGNINO1**).

$$BE_0 = \frac{P_{1,11} + P_{2,10}}{(1 + r(0,1))} + \frac{P_{2,11}}{(1 + r(0,2))^2} \rightarrow$$
 For the first year we can use the spot rate to discount. (**DISEGNINO2, updated with 4 rows**)

$$BE_1 = \frac{P_{2,11}}{(1 + f(0,1,2))} + \frac{P_{3,10}}{(1 + f(0,1,2))} + \frac{P_{3,11}}{(1 + f(0,1,2)) \cdot (1 + f(0,2,3))} \rightarrow \text{In this case}$$

we have to use the forward rates, with a product of them in order to pass from the CY 13 to the CY 14.

Why do we have to use the forward rates? Because we don't know the spot rate at the end of the following year.

$$BE_3 = \frac{P_{3,11}}{(1 + f(0,2,3))}$$

With this approximation, we are assuming that there will be a reduction in the same proportion of a reduction of the BE: **a possible drawback is related to the risk move as average.**

- 3) Estimate the expected present value of future SCR based on duration (more common in Life).
- 4) Approximate the RM as a percentage of the BE, with magnitude given by the regulation: this is the one adopted by the small companies. This percentage will yield an higher RM, because it is set on market data.

$\frac{RM}{TP}$ gives us the incidence of the RM on the whole TP (**Local GAAP, SII MISSING REASONING**)

This ratio is something around 3%/4% for the Non Life Italian sector; higher for the Life sector, since the BE is higher.

4. STOCHASTIC MODEL FOR CLAIMS RESERVE

In this framework, we want to assess a **measure for the uncertainty of the results** obtained through the deterministic models (eg. $\sigma^2(R^{PC-L})$): we will see that this volatility comes both from the data and from the estimation process.

Initially we look at the volatility of the lower triangle → **Total run-off approach**: we need this approach in order to consider also the reserve amount in the impact of the volatility.

Since one of the main aim is to compute the capital requirements, we have one year as time horizon, therefore we don't have to capture the whole lower triangle, but only the first diagonal. This one year approach is an adaptation at different time horizon of the previous deterministic approaches.

An overview of the methods:

1) Closed formulae:

- (1993) **Mack formula**, used for the volatility of the reserve, using the classical PC-L; this is one of the approaches that can be adjusted for the one year time horizon.

Pro: easy to apply.

Drawbacks: it captures only the variance, not the shape of the distribution. Consequently it is not possible to obtain some confidence intervals, which are useful also for the external actuaries, in order to compare the company's reserve with the ones obtained with some other methods.

- **Merz-Wuthrich** (PC-L)

2) **GLM and bootstrapping** (PC-L)

3) **Collective Risk Model**: frequency and severity are split in two different random variables (F-L)

4) **Multivariate claim reserving**

5) **Machine Learning for Reserving**: we don't aggregate data (as for the triangles) but we consider each claim as singular; this is important since we can include the characteristic of the policyholder who makes the claims.

4.1 MACK MODEL

The Mack model uses the **C-L method**, without making any kind of assumption on the type of payments distribution (differently from the bootstrapping method).

The **aim**, as for the other total run-off approaches, is the **computation of the mean square error prediction ($msep(\hat{y})$)**, in which we have the reserve $\hat{R}$ in place of $\hat{y}$.

In general: $msep(\hat{y}) = E[(y - \hat{y})^2]$, in which y is a random variable, representing the effective realisation of the reserve, while $\hat{y}$ is the estimation of the reserve with the PC-L.

$E[(y - \hat{y})^2] = E[((y - E[y]) - (\hat{y} - E[\hat{y}]))^2]$ obtained simply with $\pm E[y]$.

Then if we suppose that $\hat{y}$ is an unbiased estimator, then:

$E[(y - \hat{y})^2] = E[((y - E[y]) - (\hat{y} - E[\hat{y}]))^2] = E[(y - E[y])^2] + E[(\hat{y} - E[\hat{y}])^2] - 2E[(y - E[y])(\hat{y} - E[\hat{y}])]$

The **first addendum** represents the **process variance**, that is the volatility of the process that depends on the **payments** and **characteristics of the claims**, embedded in each portfolio.

The **second addendum** is the so called **estimation variance**: this part of the volatility comes from the choice of the method for the estimation of the parameters.

The **third term** will be **neglected**, since its impact is small, and this choice leads often to an overestimation: this assumption can be seen as the **independence between y and $\hat{y}$** .

Finally $E[(y - \hat{y})^2] \approx E[(y - E[y])^2] + E[(\hat{y} - E[\hat{y}])^2]$; typically the overestimation coming from the positive dependency of y and $\hat{y}$ has not a strong impact.

The Mack formula is useful to compute $msep(\hat{y}) = E[(y - \hat{y})^2]$ (if we compute the square root of $msep(\hat{y})$ we obtain the prediction error).

Two steps process:

1) Focus on the **row** of the reserve: $msep(\hat{R}_i)$

2) Focus on the **total reserve**: $msep(\hat{R})$; in order to compute this second step, we have to consider the possible **dependency** between the reserve in **different AY**, and this is important because we have use the same λ_j for different rows of the same column.

Assumptions:

- $E[C_{i,j+1} | C_{i,j}] = C_{i,j} \cdot \lambda_j$ (For the **lower triangle**) → Remember: with C we indicate the realisation of the payments, whereas with $\hat{C}$ we refer to the estimation of these payments. We can check this assumption on the upper triangle: more the data are distant from this hypothesis, higher is the volatility.

We can rewrite it as $E[\frac{C_{i,j+1}}{C_{i,j}} | C_{i,j}] = E[f_{i,j} | C_{i,j}] = \lambda_j$ and we can see that the assumption of stability for the individual link ratio is already a requirement for the application of the PC-L, even though in the case of the deterministic approach I should condition on $C_{i,1}, \dots, C_{i,j}$

($E[\frac{C_{i,j+1}}{C_{i,j}} | C_{i,1}, \dots, C_{i,j}] = \lambda_j$). But it is the same because the **process is memoryless**

(Markov chain), therefore $C_{i,j}$ is the same filtration as $C_{i,1}, \dots, C_{i,j}$.

- $\text{Var}(C_{i,j+1} | C_{i,j}) = C_{i,j} \cdot \sigma_j^2 \rightarrow \sigma_j^2$ is the **common source of variability** for the different rows of a specific column and this is the variability for the process used to estimate the column.

We can rewrite it as $\text{Var}(\frac{C_{i,j+1}}{C_{i,j}} | C_{i,j}) = \text{Var}(f_{i,j} | C_{i,j}) = \frac{\sigma_j^2}{C_{i,j}}$: we can see that σ_j^2 is not the

variance of $f_{i,j}$, it must be **rescaled** for the cumulative payments involved (importance for the volatility of the cumulative payments/reserve).

If $f_{i,j}$ is exactly the same for the whole triangle, then the variability is absent and $\text{Var}(f_{i,j} | C_{i,j}) = 0$

On the other hand, if $f_{i,j}$ changes significantly for the different rows, then $\text{Var}(f_{i,j} | C_{i,j})$ is very high.

- $C_{i,0}, \dots, C_{i,n}$ and $C_{j,0}, \dots, C_{j,n}$ are independent $\forall i, j$ ($i \neq j$), which means that **payments in different rows are independent**, only for C not for $\hat{C}$ (we use the same λ to estimate the columns). This assumption is **not so realistic**, since C depends on the **settlement speed** and on the same **inflation** rate for the same diagonal.

When we apply the model (especially for capital requirements), we have to show to the supervisor authority that the assumption are respected by the data (upper triangle): to do so, the companies have few data at disposal to prove independence, therefore there is no high credibility for the test. The problem lies both in the size of the sample and in its the stability of the model: in fact, few changes in the data can change sensibly the realisation of the assumptions.

As consequence of the last assumption, the **cumulative payments** form a **Markov Chain** for each i ($0 \leq i \leq n$).

We assume as the only source of information for claims prediction the data at disposal (no external information): $D^{(n)} = \{C_{i,j} : i + j \leq n\}$ is the filtration used, updated every year.

At the moment we will work with **no tail**: the problem is in the calibration of the parameters λ and σ^2 for the tail, with a critical number of data at disposal and low stability.

How $\hat{\lambda}$ and $\hat{\sigma}^2$ are computed?

$$1) \hat{\lambda}_j = \frac{\sum_{i=0}^{n-j-1} w_{i,j} f_{i,j}}{\sum_{i=0}^{n-j-1} w_{i,j}} \text{ for } j = 0, \dots, n-1. \text{ This computation of the individual link ratio comes}$$

from a generalisation of the classical C-L: if $w_{i,j} = C_{i,j}$ then we obtain the **classical approach**.

The properties of this estimators are valid for all the choice of the weight, but with the Mack formula we have to consider the classical PC-L. One of the properties of $\hat{\lambda}$ is that it is an

unbiased estimator: $E[\hat{\lambda}_j | D^{(n)}] = \lambda_j$. This result implies also that $E[\hat{C}_{i,n} | D^{(n)}] = C_{i,n} \forall i$, meaning that also $\hat{C}_{i,n}$ is an **unbiased estimator**: this is consistent, if $\hat{\lambda}$ is unbiased, then also

the estimation for the ultimate cost shall be unbiased, and even the **reserve estimation** is **unbiased** (UC - main diagonal, which is deterministic).

- 2) $\hat{\sigma}_j^2 = \frac{1}{n-j-1} \cdot \sum_{i=0}^{n-j-1} w_{i,j} (f_{i,j} - \hat{\lambda}_j)^2$ for $j = 0, \dots, n-2$. The number $n-j-1$ is the classical observation minus 1 to obtain an unbiased estimator for the variance (**variance of the sample**); i varies from 0 to $n-j-1$, that are the same rows for which λ is computed. $(f_{i,j} - \hat{\lambda}_j)$ gives the distance between the ratio of two consecutive cumulative amounts and $\hat{\lambda}$, that is the distance between the ratio and its weighted mean: this difference is null if $f_{i,j}$ is constantly equal to $\hat{\lambda}_j \rightarrow \hat{\sigma}_j^2 = 0$.

Why do we take the weight of the difference? in order to be consistent with the computation of $\hat{\lambda}$, that involves the weights.

The more we go on in the DY, the less data we have at disposal to compute $\hat{\sigma}_j^2$: for $j=n-1$ we would have $\hat{\sigma}_{n-1}^2 = \frac{0}{0}$, but this does not make sense.

For the **last value of the variance** to estimate, we have two different possibility of computation:

- 1) $\hat{\sigma}_{n-1}^2 = \min(\frac{\hat{\sigma}_{n-2}^4}{\hat{\sigma}_{n-3}^2}; \hat{\sigma}_{n-2}^2; \hat{\sigma}_{n-3}^2)$ with the term in the middle which is useless. This way to compute the last estimator for the variance is based on the assumption that the variance decreases the more you go towards the tail, ideally with an exponential decreasing behaviour.
- 2) **Fitting a curve** on $\hat{\sigma}_0^2, \dots, \hat{\sigma}_{n-2}^2$, estimating in this way $\hat{\sigma}_{n-1}^2$.

If we want to include the tail, we have to estimate also $\hat{\sigma}_n^2$, adding an extreme variable factor.

[06/10]

Main properties of $\hat{\lambda}$:

- 1) $E[\hat{\lambda}_j | \mathbf{D}^{(n)}] = \lambda_j \forall j \rightarrow E[\hat{C}_{i,n} | D^{(n)}] = C_{i,n} \forall i$
- 2) $E[\hat{\lambda}_j] = \lambda_j \forall j$: this estimator is unbiased unconditionally, but this is not interesting for us since we will always measure the estimator given a triangle.
- 3) $E[\hat{\lambda}_j \cdot \hat{\lambda}_k] = \lambda_j \cdot \lambda_k \forall j, k \ j \neq k$: this implies that the two estimators are **uncorrelated (no linear relationship)**, which is different from independence. This is a **strong assumption** since a part of the denominator of λ_{j+1} is a part of the numerator of λ_j .

The **real dependency** between these two estimators is **negative** (since we have common terms in the numerator for one estimator and in the denominator for the other one) and it is **not linear**.

1), 2), 3) are **valid for every choice of the weights**; for every kind of C-L the estimators are unbiased.

$$\hat{C}_{i,n} = C_{i,n-i} \cdot \hat{\lambda}_{n-i} \cdot \dots \cdot \hat{\lambda}_{n-1} \rightarrow$$

$$E[\hat{C}_{i,n}] = C_{i,n-i} \cdot E[\hat{\lambda}_{n-i}, \dots, \hat{\lambda}_{n-1}] = C_{i,n-i} \cdot E[\hat{\lambda}_{n-i}, \dots, \hat{\lambda}_{n-1}] = C_{i,n-i} \cdot E[\hat{\lambda}_{n-i}] \cdot \dots \cdot E[\hat{\lambda}_{n-1}]$$

it is possible to prove that for $w_{i,j} = C_{i,j}$, then the estimators of λ_j are the ones with the **minimal**

conditional variance (and in this sense they are optimal): $\text{Var}(\hat{\lambda}_j | \mathbf{D}^{(n)}) = \min_{\alpha_{i,j} \in \mathbb{R}} \sum_{i=0}^{n-j-1} \alpha_{i,j} f_{i,j}$

with the minimum that is reached for $\alpha_{i,j} = \frac{C_{i,j}}{\sum_{i=0}^{n-j-1} C_{i,j}}$

This is the reason why Mack chose the weights of the classical C-L: for every different choice of weights, the variance will be higher (always considering unbiased estimators).

We will prove that the **variance of the estimator** is $\text{Var}(\hat{\lambda}_j | \mathbf{D}^{(n)}) = \frac{\sigma_j^2}{\sum_{i=0}^{n-j-1} C_{i,j}}$: this is similar

to $\text{Var}(\frac{C_{i,j+1}}{C_{i,j}} | C_{i,j}) = \text{Var}(f_{i,j} | C_{i,j}) = \frac{\sigma_j^2}{C_{i,j}}$. In both cases we have that the denominator of the

result that is also at the denominator of the component within the variance. The **former variance** is regarding the **volatility of the estimators**, the **latter variance** is regarding the **volatility of the process**.

If we apply another methodology instead of the PC-L, the variance of the process does not change, while the variance of the estimators obviously changes: in fact, the variance of the process depends on C and not $\hat{C}$.

Finally we can state that also $\hat{\sigma}_j^2$ are **unbiased** (conditionally and unconditionally).

As we have already said, we have two step process for the mean square error prediction.

Computing the variance for the reserve or for the ultimate cost is the same, because we can pass from one to the other adding algebraically a deterministic term (the last diagonal).

$$mse_p(\hat{R}_i | D^{(n)}) = mse_p(\hat{C}_{i,n} | D^{(n)}) = E[(C_{i,n} - \hat{C}_{i,n})^2 | D^{(n)}] = \{ \pm E[C_{i,n} | D^{(n)}] \} =$$

$$E[(C_{i,n} - E[C_{i,n} | D^{(n)}])^2 | D^{(n)}] + E[(\hat{C}_{i,n} - E[C_{i,n} | D^{(n)}])^2 | D^{(n)}] = \{ \hat{C}_{i,n} \text{ and } E(C_{i,n} | D^{(n)}) \text{ are assumed to be deterministic conditional on data} \} = \text{Var}(C_{i,n} | D^{(n)}) + (\hat{C}_{i,n} - E[C_{i,n} | D^{(n)}])^2$$

The first addendum is the process variance, while the second addendum is the estimation variance.

PROCESS VARIANCE

Proof

$$\text{Var}(C_{i,n} | D^{(n)}) = \text{Var}(C_{i,n} | C_{i,n-i}).$$

Firstly we have removed all the values outside the row i , since according to the Mack's assumption the cumulative payments are independent per row; then we have considered only the last observed value in the row i , since the cumulative payments form a Markov chain.

Between $C_{i,n}$ and $C_{i,n-i}$ we have the sum of the incremental payments in the lower triangle, therefore a sum of random variables.

Law of the total variance: $\text{Var}(X) = E[\text{Var}(X | Y)] + \text{Var}(E[X | Y])$, with X,Y random variables, variance within and variance between (EXAMPLE MISSING)

Going on with the proof:

$$\text{Var}(C_{i,n} | C_{i,n-i}) = E[\text{Var}(C_{i,n} | C_{i,n-1}) | C_{i,n-i}] + \text{Var}(E[C_{i,n} | C_{i,n-1}] | C_{i,n-i})$$

We apply the definitions of the variance and the mean according to the assumption of the

approach ($E[C_{i,j+1} | C_{i,j}] = C_{i,j} \cdot \lambda_j$; $\text{Var}(C_{i,j+1} | C_{i,j}) = C_{i,j} \cdot \sigma_j^2$):

$$E[\text{Var}(C_{i,n} | C_{i,n-1}) | C_{i,n-i}] + \text{Var}(E[C_{i,n} | C_{i,n-1}] | C_{i,n-i}) = \sigma_{n-1}^2 E[C_{i,n-1} | C_{i,n-i}] + \lambda_{n-1}^2 \text{Var}(C_{i,n-1} | C_{i,n-i})$$

Now we can consider $E[C_{i,n-1} | C_{i,n-i}] = E[E[C_{i,n-1} | C_{i,n-2}] | C_{i,n-i}]$ with

$$E[C_{i,n-1} | C_{i,n-2}] = \lambda_{n-2} \cdot C_{i,n-2} \text{ and we can iterate the process.}$$

$$\text{Finally, we obtain: } \text{Var}(C_{i,n} | \mathbf{D}^{(n)}) = \sigma_{n-1}^2 \prod_{h=n-i}^{n-2} \lambda_h C_{i,n-i} + \lambda_{n-1}^2 \text{Var}(C_{i,n-1} | C_{i,n-i})$$

in which $h=n-i$ is the last point in which I can iterate.

it is possible to stop here, with the variance of the ultimate cost given the diagonal that depends both on parameters and on the variance of the column before the one I am considering: this is a recursive formula, that starts from the variance of the main diagonal that is zero (deterministic).

It is possible to extend the formula with the inclusion of the tail (with the usual problems of reliability and volatility).

We can also rewrite the formula in a more compact way, considering $Var(C_{i,n-1} | C_{i,n,i})$ as a function of $C_{i,n-2}$ and iterate the process:

$$Var(C_{i,n} | D^{(n)}) = \sigma_{n-1}^2 \prod_{h=n-i}^{n-2} \lambda_h C_{i,n-i} + \lambda_{n-1}^2 [\sigma_{n-2}^2 \prod_{h=n-i}^{n-3} \lambda_h C_{i,n-i} + \lambda_{n-2}^2 Var(C_{i,n-2} | C_{i,n-i})] = \dots$$

$$= C_{i,n-i} \sum_{m=n-i}^{n-1} \left(\prod_{h=n-i}^{m-1} \lambda_h \cdot \sigma_m^2 \cdot \prod_{h=m+1}^{n-1} \lambda_h^2 \right)$$

The number of elements in the product depends on the value of m .

The more you go back, the more terms in λ^2 you are adding and the less terms in λ .

(EXAMPLE)

With this compact formula, it is easier to compute the variance, since we need only the parameters; the recursive formula is useful if we are interested in computing only the next diagonal (for the one year approach).

How to estimate $\hat{\lambda}$ and σ ? The assumption of Mack is to use the estimation from the classical C-L

Remember: $\hat{\lambda}, \hat{\sigma}, \hat{C}$ are numbers, not random variables.

So we estimate the process variance using the estimator for λ_j and σ_j^2 :

$$\widehat{Var}(C_{i,n} | C_{i,n-i}) = C_{i,n-i} \sum_{m=n-i}^{n-1} \left(\prod_{h=n-i}^{m-1} \hat{\lambda}_h \cdot \hat{\sigma}_m^2 \cdot \prod_{h=m+1}^{n-1} \hat{\lambda}_h^2 \right) \text{ we can take the cumulative value}$$

inside the summation, since $C_{i,n-i} \cdot \prod_{h=n-i}^{m-1} \hat{\lambda}_h = \hat{C}_{i,m}$ for the classical C-L.

$$C_{i,n-i} \sum_{m=n-i}^{n-1} \left(\prod_{h=n-i}^{m-1} \hat{\lambda}_h \cdot \hat{\sigma}_m^2 \cdot \prod_{h=m+1}^{n-1} \hat{\lambda}_h^2 \right) = \sum_{m=n-i}^{n-1} \left(\hat{C}_{i,m} \cdot \hat{\sigma}_m^2 \cdot \prod_{h=m+1}^{n-1} \hat{\lambda}_h^2 \right)$$

$$\text{Then we multiply and divide for } \hat{\lambda}_m^2: \sum_{m=n-i}^{n-1} \left(\hat{C}_{i,m} \cdot \hat{\sigma}_m^2 \cdot \prod_{h=m+1}^{n-1} \hat{\lambda}_h^2 \right) = \sum_{m=n-i}^{n-1} \left(\hat{C}_{i,m} \cdot \frac{\hat{\sigma}_m^2}{\hat{\lambda}_m^2} \cdot \prod_{h=m}^{n-1} \hat{\lambda}_h^2 \right)$$

$$\text{Again } \hat{C}_{i,m}^2 \cdot \prod_{h=m}^{n-1} \hat{\lambda}_h^2 = \hat{C}_{i,n}^2 \text{ and finally:}$$

$$\widehat{Var}(C_{i,n} | C_{i,n-i}) = \hat{C}_{i,n}^2 \cdot \sum_{m=n-i}^{n-1} \frac{\hat{\sigma}_m^2}{\hat{\lambda}_m^2 \cdot \hat{C}_{i,m}} \rightarrow \text{The process variance of the reserve of row } i.$$

For $m = n - i$, $\hat{C} \equiv C$; for all the other values $\hat{C}$ is estimated with the C-L.

In the proof we did not fix the weights, therefore this formula holds for **every choice of the weight**. Obviously the values for $\hat{\lambda}$ and $\hat{\sigma}$ change for the different choices of $w_{i,j}$.

This result is in line with the fact that we are computing the **process variance**: it is quite trivial to notice that the variability of the data **should not depend on the weights** used in the estimation of the lower triangle.

For each row we consider the volume of the ultimate cost ($\hat{C}_{i,n}^2$): we are connecting the risk measure with the ultimate cost; the number of elements in the summation is the number of elements to estimate, namely the number of random variables involved (and coming from $\hat{C}$).

There is one lag regarding the column that I want to estimate and the parameters: this idea comes from the CL, which uses the information of the previous column to compute the following one.

$$\text{What is } \frac{\hat{\sigma}_m^2}{\hat{\lambda}_m^2 \cdot \hat{C}_{i,m}^2}?$$

$$\frac{\sigma_m}{C_{i,m}} = Var(f_{i,m} | C_{i,m}) \rightarrow \frac{\hat{\sigma}_m}{\hat{C}_{i,m}} = \widehat{Var}(f_{i,m} | C_{i,m}): \text{this term is composed by the } \mathbf{common} \\ \mathbf{variance} \text{ for the } \mathbf{column}, \mathbf{rescaled} \text{ according to the } \mathbf{cumulative payments} \text{ of the cell.}$$

$\hat{\lambda}^2$ is the **squared value of the mean of f** , its weighted average.

Therefore $\frac{\hat{\sigma}_m^2}{\hat{\lambda}_m^2 \cdot \hat{C}_{i,m}^2}$ can be seen as $\frac{Var^2(f_{i,m})}{E^2[f_{i,m}]} = CoV^2$: it is a **sort of measure of relative**

volatility of f .

[10/10]

An example (15)

The triangle in the Mack's example comes from reinsurance: this means that it involves high volatility (also negative value) that is reflected on the individual link ratios (from 1 to 40 in the example). This is pretty common for the reinsurance sector, especially for the non proportional part of the treaties. In addition we are in a long tailed LoB (GTPL).

We can appreciate a typical **trade-off** of the stochastic methods: on one hand, if we use the **classical C-L**, we have to take into account all the value in the triangle (also possible **outliers**), and we know that we will obtain the minimum variance for an unbiased estimator; on the other hand, if we decide to consider only some values (to omit the outlier), we should obtain a more consistent value for the reserve, but the variance will be higher than in the previous case.

Using the Mack formula, we don't have to guarantee positive amount for the incremental payments: this assumption will be crucial for all the methodology that involves a fixed probability distribution; in that case it is possible to consider some distributions that require positive amounts.

(17) We can see the results obtained from the classical C-L, compared with some other choices for the weights: higher the volatility of the link ratios, higher the difference between different estimators. For example, in this case, the factor that introduces high volatility is the '40': with different choices for the weights, we can give more or less importance to this value, and this possibility allows the results to sharpen their difference. In the example, the weighted average is preferable to the arithmetic mean.

(18) Application of the Mack formula: λ_j is based on the classical C-L; σ_j^2 is based on the formula that takes into account the squared difference of λ_j and $f_{i,j}$, weighted by the cumulative payments (for all the values, except the last one). The last value of the variance is obtained with

$\hat{\sigma}_{n-1}^2 = \min(\frac{\hat{\sigma}_{n-2}^4}{\hat{\sigma}_{n-3}^2}; \hat{\sigma}_{n-2}^2; \hat{\sigma}_{n-3}^2)$, but also with the interpolation in this case we would obtained a similar result.

in terms of absolute value, we cannot say if σ_j^2 is high (27883.48), it depends on the parameters.

The behaviour of σ_j^2 is in line with the assumption of Mack of exponential decreasing behaviour. In fact there is a more or less linear behaviour in the logarithm of σ_j^2 .

(19) Process variance: $1.5024 = \frac{Var(f_{9,0})}{\hat{\lambda}_0^2}$ is a sort of measure of relative volatility of f (CV^2).

Why the CV's are decreasing towards the tail? The difference between $f_{i,j}$ and λ_j reduces and $C_{i,j}$ is higher.

We compute the process variance for each row, as product of volume measure (cumulative payments) and sum of the relative volatilities; then, since the cumulative payments are assumed to be independent by row, the process variance for the total reserve is simply the sum of the process variances of the different rows.

Total CV=0.48 is quite high number: in fact, in terms of capital requirements we would have

$$\frac{CaR}{E[R]} = 3 \cdot 48 \%$$

More in general, if we consider a generic percentile of the Gaussian X_ϵ and percentile of the

standard Normal at the same level ϵ , $Y_\epsilon: X_\epsilon = Y_\epsilon \cdot \sigma(X) + E[X] \rightarrow Y_\epsilon \cdot \sigma(X) = X_\epsilon - E[X]$

meaning that $k \cdot \sigma(R)$ represents the distance between the expected value of the reserve and the percentile of a Gaussian distribution.

Therefore $3 \cdot 48\%$ gives us an idea of the ratio between the capital requirement and the mean: since this is a value greater than 1 (namely 1.44), this indicates that the capital requirement is 1.44 times the reserve, and if we think about GTPL, the reserve is higher than premiums (4/5 times). The number 3 is present also in SII (Non-Life underwriting risk), since it is higher than the 2.58 coming from the Gaussian distribution, indicating the presence of the positive skewness.

Lognormal distribution: Y_ϵ depends on the confidence level, on the skewness of the distribution and also on the CV $\rightarrow$ I cannot have a formula like for the Normal, since the value is not constant given ϵ (higher the skewness, higher the percentile).

Of course, the fact that the multiplier is fixed at 3 for the Non-Life underwriting risk leads to an overestimation of the capital requirement for the low skewed businesses and an underestimation for the highly skewed businesses (problem for small companies).

(20) We can compute the process variance also in a recursive way: I start from the diagonal and I compute one column at a time.

The formula is based on two components:

- 1) The **first addendum** consider the **additional volatility** coming from the **payments of new claims**. If I knew in advance the new claims, I would have no volatility.
- 2) The **second addendum** consider the fact that we are working with cumulative payments: it consider the variance of the previous cell. I multiply the variance of the previous cell for $\hat{\lambda}$, why? **In order to consider how the variance increases if the payments go as expected. In this case the variance increases because the volume of the payments increases.**

In the example:

- 1) $\sigma_0^2 \hat{C}_{9,0}$ variance coming from the new payments in the column, if they don't go as expected.
- 2) $\widehat{Var}(C_{9,0} \cdot \hat{\lambda}_0)$ variance of what we expect to obtain for $C_{9,1}$ in terms of cumulative amounts.

$C_{9,0}$ is deterministic, therefore the second component is null.

The last column of this triangle is the process variance, since it is the variance of the ultimate cost.

ESTIMATION VARIANCE

We start from $(\hat{C}_{i,n} - E(C_{i,n} | D^{(n)}))^2$; then applying the usual transformation of the conditional

mean ($E[C_{i,n} | D^{(n)}] = E[E[C_{i,n} | C_{i,n-1}] | D^{(n)}]$) we can obtain $C_{i,n-i}^2 \left(\prod_{t=n-i}^{n-1} \hat{\lambda}_t - \prod_{t=n-i}^{n-1} \lambda_t \right)^2$.

In this case we cannot replace λ with $\hat{\lambda}$ as we have done with the process variance, because we want to compute the measure of volatility for the estimation: therefore $\hat{\lambda}$ is a random variable.

We introduce a random variable S_k such that $\left(\prod_{t=n-i}^{n-1} \hat{\lambda}_t - \prod_{t=n-i}^{n-1} \lambda_t \right)^2 = \left(\sum_{k=n-i}^{n-1} S_k \right)^2$.

This new random variable can be defined as $S_k = \prod_{t=n-i}^{k-1} \hat{\lambda}_t \cdot (\lambda_k - \hat{\lambda}_k) \cdot \prod_{t=k+1}^{n-1} \lambda_t$, in which the

extremes depends on k .

How we can pass from a product to a summation? **Telescopic sum**: only the first and the final terms remains in the end, the rest of the terms cancel out.

We can rewrite $(\hat{C}_{i,n} - E(C_{i,n} | D^{(n)}))^2$ as $\hat{C}_{i,n-i}^2 \left(\sum_{k=n-i}^{n-1} S_k^2 + 2 \sum_{n-i \leq h < k \leq n-1} S_h S_k \right)$ and the second

term is simply a double summation $\sum_{h=n-i}^{n-2} \sum_{k=h+1}^{n-1} (\cdot)$ (n-2 for the first summation since we cancel

out the case of $k=h$).

Assumption of Mack: approximation of a random variable with its expected value conditional to a specific set $B^{(k)}$ (set of cumulative amounts).

$B^{(k)} = \{C_{i,j} : i+j \leq n, j \leq k\}$ therefore $B^{(k)} \subseteq D^{(n)}$ **(DISEGNINO)**

We compute now the conditional expectation on $B^{(k)}$ for both of the values in the last formula. S_k is not deterministic in $B^{(k)}$, because it contains $\hat{\lambda}_k$ that it is obtained by C_{k+1} , which value is not present in $B^{(k)}$.

[12/10]

We want to understand which part of $E[\sum_{n-i \leq h < k \leq n-1} S_h S_k | B^{(k)}]$ is a **random variable**: given

$B^{(k)}$, the part of S_h regarding the product of the $\hat{\lambda}$ from $n-i$ to $h-1$ is deterministic, as well as $\hat{\lambda}_h$, since it depends on C_{h+1} with $h < k$ (meaning that $h + 1 \leq k$) $\rightarrow$ All the terms of S_h are **deterministic**.

For S_k , given $B^{(k)}$, the part of the product of the $\hat{\lambda}$ from $n-i$ to $k-1$ is deterministic, whereas $\hat{\lambda}_k$ is a **random variable** since it involves C_{k+1} , which is not present in $B^{(k)}$.

Having said that,

$$E[\sum_{n-i \leq h < k \leq n-1} S_h S_k | B^{(k)}] = \sum_{n-i \leq h < k \leq n-1} S_h \prod_{t=n-i}^{k-1} \hat{\lambda}_t \cdot E[(\lambda_k - \hat{\lambda}_k) | B^{(k)}] \cdot \prod_{t=k+1}^{n-1} \lambda_t$$

Since we have seen that the estimator for λ is **unbiased**, $E[\hat{\lambda}_k] = \lambda_k \rightarrow$

$$\sum_{n-i \leq h < k \leq n-1} S_h \prod_{t=n-i}^{k-1} \hat{\lambda}_t \cdot E[(\lambda_k - \lambda_k) | B^{(k)}] \cdot \prod_{t=k+1}^{n-1} \lambda_t = 0$$

Therefore we can rewrite the estimation variance as $\hat{C}_{i,n-i}^2 \sum_{k=n-i}^{n-1} S_k^2$, from which

$\hat{C}_{i,n-i}^2 E[\sum_{k=n-i}^{n-1} S_k^2 | B^{(k)}]$. Following all the passage of the slide 25, Mack decided to use the

weights of the classical C-L, since they assure the **minimum variance** between the **unbiased** estimators.

The variance of the summation of independent random variable is equal to the sum of the variance: this is true in this case since we have supposed that the cumulative payments are independent per row.

Why it is sufficient to consider $B^{(k)}$ in place of $D^{(n)}$? Both for the property of **independence** of cumulative payments per **row** and for the property of **markovianity** of these cumulative amounts.

For a single value of k : $Var(C_{i,k+1} | B^{(k)}) = Var((C_{i,k+1} | C_{i,k}) | B^{(k)}) = \sigma_k^2 C_{i,k}$

For the whole summation we obtain the final result:

$$(\hat{C}_{i,n} - E(C_{i,n} | D^{(n)}))^2 = C_{i,n-i}^2 \sum_{k=n-i}^{n-1} [\prod_{t=n-i}^{k-1} \hat{\lambda}_t^2 \frac{\sigma_k^2}{\sum_{i=0}^{n-k-1} C_{i,k}} \prod_{t=k+1}^{n-1} \lambda_t^2]$$

Now, as for the process variance, **we use the estimators for λ_j and σ_j^2** (since we don't have r.v.'s anymore) and we multiply and divide for λ_k^2 (in order to obtain the ultimate cost) : given

$$C_{i,n-i}^2 \cdot \prod_{t=n-i}^{k-1} \hat{\lambda}_t^2 \cdot \hat{\lambda}_k^2 \cdot \prod_{t=k+1}^{n-1} \lambda_t^2 = \hat{C}_{i,n}^2 \rightarrow (\hat{C}_{i,n} - E(C_{i,n} | D^{(n)}))^2 = \hat{C}_{i,n}^2 \sum_{k=n-i}^{n-1} [\frac{\sigma_k^2}{\hat{\lambda}_k^2 \sum_{i=0}^{n-k-1} C_{i,k}}]$$

In the meantime, we have **demonstrated** that $Var[\hat{\lambda}_k | B^{(k)}] = \frac{\sigma_k^2}{\sum_{i=0}^{n-k-1} C_{i,k}}$, which is **similar** to

$Var[f_{i,k} | B^{(k)}] = \frac{\sigma_k^2}{C_{i,k}}$: in both cases there is a **common variability** σ_k^2 , which depends on the

empirical estimation of the variance of $f_{i,k}$, meaning that the variance of $f_{i,k}$ affects also the variance of λ_k .

This **common variance** is **rescaled** for the amounts involved in the computation of $f_{i,k}$ and λ_k ,

respectively $C_{i,k}$ and $\sum_{i=0}^{n-k-1} C_{i,k}$.

Looking at the formula, the **EV** is very **similar** in the structure to the **PV**:

- 1) **Volume measure** outside the summation ($\hat{C}_{i,n}^2$)
- 2) **Summation** with the number of elements which is related to the number of cells to be estimated.
- 3) **Index** of the summation is the same: **lag** of one period between the index to consider and the time of cell to be estimated.

The main **difference** between the process variance and the estimation variance is that for the former we are focusing on the **f**, whereas for the latter it is involved the λ_k : in fact, similarly to

$$\frac{Var(f)}{E[f]^2} \approx CV^2, \text{ here we have } \frac{Var(\hat{\lambda}_k)}{E[\hat{\lambda}_k]^2}, \text{ that is again a measure of relative volatility.}$$

Observation: the value within the summation in the EV ($\sum_{k=n-i}^{n-1} [\frac{\sigma_k^2}{\hat{\lambda}_k^2 \sum_{i=0}^{n-k-1} C_{i,k}}]$) **does not**

depend on the row i : indeed this is a common factor for all the values in the same column.

Why there is a common component? Because the error coming from the estimation depends on the $\hat{\lambda}$ used, and this is the **same** one for the whole **column**; the **cumulative payments** amount is used to **rescale** for the magnitude of the specific row.

To obtain the common factor it is possible simply to compute the relative variance for the last row: in this way, I have all the values for the lower triangle.

Total mean square error process for the row i :

$$mse_p(\hat{R}_i | D^{(n)}) = \hat{C}_{i,n}^2 \sum_{j=n-i}^{n-1} \frac{\hat{\sigma}_j^2}{\hat{\lambda}_j^2 C_{i,j}} + \hat{C}_{i,n}^2 \sum_{j=n-i}^{n-1} \frac{\hat{\sigma}_j^2}{\hat{\lambda}_j^2 \sum_{i=0}^{n-j-1} C_{i,j}} = \hat{C}_{i,n}^2 \sum_{j=n-i}^{n-1} [\frac{Var(f_{i,j}) + Var(\hat{\lambda}_j)}{\hat{\lambda}_j^2}]$$

Since we are using the same estimator for the column, it is not possible to assume that the mse of the total reserve is simply the sum of the mse for the different rows.

in fact, only for the estimation variance, it is considered a positive covariance:

$$mse_p(\hat{R} | D^{(n)}) = \sum_{i=1}^n mse_p(\hat{R}_i | D^{(n)}) + 2 \sum_{i=1}^{n-1} [\hat{C}_{i,n} (\sum_{h=i+1}^n \hat{C}_{h,n}) \sum_{j=n-i}^{n-1} \frac{\hat{\sigma}_j^2}{\hat{\lambda}_j^2 \sum_{i=0}^{n-j-1} C_{i,j}}]$$

We start **from $i=1$** , because there is **no tail**. The term of the covariance considers the covariance between all the couples of rows.

Single covariance (equal both for the reserve and for the ultimate cost):

$$Cov(\hat{C}_{h,n}, \hat{C}_{i,n} | D^{(n)}) = \frac{\hat{C}_{h,n}}{\hat{C}_{i,n}} Var(\hat{C}_{i,n} | D^{(n)}) \text{ with } h > i; \text{ this factor depends on both the}$$

ultimate cost of the row h and on a factor similar to EV of the row i (Var over ultimate cost).

Higher the contribution of EV, higher the effect of the dependance.

Now we can compute the correlation coefficient between reserves of a couple of rows: ($h > i$)

$\rho(\hat{R}_i, \hat{R}_h | D^{(n)}) =$ we consider only the estimation component, neglecting the process variance =

$$\frac{Var(C_{i,n} | D^{(n)}) \cdot \frac{\hat{C}_{h,n}}{\hat{C}_{i,n}}}{\sqrt{Var(C_{i,n} | D^{(n)}) \cdot Var(C_{h,n} | D^{(n)})}} = \text{(with the process variance at the denominator we would have the product of the mse, both for } i \text{ and for } h) =$$

$$\frac{\sqrt{Var(C_{i,n} | D^{(n)})} \cdot \frac{\hat{C}_{h,n}}{\hat{C}_{i,n}}}{\sqrt{Var(C_{h,n} | D^{(n)})}} = \frac{\sqrt{Var(C_{i,n} | D^{(n)})}}{\sqrt{Var(C_{h,n} | D^{(n)})}} \cdot \frac{\hat{C}_{h,n}}{\hat{C}_{i,n}}$$

$$\rho(\hat{\mathbf{R}}_i, \hat{\mathbf{R}}_h | \mathbf{D}^{(n)}) = \frac{\sqrt{\sum_{j=n-i}^{n-1} \left(\frac{\sigma_j^2}{\lambda_j^2 \cdot \sum_{i=0}^{n-j-1} C_{i,j}} \right)}}{\sqrt{\sum_{j=n-h}^{n-1} \left(\frac{\sigma_j^2}{\lambda_j^2 \cdot \sum_{i=0}^{n-j-1} C_{i,j}} \right)}}$$

If the process variance is included, the only difference between $msep(i)$ and $msep(h)$ is in the number of the elements in the summation: under the assumption $h > i$, the number of elements at the denominator is higher than the number of elements at the numerator → **The ratio is always lower than one.**

Given the previous characteristics of the **correlation coefficient**, we can say that it is **always positive** and its value varies **between 0 and 1**.

The **dependency is not constant throughout the rows**: in fact the **closer** they are (**AY**), the **more dependent** they are. Why? because there are **more elements** (of the lower triangle) that are computed with the **same lambda** (**DISEGNINO**).

For the whole process, the value of the correlation coefficient will be lower, since we have the same numerator with an higher denominator.

One of the aim of the stochastic approaches is to build an **interval of confidence**: one possibility to do it is to make some an **assumption** regarding the **distribution of the reserve**, usually the reserve is considered to be distributed as a **Log-Normal**. In this way I can compute, beside the mean and the standard deviation, all the moments (with the MM) and also the confidence interval (**typically 25%-75%**). (**STAKEHOLDERS DISTRIBUTION**)

The recursive formula, present for the process variance, holds also for the estimation variance; it holds also for the the tail, but with some complications. (**MISSING COMMENT**)

How to deal with the presence of the tail?

- **Apply Mack's formula without the tail**: I shall justify this choice to the supervisor authorities (since it is important regarding the capital requirements), particularly I shall demonstrate that in this way I don't have biased volatility. in this case I compute $msep(\hat{R}^{NoTail})$, then $msep(\hat{R}^{Tail})$ with another algorithm, finally $msep(\hat{R}) = CV^{NoTail} \cdot \hat{R}^{tail}$. Here the underlying assumption is that the **relative volatility doesn't change** with the presence/absence of the **tail**.
- I can include the **tail in the year n**: $C_{0,n}^{NEW} = C_{0,n} + R_{0,n}$ (THIS IS NOT THE RESERVE OF C-L WITH TAIL) and $\hat{\sigma}_n^2$ is **included in** $\hat{\sigma}_{n-1}^2$. The problem here is in the last assumption: we have already seen all the problems related to $\hat{\sigma}_{n-1}^2$ and estimating also the tail in this way can produce an higher error.
- I can **enlarge the triangle**, with the usual drawbacks regarding the reliability of the data.

Until now we are not including the discounting: **CV = 10% in Local GAAP is the same as CV = 10% in SII? No**, since the amount on the **different diagonals** are discounted with different terms and in Local GAAP I can introduce some judgement . Notwithstanding this difference, the **common assumption** is to consider them **equal**.

It is quite complex to include in a closed formula the discounting, if we are not considering a flat rate.

FINAL PART LISTENING.

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(37)

From A to D, we have that the individual link ratio of the first column is more volatile (since the maximum value is higher and the lower value is lower); the resulting triangle is similar to the starting one but the volatility is higher.

From A to E, change only in the last column: even though here the change is lower than the one on D, the total variability is higher since the effect on the last value is more impactful, since the magnitude is higher.

In D and in E we have effect on σ , so both on the process and on the estimation variance, but I am affecting directly f , therefore I expect an higher impact on the process part.

4.2 ODP MODEL

Now we want to introduce the bootstrapping methodology: we keep using C-L, but we want to obtain the distribution underlying the process, specifying the distribution of the payments. In this framework we make the assumption to have **payments** distributed as **ODP**: even though we are using a discrete model to describe a continuous variable, in several practical cases it works.

The main **limit** of the **Poisson** is that the **mean** is equal to the **variance**: in reality, we usually observe an higher value for the variance than the one of the mean.

With the ODP we allow the variance to be different than the mean; conceptually it is similar to the mixed Poisson, since we introduce an element (over-dispersion parameter/ $\tilde{q}$) in order to add some contribution for the variance.

4 main assumptions:

- 1) $P_{i,j} \sim \text{ODP}$ and **independent**, not only by **rows** as in Mack (even though in that case we are dealing with cumulative amounts), but also by **columns**. This is a critical assumption. **IMPORTANT**: we are **not assuming IID** variables.
- 2) $E[P_{i,j}] = x_i y_j$. We introduce two parameters, one that depends on rows and one that depends on columns. x_i takes into account that **different rows** can have **different** volumes for **UC**, in fact if the UC are equal for all the rows, then it is not important to consider it; y_j is similar to SS.

- 3) $\sum_{j=0}^n y_j = 1 \wedge y_j > 0$. Since the summation stops in n , we are **neglecting the tail**.

Why it is important this assumption? In this way $E[C_{i,n}] = E[\sum_{j=0}^n P_{i,j}] = \sum_{j=0}^n E[P_{i,j}] = x_i \sum_{j=0}^n y_j = x_i$

This means that if the model is a perfect **replication** of the **C-L**, then $x_i = UC_{CL}$.

y_j can be interpreted as **incremental settlement speed**, since it is the proportion of UC paid in the year ($E[P_{i,j}] = E[C_{i,n}]y_j$).

Not only $P_{i,j} \sim ODP$, but also $C_{i,j} \sim ODP$, because it is sum of independent poisson (over-dispersed).

Another important result: $E[\frac{P_{i,j}}{P_{i,h}}] = \frac{y_j}{y_h} \forall j, h \rightarrow$ The **incremental ratio of two columns** does

not depend on the **row**: if we suppose **two adjacent columns**, we have an **assumption similar** to the one for **CL**, but for the incremental payments.

- 4) $\text{Var}(P_{i,j}) = \phi x_i y_j$. ϕ is the **over/under-dispersion parameter**; it can be estimated, so it is an **output** for the model.

We can see that ϕ does not affect the estimation (expected value) of x_i, y_j ; this value is involved only in the variability: therefore, in order to estimate x_i, y_j we can use the simple poisson.

ϕ is **not** involved in the **estimation variance**, but it is **certainly** in the **process variance**.

To solve this model we can use **GLM**: this implies a **multiplicative structure**, linear in the logarithm $\rightarrow \log(P_{i,j}) = c + \alpha_i + \beta_j$, with $\alpha_i = \log(x_i)$ and $\beta_j = \log(y_j)$. The result is the estimation of the parameters and the Var/Cov matrix.

(41/42) for the Max Likelihood

Mack and ODP have the same mean, but it is possible to observe **different variance**, even though they are both based on C-L: this difference lies in the **different assumptions**.

This difference in the results is a problem also from the supervisor point of view: it is possible that the companies apply some degree of arbitrage in order to fit their data in the assumptions of the model which compute the lower capital requirements $\rightarrow$ **Importance of justification for the assumption**.

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(43)

Last time we have proved that the estimation of the reserve with the C-L coincide with the estimation given by the ODP: therefore **apply one of these two models yields the same result**.

x_i represents the estimation of the ultimate cost based on the C-L; y_i representation is obtain by

$$\text{fixing } \hat{C}_{i,n}^{ODP} = \hat{C}_{i,n}^{PCL} \rightarrow \sum_{j=0}^{n-i} \hat{y}_j = \frac{1}{\prod_{j=n-i}^{n-1} \hat{\lambda}_j} = \frac{1}{\widehat{CDF}_{n-i}}.$$

The important aspect is that we have a link between the parameter y_i and the CDF, with consistency between the index of the summation and the subscript of the CDF (last diagonal).

Knowing the CDF from the CL allows us to go back to y_i : starting from $i=n$, $\hat{y}_0 = \frac{1}{CDF_0}$ and so

on. y_i **is an incremental settlement speed**, but for the first year it is exactly equal to a cumulative

settlement speed. $i=n-1$, $\hat{y}_0 + \hat{y}_1 = \frac{1}{CDF_1} \rightarrow \hat{y}_1 = \frac{1}{CDF_1} - \frac{1}{CDF_0}$ (from cumulative settlement speed to incremental one).

$$\text{In general } \hat{y}_j = \frac{1}{CDF_j} - \frac{1}{CDF_{j-1}}.$$

Remember: this results are based on the assumption that I can obtain the values of the estimators from GLM models. Knowing that I can apply GLM for Poisson, I can change the underlying distribution: of course, in this way we don't have the link with the C-L result.

4.3 BOOTSTRAP FOR CLAIMS RESERVING

The power of this approach is that we can obtain the **distribution of the reserve**, in addition to the usual expectation and volatility for the C-L results.

Main assumption: underlying model is an **ODP**.

The idea is that is possible to obtain the distribution of an estimator just having a disposal a sample.

eg. $X \text{ r. } v \rightarrow x_1, \dots, x_n$ observation; now we assume to have an estimator that is a function of the observations $f(x_1, \dots, x_n)$, such as the mean. We want to find the **distribution of this estimator**.

Bootstrapping: if we can assume that the observation are generated from **iid random variables**, then we can extract a number of samples (H) from the original one, using a **sampling with replacement** procedure (otherwise I would obtain the same samples with different order, since we are not changing the size).

For each sample is possible to compute a mean $\bar{x}_1, \dots, \bar{x}_H$, and if we consider all of them then we have a distribution (we can use also different functions).

One **possible drawback** of this method is that it **relies only on the original observation**, therefore, if the initial sample has low variability, then also in the samples originated by the bootstrap we will have low variability (**problem with extreme value/outliers**).

In our application, we want to obtain the claim reserve from the triangle, so the result relies very much on the triangle used.

We can apply this approach, only if we are iid observation: we should select which observations can respect this assumption.

We **cannot** work on the **usual triangle** (problem with the **identically distribution** part, which implies same mean and variance): we move from classical triangle (payments) to **triangle of residuals**, which are usually assumed to be **iid**.

In order to work with residual, we have to specify which is the model from which they come: in this framework we want to apply the PC-L to the triangle and then see how the result differ from the observation (difference between observation and values predicted by the model). The problem is that we can compute the residuals only on the upper triangle

Main steps:

- 1) Estimation of the **development factors** of PC-L: $\hat{\lambda}_j = \frac{\sum_{j=0}^{n-j-1} C_{i,j+1}}{\sum_{j=0}^{n-j-1} C_{i,j}}$
- 2) Apply the **C-L on the upper triangle**: we build a new triangle, composed by **pseudo-cumulative amounts** (theoretical values). If we are in the last diagonal of the upper triangle, then the values come from the original triangle; for all the other values we are applying the inverse of the logic of the C-L (possible also with a recursive formula) $\rightarrow$

$$\dot{C}_{i,j} = \begin{cases} C_{i,n-i} & \text{if } j = n - i \\ \frac{C_{i,n-i}}{\prod_{h=j}^{n-i-1} \hat{\lambda}_h} = \frac{\dot{C}_{i,j+1}}{\hat{\lambda}_j} & \text{if } j < n - i \end{cases}$$

If the **individual link ratio** is exactly **equal to** λ , then we obtain the **same values** for the new triangle as the **C-L**.

Farther to the main diagonal, more the possible error is propagated, since we divide for more than one values of λ : in this case, even if one link ratio is equal to λ , but not the other ones, the new triangle is not equal to the original one.

- 3) From **cumulative to incremental**: in this case I can compute also the variability for the last diagonal, otherwise the difference between $\dot{C}$ and C on this diagonal would be null for

$$\text{construction} \rightarrow \dot{P}_{i,j} = \begin{cases} \dot{C}_{i,j} & \text{if } j = 0 \\ \dot{C}_{i,j} - \dot{C}_{i,j-1} & \text{otherwise} \end{cases}$$

Once we have obtain the incremental triangle, we compare the new one with the original one: the volatility will be lower in the cases in which the original triangle satisfies the assumption of the C-L

- 4) **Building the unscaled Pearson residuals** (even though there are different types of residuals),

$$\text{based on the difference between the expectation and the observation: } r_{i,j} = \frac{P_{i,j} - \dot{P}_{i,j}}{\sqrt{\dot{P}_{i,j}}}.$$

These residuals are **standardised**, in order to obtain mean 0 and std 1; moreover, for very large sample, we can assume iid and distributed as Standard Normal.

The definition of **residual** as observation minus expectation over the standard deviation comes from the assumption of **ODP as underlying model** (ODP and C-L are the same in terms of estimation of the reserve; since the over-dispersion parameter does not affect the estimation of the reserve, we can neglect this parameter $\rightarrow$ Simple poisson: mean $\dot{P}_{i,j}$, std $\sqrt{\dot{P}_{i,j}}$, therefore

$$P_{i,j} \sim \text{Poisson}(\dot{P}_{i,j}).$$

When the residuals are null? When the original triangle **satisfies** perfectly the **assumption** of the C-L.

Example (47 and so on)

In the unscaled Pearson's residual triangle we have all value different from 0 in all the cells except two: therefore the assumption of C-L in the original triangle are not perfectly satisfied.

The elements of this triangle are assumed **iid**: if this is true we can apply the bootstrapping methodology; of course I can use for this method only the residual that I observed in the upper triangle.

The proof of residuals iid is quite challenging, since we have few data at disposal, so all the **statistical tests assume low reliability**: a bit of bias could be present.

Two residual, for construction, will be always equal to 0: columns 0 - row n is null because $\dot{C}=C$ (for construction) and the same for the incremental payments (since we are in the first row); columns n - row 0 because the link ratio is equal to λ (for construction, using the traditional method for the computation).

Even if we introduce the tail, the residuals will be always null in columns $n - \text{row } 0$ because we have only one individual ratio, and we use it to estimate $\lambda_\infty \rightarrow$ Also the added column (columns) will have residual 0. This is not true if we interpolate to obtain λ_∞ . Typically the bootstrapping is applied **without tail**.

- 5) We **adjust the Pearson residuals** (it is possible to apply this passage at the end of the bootstrapping): the idea is to rescale (mainly the variance of) the residuals, in order to **correct the bias** related to the **problem of the size of the sample**.

$$r_{i,j}^{\text{adj}} = r_{i,j} \sqrt{\frac{\frac{N(N+1)}{2}}{\frac{N(N+1)}{2} - (2N-1)}} \text{ with } N=n+1 \text{ (number of rows or columns, w/o tail); this}$$

constant factor depends both on the number of the observations (size of the sample, of the cells of the upper triangle) and the number of parameters.

$\frac{N(N+1)}{2} = \# \text{ cells in the upper triangle.}$

$(2N-1) = \# \text{ parameters.}$ Why? The mean of ODP, as we have seen, depends on a product of a term regarding the row (x_i) and a term regarding the columns (y_j) $\rightarrow 2N$ parameters but we have

the constraint on y_j ($\sum_{j=1}^n y_j = 1$), therefore it is sufficient to estimate $n-1$ parameters and obtain

the last one without estimation $\rightarrow 2N-1$.

Notice: the **denominator** is observation minus parameters $\rightarrow$ **Degree of freedom**.

Finally, the **multiplicative factor** can be considered as the **square root** of the **ratio** between **number of observations** and **degree of freedom**.

The value of this coefficient varies a lot depending on the triangle considered: in general, **larger the triangle, lower the effect** of the coefficient; this is coherent with the fact that with more observation in the sample, the approach is more reliable.

Now we can start applying the bootstrapping.

[19/10]

The bootstrapping can be applied for every formula of $\hat{\lambda}$, not only for the classical C-L.

Asymptotically the residuals are assumed to be distributed as a Standard Normal; since we are dealing with no large samples (triangles), the residuals are adjusted (or scaled) in order to have consistent results between bootstrap and ODP in terms of variance.

At the moment we are focused only on the **estimation component**, and that is why the ϕ is **neglected**.

Triangle for the adjusted residuals we have a sample of **iid observations**: we can finally apply **bootstrap**.

From $r_{i,j}^{\text{adj}}$ we can build iteratively H new triangles, sampling with replacing from the starting

triangle (each cells has **probability** $P = \frac{1}{\frac{N(N+1)}{2}}$ to be picked up): we obtain $r_{i,j}^h$ with

$h = 1, \dots, H$.

The **residuals are a measure of error**, since they compare the observations with the results that should be obtained in theory with a specific methodology: with the bootstrap the error that I measured in a specific cell originally can be observed in another cell in the new triangles; moreover I can only sample from errors from the original triangle, because I can only consider the empirical observation and this is one of the main limitation of the approach.

Going back on the points of the methodology, with passages that must be repeated H times (higher the value of H higher the convergence, more reliable results):

- 7) **Resample** the adjusted residuals with **replacement**, creating new triangles $r_{i,j}^h$.

8) For each iteration we want to compute the **pseudo incremental payments**:

$P_{i,j}^h = \dot{P}_{i,j} + r_{i,j}^h \sqrt{\dot{P}_{i,j}}$. Higher the standard deviation for $r_{i,j}^{adj}$, higher the volatility of $P_{i,j}^h$; in general all the characteristics of $r_{i,j}^{adj}$ affect $P_{i,j}^h$.

For each $P_{i,j}^h$, we can apply whatsoever deterministic approach: the important part is that there must be **consistency** between the way in which we compute $\hat{\lambda}$ in the first passage of the process and how it is compute $P_{i,j}^h$, because we are calculating the **EV of a specific methodology**.

9) We create the associated **set of pseudo-cumulative data** and **fit the classical C-L** to these data; obtaining the future triangle (lower one) of the cumulative payments, it is immediate to compute the reserve for the specific AY (R_i^h) and the total reserve (R^h).

In this way we can obtain the **distribution of the reserve**, considering only the estimation variance component; in addition we have also distribution for the reserve according to the row (R_i^h) and **distribution for the incremental payments** (for the single cell).

Characteristics of R^h :

1) When **h increases** and the **expected value** for the **residuals is exactly equal to 0** —>

$E[R^h] = \text{asymptotically} = E[ODP] = E[C - L]$. the behaviour is similar to the Montecarlo Simulation: increase in the number of simulation corresponds to an increase in convergence.

In **claim reserve** the **speed of convergence** is **faster** than in **premium risk**, because in the latter the distribution of the severity is **highly skewed**: more simulation are required (100.000 VS 1.000/10.000).

Obviously, the simulation required depends also on the type of triangle (LoB) considered.

If $E[Res] \neq 0$, increasing the number of simulations is not sufficient to obtain

$E[R^h] = E[ODP] = E[C - L]$, meaning that there will be an **over/under estimation of the reserve**.

2) $\sigma(R) \rightarrow \sigma(ODP)$ estimated with GLM, including the zeros in the triangles of residuals.

In general: the mean should be compared to the classical C-L reserve estimates to check for bias, while the standard deviation gives an estimate of the estimation error.

It is possible to have **only positive values** for the **starting triangle (incremental payments)** and obtain some **negative values in the pseudo-incremental triangle**: this could create **problem** for the **process variance**.

Example (54)

Mean of Bootstrap $\neq$ Mean of C-L (check for the residuals).

Now we want to **include the process variance**: initially through a **closed formula**, then via **simulation** (historical development).

With the **closed formula** we **lose** the information regarding the **shape of the distribution** (which is one of the main reason to apply the bootstrap):

$$mse_p(\hat{R} | D^{(n)}) = \phi \cdot \hat{R} + Var_{bs}(\hat{R}) \left(\frac{\frac{N(N+1)}{2}}{\frac{N(N+1)}{2} - (2N-1)} \right)$$
, in which the **first term** is the

process variance, while the **second term** is the **variance of the bootstrapping** (EV).

$$\frac{\frac{N(N+1)}{2}}{\frac{N(N+1)}{2} - (2N-1)}$$
 is present because it is possible to **use** directly the **Pearson residuals**

without any adjustment: in that case we adjust here for the size of the sample.

The result in terms of variance is the same, both for the adjustment made directly on the residuals and for the adjustment made in the mse_p.

One of the **assumption of the ODP models** is $Var(P_{i,j}) = \phi E[P_{i,j}]$ (ϕ that is the term from which comes the over-dispersion), with $P_{i,j}$ independent (but not identically distributed).

We want to compute $\text{Var}(\mathbf{R} | \mathbf{D}^{(n)})$, that is the **variance** of the **sum** of the **future incremental** payments (lower triangle) $\rightarrow \text{Var}(R | D^{(n)}) = \text{Var}(\sum P_{i,j} | D^{(n)})$, then for independency,

$$\text{Var}(\sum_{i+j>n} P_{i,j} | D^{(n)}) = \sum_{i+j>n} \text{Var}(P_{i,j} | D^{(n)}) = \sum_{i+j>n} \phi E[P_{i,j}] = \phi \sum_{i+j>n} E[P_{i,j}] = \phi \hat{R}$$

$\hat{R}$ is the **mean of the reserve**: **theoretically** it should be the mean given by the **C-L**, since we are considering the ODP model; in **reality** we use the **mean of the bootstrapping**.

When we are not using a closed formula (but **simulation**) it is necessary to **include the adjustment** directly in the **residuals**, since there is no further possibility to modify these values, because the result of the simulation is a distribution and not variance (as in the closed formula). Using **simulation** approaches, it is possible to **include** in the computation of the variability the **discounting**, while using closed formula is not feasible: in the latter case we include the discounting as $CV \cdot \hat{R}^{\text{Discounted}}$ (**variability of the discount not present in CV**, it is fixed), while in the former case I discount directly from $P_{i,j}^h$, appreciating the effect on the variance of the discounting.

Estimation of the over-dispersion parameter:
$$\hat{\phi} = \frac{\sum_{i=0}^n \sum_{j=0}^{n-i} r_{i,j}^2}{\frac{N(N+1)}{2} - (2N-1)}$$

This computation comes from the usual assumption on the residuals that they are distributed as

$$N(0,1) \rightarrow \sum \left(\frac{P_{i,j} - \dot{P}_{i,j}}{\sqrt{\dot{P}_{i,j}\phi}} \right)^2 \sim \chi^2 \text{ with } \frac{N(N+1)}{2} - (2N-1) \text{ degree of freedom; applying the}$$

MM we obtain the previous formula for $\hat{\phi}$: $\sum res^2 = E[\chi^2]$ given by the degree of freedom,

namely
$$\sum \left(\frac{P_{i,j} - \dot{P}_{i,j}}{\sqrt{\dot{P}_{i,j}\phi}} \right)^2 \approx dgf.$$

The **volatility of the payments**, for the **process** component, is related to the **volatility of the residuals**: higher value for the residuals (in absolute term), higher value for the over dispersion parameter. There is a sort of **parallelism** between the **Mack's σ^2 common** parameter for **EV** and **PV** and the **bootstrapping residuals** (which affects both of the components); both σ^2 and residuals are two ways in which is possible to see how much the individual link ratio are different from λ

Observation: we introduce the adjustment term in the residuals in order to obtain the same value for the variance as in the ODP; this operation introduces some bias on the mean of the distribution only when the mean of the residuals is different from zero, but our assumption is to have mean of the residual null. Therefore there is an error in doing that, but it will be low since the expected value of the residual is similar to 0.

EXCEL

- Positive residuals when $f < \lambda$, because the ratio between payments is lower that the λ used to estimate.
- Negative residuals when $f > \lambda$, because the ratio between payments is higher that the λ used to estimate.
- “Large” considers, in a set of values, the value in a specifically given position
- “Random”: random choice in the interval [0,1)
- With the whole code of Resampled Pearson we obtain integer values between 1 and $\frac{N(N+1)}{2}$

- $E[\text{res}] = 0.006 \neq 0$: we expect in the result an over estimation in the reserve with respect to the one given by C-L and there will no be exact convergence, disregarding the number of simulations.
- Negative skewness. Why? Because the speed of convergence for the skewness is lower, therefore there are required more simulation to obtain the real value.

Now we introduce the alternative approach for obtaining the **PV: simulation** for a complete predictive distribution.

The methodology proceeds by interrupting the bootstrapping procedure at each iteration and drawing a random observation from the underlying process distribution, conditional on the bootstrap value.

From 1-9 same steps as before, in order to obtain $\hat{C}_{i,j}^h$ ($i+j > n$):

DISEGNINI

10) Obtain the **H corresponding future triangles of incremental payments** $\hat{P}_{i,j}^h$, de-cumulating $\hat{C}_{i,j}^h$ for $i+j > n$. These **values** are used as the **mean** when simulating from the **process distribution** (to obtain the PV).

If we consider all the values for the **same cell** in the **different triangles**, we obtain the **distribution** of the **incremental payments** for the specific cell: in this case we are considering only the EV.

So we are starting from a triangle that includes already variability,

For the **PV**, we are considering ODP with **mean** $\hat{P}_{i,j}^h$: on average we expect to pay what I have obtained from the bootstrap in the triangle. The **variance** is $\phi \hat{P}_{i,j}^h$, again related to the values coming from the bootstrap.

Even If we considering the ODP as underlying distribution, we know how to calibrate the parameters of the distribution, through the MM.

In the triangles we have the same underlying distribution, but for each cell we have different parameters, since they depend on $\hat{P}_{i,j}^h \rightarrow$ **The mean of the distribution as its own distribution.**

11) **From each cell** i, j in the lower triangle, and for each h , we **simulate a payment** from the process distribution with **mean** $\hat{P}_{i,j}^h$ and **variance** $\phi \hat{P}_{i,j}^h$.

Particularly, we obtain $\hat{P}$ values **simulating** from ODP, with **mean** $\hat{P}_{i,j}^h$ and **variance** $\phi \hat{P}_{i,j}^h$: we want to see how much the new value is distant from the mean.

Now we have identified **two levels of volatility** (decomposition of the volatility):

- $\hat{P}$: volatility due to **bootstrapping** $\rightarrow$ **EV**
- $\hat{P}$: volatility due to the volatility of the parameter used in the underlying distribution and for the fact that we are simulating from a distribution (due to the presence of ϕ) $\rightarrow$ **Prediction variance** for the incremental payments

If we suppose to have **independence between EV and PV**, simply **subtracting** from the prediction variance the EV it is possible to obtain the **PV**.

12) For each triangle: $\hat{R}^h$ is the **reserve** for the triangle h , which includes both **PV and EV**; the variance of the vector with $\hat{R}^h$ is the prediction variance.

[20/10]

Criticism: sampling only from the observation, crucial problem with the presence/absence of **outliers**, since the volatility comes only from the observation.

The observation is certainly important, but this problem is present also in other methods, which rely only on the data coming from the triangle. There is always the trade-off between the size of the sample and the reliability of data: this balance is present in all the methods which use **aggregated data**.

Advantage of individual data: size of the sample; **Disadvantage of individual data:** no regularity.

Problems with ODP for the generation of the observation ($\ddot{P}_{i,j}$): **no closed formula for the distribution of ODP**—> When we compute the MLE we don't have the likelihood but the quasi-likelihood.

Solution 1: introduce $\mathbf{Y}_{i,j}^h \sim \mathbf{P}\left(\frac{\hat{\mathbf{P}}_{i,j}^h}{\phi}\right)$ and compute $\ddot{\mathbf{P}}_{i,j}^h = \phi \mathbf{Y}_{i,j}^h$; in this way $E[\ddot{P}_{i,j}^h] = \hat{P}_{i,j}^h$ and $Var(\ddot{P}_{i,j}^h) = \phi \hat{P}_{i,j}^h$.

Briefly, we simulate from a simple poisson, we multiply for the over-dispersion parameter ϕ for rescaling, and we obtain the ODP result.

Solution 2: negative binomial as underlying distribution, with the parameters calibrated with the MM, in order to obtain the mean equals to $\hat{P}_{i,j}^h$ and the variance equals to $\phi \hat{P}_{i,j}^h$.

in this case we don't have to do two different passages, because the mean is already different from the variance.

In terms of **mean** and **variance** we have the **same result**, but the **time for the convergence is different**.

In both of the two solutions, it is still valid the fact that **asymptotically** the **mean** of the reserve will be equal to the estimation made by the **C-L** and the **variance** tends to the **estimation variance** obtained by ODP, because we are simulating assuming, which is the mean of the bootstrap —> The mean of the **final distribution** is equal to the mean of the bootstrap procedure (if residuals have mean equals to zero and the number of simulation is enough high).

The same holds for the variance, which is equal to the variance of ODP compute via GLM: the difference is that in step 9 we have only EV, while in step 12 we have the prediction variance, but both are equal to the analytical variance. This comparability is obtained through the scaling factor for the degree of freedom (adjusted).

On the contrary, it is possible to observe some **difference** in the **skewness**: with the **ODP** (via GLM), there is **no skewness** since we are considering a **closed formula**.

The choice of the **underlying model** for the simulation is crucial for the **skewness**, since Negative binomial and ODP yield two different results, while it is not important for mean and variance.

Even if in theory it is possible to use two different distribution to obtain the EV and the PV, this is not fully consistent.

For traditional triangles the skewness of the total reserve is typically low: this changes sensibly for more volatile LoB.

If we consider **one cell**: skewness of $Y_{i,j}^h$ is $\frac{1}{\sqrt{\frac{\mathbf{P}_{i,j}^h}{\phi}}} \rightarrow 0$ depending both on the value of $P_{i,j}^h$ and

on the value of ϕ .

Putting together the cells, which means summing all the $\hat{P}_{i,j}^h$ (lower triangle), the skewness tends to 0: this happens because we have supposed that the **payments** are **independent**, therefore we are simulating separately their values; if they were iid, the skewness would tend to 0 more rapidly, under the effect of the **CLT**.

We can say that for the final result (the whole reserve), the choice of the distribution is not crucial, because the skewness (which is the only value that varies between different distribution) tends to 0; it remains a crucial choice for the single cells.

We said that we suppose to have independent payments, but in reality there is some implicit dependency on the parameters. Why implicit? Because, differently from Mack's formula, in which there is an explicit term for the dependency between rows for the EV part, here the dependence is included in steps 1-9.

The **dependence** is present because we use the same λ in the **column**, but in this case we apply the C-L in the triangles that are used both for EV and for PV --> The dependency is present in both components, but we simulate independently, therefore we miss the dependency for the PV. Finally, this reserve takes into account also the presence of the dependence for the process variance.

Observation: $P_{i,j}^h = \dot{P}_{i,j} + r_{i,j}^h \sqrt{\dot{P}_{i,j}}$ as we have already seen could lead to negative values even if we start from only positive values; this could be a problem for the PV: if the negative values are such that $\hat{P}_{i,j}^h < 0$ (meaning that $P_{i,j}^h < 0$ leads to $\lambda < 1$, eg last column, for which there is only one value for the computation of the whole column)--> The problem is that $P_{i,j}^h$ is the variance of the distribution.

How can we solve this point?

$$\hat{Y}_{i,j}^h \sim P\left(\frac{|\hat{P}_{i,j}^h|}{\phi}\right) \text{ and } \hat{P}_{i,j}^h = \phi Y_{i,j}^h - 2 \max(-\hat{P}_{i,j}^h, 0)$$

Since I am introducing a distortion on the mean (for the negative values), I correct (ex post) the value with the second part of the formula to compute $\hat{P}_{i,j}^h$; the distortion is exactly equal to 2 times the value of the negative incremental payment.

I have a bit of distortion also for the variance, but this is fair when the number of negative values is not high.

(58)

The PV is simulated and the tail is included in the first cell (0 as residuals): this is done in order to have comparable values with Mack.

50000 simulations.

Mean: close to the Mack's one but not equal, because the mean of the residuals is different from 0.

CV: higher for ODP, even though there is always the PC-L as underlying process —> This implies an higher capital requirement.

The choice of the model is crucial both in terms of variance and in terms of capital requirements. It is important to justify the choice of the model on the data, in order to avoid any kind of arbitrage for save some capital but enlarging the risk for the company.

Often the volatility is higher for CV than for ODP. Why there is this difference? Because even though the deterministic method is the same, we have different stochastic part.

With the Mack formula the larger part of volatility is in the PV, while with Bootstrap and ODP the greater part is related to the EV: this difference lies in the parameters on which they relies (σ^2 and residuals).

LAST PART

[24/10]

INITIAL PART

4.4 COLLECTIVE RISK MODEL

This will be the last method for which the validation in the **total run-off approach**.

The idea is to provide a stochastic methodology based on **frequency-severity** logic (Risk Theory), common in premium computation (rating). Therefore, it will relies on the deterministic approach that split the frequency and the severity, namely **F-L** and **Frequency-Severity**.

This approach was proposed in 2004, during the first years of the creation of SII, in order to compute the capital requirements both for the premium risk and for the reserve risk.

In this application (reserve), we don't consider the aggregate claim amount for the future contracts (as in pricing), but the **total amount** to be paid in future only for **existing claims**: this is the meaning of the r.v. **X**.

$X = \sum_{h=1}^K Z_h$; **K** is the r.v that describes the **number of claims** and **Z_h** is the r.v for the **average cost**.

E[X] is equal to the **estimation** of the **claims reserve**.

Main problem —> in RT we introduce **three main assumptions**:

- 1) **K** independent by $Z_h \forall h$
- 2) Z_h independent $\forall h$
- 3) Z_h identically distributed $\forall h$

In this framework we cannot say that the cost of claims does not depend on the column position; a possible extension is based on the assumption that the amount of incremental payments $X_{i,j}$ of a **single cell** of the future triangle is described by CRM.

$$X_{i,j} = \sum_{h=1}^{K_{i,j}} Z_{i,j,h} \rightarrow \text{In this way, we don't model directly } X \text{ but we model the amount for single}$$

cells: in this way we can **assume** that **Zs are iid**.

The problem now is to find a methodology to aggregate the single amounts.

Each cell is a realisation of $X_{i,j}$ (in the triangle of incremental payments): if we need to introduce some assumptions on K and Z , now it is not anymore sufficient to use the information from the triangle, because in the cell we have only one value $\rightarrow$ the **data** are **not** anymore **sufficient** to calibrate the model.

MISSING SENTENCE

Now we assume that **K and Z are independent** and $Z_{i,j,h}$ **iid in a single cell**.

Number of claims

$K_{i,j} \sim P(\hat{n}_{i,j} \cdot q) \rightarrow$ **Mixed Poisson**, with the usual assumption of $E[q]=1$ and $q \in [0, +\infty)$, s.t. $E[K_{i,j}] = \hat{n}_{i,j}$.

If we calibrate $\hat{n}_{i,j}$ in the lower triangle using the number of payments given by the **deterministic methodology**, then $E[K_{i,j}] = E[\text{Det approach}]$. This is important since we are estimating the volatility and the reliability of the specific deterministic methodology.

This holds also for the summation of the lower triangle, because the summation is a linear operator, so $E[\sum] = \sum E[]$.

Introducing q, we are including uncertainty on the estimation of the number of claims, in particular **two levels of stochasticity**:

- **q** random variable
- **Poisson** random variable

These two levels of stochasticity **replace** the **EV** and **PV** of the previous approaches (as we will see).

How to define q? We display the two most extreme cases, but there are some intermediate degree in between.

[1] $\tilde{K}_{i,j} \sim P(\hat{n}_{i,j} \cdot \tilde{q}_{i,j})$ with $\tilde{q}_{i,j}$ **independent** one to the others; [2] $\tilde{K}_{i,j} \sim P(\hat{n}_{i,j} \cdot \tilde{q})$

[1]: in each cell the number of claims is modelled separately, without any kind of dependency between cells.

[2]: only one random variable that affects all the lower part of the triangle.

What changes? Not the mean (which remains the same for both the approaches), but the volatility related to the sum of different cells: **DISEGNINO** [1] : $Var(K_1 + K_2) = Var(K_1) + Var(K_2)$;

[2] : $Var(K_1 + K_2) \neq Var(K_1) + Var(K_2)$ because we are introducing dependence given the common $\tilde{q}$.

With $\tilde{q}$ we have a common uncertainty that affects the cells (as σ^2 for Mack)

The **model depends** on the **deterministic method** and on the **dependency** present in the deterministic approach: in **F-L** we have **dependency** both for the **rows** and for the **columns** (because we estimate α for the column but it is used for the rows and we use something similar to settlement speed for the columns); with **F-S** there is higher **dependency** between **columns** (similar to C-L) $\rightarrow K_{i,j} \sim P(\hat{n}_{i,j} \cdot \tilde{q}_j)$

$$\sigma(K_{q_{i,j}}) < \sigma(K_{q_j}) < \sigma(K_q)$$

If we stop here, for the single cells we know already mean, variance and skewness (for the X) and all the other moments, because we have the moment generating function of X.

The problem, again, is related to the aggregation of the cells.

How to simulate the two processes?

[1] We have to **simulate** the $q_{i,j}$ (it is created a vector) separately **for each cell** and then we have to **multiply** for $\hat{n}_{i,j}$ (obtained in the deterministic approach), to have the parameter from which simulate with the **Poisson**. The process follows **two steps** since we are assuming independence between cells. **Asymptotically** we expect $\text{Var}(K_1 + K_2) = \text{Var}(K_1) + \text{Var}(K_2)$.

[2] We cannot separate the process for the cell, because there is dependency; then I have to **simulate** q , equals for all the cells of the lower triangle, and then **multiply** for the single values of $\hat{n}_{i,j}$; varying q , we rescale all the parameters of the Poisson for the lower triangle. Here

$\text{Var}(K_1 + K_2) \neq \text{Var}(K_1) + \text{Var}(K_2)$.

If $q \sim \gamma(h, h)$, the **mixed poisson** is a **negative binomial**, therefore it is not necessary to simulate q and the Poisson, but this is done in **one passage**.

Single claims size(severity)

It is possible to introduce also in Z a degree of uncertainty (through structured variables).

We suppose to have $Z_{i,j}$ with **mean** $\hat{m}_{i,j}$ (incremental average cost coming **from** the underlying **deterministic methodology**; for F-S we have to de-cumulate the result with the particular formula we have seen).

With the previous assumption we can assure that the **mean** for the **reserve** of the **stochastic** methodology is exactly **equal** to the **mean** of the **deterministic** one.

How do we add **volatility** to the single claim severity? $Z_{i,j}$ such that $C_{z\ i,j} = \hat{C}_{z\ j} \cdot r_j \ \forall i$; in this formula we have the **first factor** that is the empirical estimation from the data of the C_z for the **column** j (as **average value**), while the **second factor** is the one that introduces **uncertainty** on the C_z .

[26/10]

The **choice** on the dependency of q (if it is independent cell by cell or if it is dependent by column, row or both), will **affect only** the value of the variance when cells are **aggregated**.

There should be correspondence between the underlying deterministic approach and the dependency of the values: with F-L I expect to see q as value because there is dependency both on columns and on rows, while with F-S I expect to see q_j , because there only dependency by columns, since it is based only the C-L.

This reasoning is equal for the single claim amount (Z): here we are introducing uncertainty for Z , in particular, on the value of $C_z \rightarrow C_{z\ i,j}$ is assumed to be a r.v.

If we assume the distribution for Z with two parameters, and if we calibrate the parameters with the method of moments, then the parameters of the distribution becomes random variables (at least one): eg. If we consider the Log Normal distribution, we can exploit a relationship between σ (which is one of the two parameters) and $C_z \rightarrow \sigma$ r.v. But also μ (the other parameter) depends on $\sigma \rightarrow$ Also μ r.v.

$C_{z\ i,j} = \hat{C}_{z\ j} \cdot r_j$ with some **common assumptions**: $E[r_j] = 1$; $r_j > 0 \rightarrow E[C_{z\ i,j}] = \hat{C}_{z\ j}$ the presence of the structured variable does not alter the mean (something similar to what we have done for ODP and mixture Poisson).

With this additional source of stochasticity, we have now **two levels of volatility**: one coming from the presence of r_j and one because we are **drawing from a distribution**.

For r_j we assume **dependency** only **between** different **rows** of the same column; another assumption is that r_j is **independent** from q , because we have to **assure** the **independency between K and Z**.

Even though it is possible to consider different distribution for different r_j , in the analytic formula we will always assume r_j **iid**: this assumption is not necessary to simulate, but it is still reasonable.

Why $C_{z\ i,j}$ (on lower triangle) of the single cells does not depend on ' i '? (*)

One of the crucial aspect is how to estimate $\hat{C}_{z,j}$.

Until now all the computation are made with all the information in the triangle (also for $\hat{n}, \hat{m}$); for $\hat{C}_{z,j}$ is not anymore possible.

DISEGNINO

In order to have $\hat{C}_{z,j}$, I have to consider the individual claims underlying a single cell of the triangle: this happens because the results present in the cells of both the triangle $P_{i,j}$ and the triangle $m_{i,j}$ come from the sum of different Z, sum of different claims.

We **introduce** another **triangle**, in which we can collect all the **empirical CV** of the underlying distribution of **Z**.

From the empirical underlying distribution of Z we can compute $C_{z,i,j}$ for the triangle.

In theory, we can also find the theoretical distribution of the severity, given mean and CV, which is the one that fits best the data: in some cases, it is possible to have different distribution for each cell.

At the moment we are focusing on the upper triangle, but what I assume in the lower triangle?

How can I obtain $C_{z,i,j}$ in the lower triangle?

(*) Usually we **assume** that the **columns** present **regularity**, in line with the **F-L** (common characteristics of the cost of claims through the columns). This is why we consider

$C_{z,i,j} = \hat{C}_{z,j} \cdot r_j \forall i$, in this case the estimation of the average CV depends on the different column.

It appears that we can apply this approach only if we have data regarding the single claims: this could be problematic for external actuaries, if they don't receive further information on each of the claims.

Assuming $r_j \stackrel{iid}{\sim} \Gamma(\mathbf{h}, \mathbf{h})$ we obtain two important result $\rightarrow$

$$\begin{cases} E[X_{i,j}] = \hat{n}_{i,j} \hat{m}_{i,j} \\ \sigma^2(X_{i,j}) = \hat{n}_{i,j} \hat{a}_{2,Z_{i,j}} + \hat{n}_{i,j}^2 \hat{m}_{i,j}^2 \sigma_q^2 + \hat{n}_{i,j} \sigma_{Z_{i,j}}^2 \sigma^2 r_j \end{cases}$$

The **first addendum** are the **variance of payments in case of pure poisson**; the **second** one is the **additional variance given by q**; the **last addendum** is the variance coming from the **uncertainty** on the **Z**, introduced by r_j : here $\sigma_{Z_{i,j}}^2$ is the **average value of the variance** of the cell

$$C_{z,i,j} = \hat{n}_{i,j}^2 \hat{m}_{i,j}^2 \text{ and } \hat{a}_{2,Z_{i,j}} = \sigma_{Z_{i,j}}^2 + \hat{m}_{i,j}^2.$$

The choice of the distribution for r_j is important for the value of the variance, while the mean is invariant (as expected).

Comments:

- 1) Trivially, the introduction of the random variables r and q we are adding volatility.
- 2) The **variance** of q is multiplied by $\mathbf{n}^2$, while the **variance** of r is multiplied by $\mathbf{n}$: this difference is important for the **CV**.

$$CV(X_{i,j}) = \sqrt{\frac{1 + \hat{C}_{z,i,j}^2}{n_{i,j}} + \sigma_q^2 + \frac{\hat{C}_{z,i,j}^2 \sigma_{r_j}^2}{n_{i,j}}} \rightarrow \text{The structured variable } \mathbf{q} \text{ cannot be}$$

diversified, while the variability for r can be **diversified**: this is why the structured variable r is **not systematic** component!

Why can I diversify?

For volatility, given $\frac{1 + \hat{C}_{z,i,j}^2}{n_{i,j}}$, C_z is **important** for **small company** (importance of strategies

such as reinsurance, retention...), while it is not important for **big companies**, because they can **diversify** better the risk linked to the size of the claims. For the same reason, if C_z is **relevant** then r is **relevant (and vice-versa)**.

This reasoning is only related to the relative measure of volatility, **not** for the **absolute values**.

$$CV(X_{i,j}) = \sqrt{\frac{1 + \hat{C}_{z,i,j}^2}{n_{i,j}} + \sigma_q^2 + \frac{\hat{C}_{z,i,j}^2 \sigma_{r_j}^2}{n_{i,j}}} = [1 + C_z^2(1 + \sigma_r^2) = 1 + C_z^2(E[r] + \sigma_r^2) = 1 + C_z^2 a_{2,r_j}] = \sqrt{\frac{1 + \hat{C}_{z,i,j}^2 a_{2,r_j}}{n_{i,j}} + \sigma_q^2}$$

Important: the parameter **n** of RT (for the whole portfolio) is much **higher** than the parameter **n_{i,j}**, because in the former case we are considering the total number of claims (paid + reserved) in the observe period; on the other hand, as **n_{i,j}** we consider how many claims are paid in one single cell of the triangle.

Having said that, in **pricing** the **diversification** is **higher** because is higher the number of **n** considered and the **C_z** is not proportionally larger.

In **reserving**, the effect of **diversification** can depend on **location** in the triangle, because, considering MTPL, it is very likely to observe **n_{i,j}** (first DYs) > **n_{i,j}** (last DYs).

Finally: for very large company **CV** $\approx$ **σ_q** (premium/rating), while **CV(X_{i,j})** can be higher than **σ_q** (reserving).

Now we **aggregate** the single cells, in order to obtain the reserve:

- **Mean:** this is a linear operator, therefore it is quite trivial to obtain the result

$$\hat{R} = E\left[\sum_{i=0}^n \sum_{j=n-i+1}^{n^+} X_{i,j}\right] = \sum_{i=0}^n \sum_{j=n-i+1}^{n^+} \hat{n}_{i,j} \hat{m}_{i,j}$$

Before to introduce the formula for the standard deviation, we can compute the variance for only two cells of the lower triangle (**QUADERNO**) →

$$\sigma^2(\mathbf{K}_A + \mathbf{K}_B) = \sigma^2(\mathbf{K}_A) + \sigma^2(\mathbf{K}_B) + 2\mathbf{n}_A \mathbf{n}_B \sigma^2 \mathbf{q}, \text{ in which } \mathbf{n}_A \mathbf{n}_B \sigma^2 \mathbf{q} = \mathbf{Cov}(\mathbf{K}_A, \mathbf{K}_B).$$

Therefore, if we **increase** **σ_q²**, we **increase** the **variance** of all the cells and also the **dependence**

between cells. To pass from **correlation** to covariance: $\mathbf{Corr}(\mathbf{K}_A, \mathbf{K}_B) = \frac{\mathbf{n}_A \mathbf{n}_B \sigma^2 \mathbf{q}}{\sigma_{\mathbf{K}_A} \sigma_{\mathbf{K}_B}}$.

It is possible to prove this equation also for **Z** → $\mathbf{Corr}(\mathbf{Z}_A, \mathbf{Z}_B) = \frac{\mathbf{m}_A \mathbf{m}_B \sigma^2 \mathbf{q}}{\sigma_{\mathbf{Z}_A} \sigma_{\mathbf{Z}_B}}$

$$\text{Variance, assuming } \mathbf{r}=0: \sigma^2[\mathbf{R}] = \sum_{i=0}^n \sum_{j=n-i+1}^{n^+} \hat{n}_{i,j} \hat{a}_{2,Z_{i,j}} + \left[\sum_{i=0}^n \sum_{j=n-i+1}^{n^+} \hat{n}_{i,j} \hat{m}_{i,j} \right]^2 \sigma_q^2$$

The first component is the sum (with respect to the lower triangle) of the part of variance coming from the pure poisson compound process ($K \sim Pois$).

The second component, if we hypothesise only two cells, it would be

$$[\hat{n}_A^2 \hat{m}_A^2 + \hat{n}_B^2 \hat{m}_B^2 + 2\hat{n}_A \hat{m}_A \hat{n}_B \hat{m}_B] \sigma_q^2, \text{ part of}$$

$$\sigma^2(\mathbf{X}_A + \mathbf{X}_B) = \hat{n}_A \mathbf{a}_{2,Z_A} + \hat{n}_B \mathbf{a}_{2,Z_B} + [\hat{n}_A^2 \hat{m}_A^2 + \hat{n}_B^2 \hat{m}_B^2 + 2\hat{n}_A \hat{m}_A \hat{n}_B \hat{m}_B] \sigma_q^2 = \sigma^2(\mathbf{X}_A) + \sigma^2(\mathbf{X}_B) + 2\mathbf{Cov}(\mathbf{X}_A, \mathbf{X}_B)$$

As already said, if we increase **σ_q²**, the variance of the reserve will be higher also because the covariance of the cells is higher.

EXAMPLE R

How to simulate the numbers in two cells and which is the effect of different **q**.

n_A; n_B: estimation of the numbers of claims in two cells (eg. F-L)

σ_q: estimated (we will see how).

Different parametrisation of Gamma: in this parametrisation, in order to have mean equal to 1, the parameters shall be equal. Here the mean is the ratio of the parameters and the variance is given by shape over rate.

In the first case we simulate with different random variables for **q**, while in the second case we use the same **q**.

It is quite problematic to include **r** and obtain the skewness; typically this model is useful both for closed formula for variance and for simulation of the distribution of **X** (MonteCarlo simulation) and **R**.

The simulated variance in case of independence (Var) is quite similar to the value of the theoretical variance and this is quite lower than variance with dependence included (Var Dep).

With this approach we **don't explicitly decompose EV and PV**: the idea is that the structure variable captures the variability for EV, while the PV is considered in the underlying distribution. Important: for the specific cells (A and B), the variance in the case of independence is equal to the variance in case of dependence, because the dependence does not affect the variance within the cells.

In the formula of $\sigma^2[\mathbf{R}]$ we are not including r , for sake of simplicity: if we want to add also this component, we have to include $\hat{n}_{i,j} \sigma_{Z_{i,j}}^2 \sigma^2 r_j$ this component (with double summation with respect of the lower triangle), even if there is no additional dependency. Then, if present (as in this case), we have to add the dependency between cells in the same column.

All this reasoning under the assumption of $\mathbf{r}_j \sim \Gamma(\mathbf{h}, \mathbf{h}) \forall j$

In this approach we don't split explicitly PV and EV: the idea is that the **structured variables** measure the uncertainty (**EV**), while the **volatility** of the process is described by the characteristic of the underlying distribution → Eg. Mixed poisson: q for EV and Poisson for PV.

[27/10]

Trick for excel

Obviously it is possible to compute the variance for one single row, omitting $\sum_{i=0}^n$.

If we **omit** $\mathbf{r}$, then $\mathbf{C}_{z \ i,j} = \hat{\mathbf{C}}_{z \ j}$ is **deterministic**.

$Cov(X_{i,j}, X_{h,k}) = \hat{n}_{i,j} \hat{m}_{i,j} \hat{n}_{h,k} \hat{m}_{h,k} \sigma_q^2 \rightarrow n$'s components for K , m 's components for Z .

Example (from 69)

SIFA and AMASES are two companies, with different size (AMSES > SIFA): in fact,

$BE_{CHL}^{SIFA} = 230 mln$ while $BE_{CHL}^{AMSES} = 2566 mln$, circa 10 times the first one.

Changing company, how the capital requirements will change? (even if for now we are in run-off perspective).

(70)

Plot of the upper triangle of cumulative amount: each bar is a row (AY/RY), until 11; each colour is the contribution of each DY to the final amount. The idea is to analyse the evolution of the volume over the time: SIFA has an increasing behaviour, with less DY considered we have higher bars.

Similarly for AMASES.

If we observe the development of one colour, we see regularities for the incremental payments in one specific DY.

The magnitude of the increase in the ratios can be appreciated in the box-plots (lower part of the same slide).

(71)

Similar indicator, for numbers $(n_{i,j}^c)$, so more important for CRM: here we can see the important difference between SS^N and SS^A (ratio between the blue bar and the whole height), with $SS^N > SS^A$.

It is likely that the height of the bars will increase if the number of reported claims.

Average cost of claims: how much on average I have to pay, taking into consideration the evolution in time (with respect to the columns); each colour is a row (AY/RY).

From the figure we can see the volatility between different AY: of course, in higher DYs we can expect to observe more volatility for the absence of many data.

In this case the computation is made on the original triangle, therefore there some additional volatility in the numbers regarding the inflation: if we remove this inflation, the results should be more or less equal, if and only if the hypothesis of F-L are respect; otherwise there is some volatility due to this aspect.

Parameters' calibration

I have obtained **from F-L** $\hat{n}$ and $\hat{m}$; now I have to compute other parameters:

- 1) I need $\hat{C}_{Z_j}$ (from 0 to 11, not as written in the slide): in this simple example these values are fixed and not estimated on data; in reality we have to fit the data to obtain a fair estimation for $\hat{C}_{Z_j}$ and we can see that empirically at a certain point, this component will tend to stabilise.
- 2) I need **estimation** for σ_q^2 and a distribution for $\tilde{q}$: the problem is that we don't have data on which to calibrate this structured variable. A possible solution is to apply F-S: this is something similar to C-L, so, since σ_q^2 is strictly related to the dependence (both on number and on severity, but here we are interested in numbers), we can apply C-L to numbers, through Mack we can isolate the covariance between couple of numbers and then with the formula we can obtain σ_q^2 . **Main drawback**: inconsistency when we have F-L as underlying methodology, because it is not related to C-L. A **common assumption** is $q \sim \Gamma(h, h)$
- 3) I need estimation for σ_r^2 and a distribution for $\tilde{r}_j$: also in this case, the **problem** is that we don't have **data** on which to **calibrate** this structured variable. A possible solution is to calibrate $\sigma_{r_j}^2$ on the triangle C_z , measuring the variance of the column. A **common assumption** is $r \sim \Gamma(h, h)$.
- 4) I need a distribution for $Z_{i,j}$: I could select different distribution for each cell, but this would reduce the stability and regularity of that triangle. A **common assumption** (that is quite realistic) is to fix a common distribution for all the rows in the same column.

Here $\sigma_{r_j} = 3\%$, same for all the columns and the same gamma distribution for all the cells.

$\sigma_q(SIFA) = 8\%$; $\sigma_q(AMASES) = 3\%$: this is in line with the size of the company; whereas $\sigma_{r_j} = 3\%$ for both companies.

(73)

Comparing CoV: this object is **important** when we want to compare two different approaches because we are dealing with **different mean** (for F-L and C-L), so it is **better** to use a **relative measure** for the volatility.

In both cases (CRM and Bootstrap with ODP) we have the **diversification effect**: in fact, bigger is the company and lower is the CoV: in **CRM** the diversification (at least at single cell level) is **proved** (67), only because we are assuming to have same C_Z for both the companies. Also when we aggregate we have a similar effect, because we are summing r.v.'s and the sum tends to benefit when the size grows (also in terms of CoV for the reserve). The limit value is given by σ_q , so for SIFA we have that only $8.7 - 8 = 0.7$ is the part that can be diversified, while for AMASES only $3.3 - 3 = 0.3$ is the only part that can be diversified.

In **Bootstrap** we **don't** have a **mathematical proof** of diversification (as seen in scaling of triangle): because bootstrap depends on the volatility of the link ratio (it is based on residuals), we expect that the volatility of $f_{i,j}$ can benefit of diversification (eg. More stable settlement speed), but this is not a mathematical proof.

This comparison is useful to see that to different volatilities correspond different capital requirements (even though we are still in run-off perspective, not 1 year time horizon).

Larger CoV doesn't imply larger capital requirements but larger $\frac{Cap Req}{Mean}$ (we have seen this passage).

Comparing skewness: for Bootstrap in 'quiet' triangles, the skewness is quite low, and it benefits from the sum of random variables (similar to CTL, but not equal since we are assuming only independence and not iid). For CRM, there is an implicit dependence included in skewness; neglecting r , in the single cell we have the same model of RT $\rightarrow$ Larger n , the skewness tends to the skewness of q ($= 2\sigma_q$ iff $q \sim \Gamma(h, h)$), in fact $0.15 \approx 2 \cdot 8\%$ and $0.06 \approx 2 \cdot 3\%$.

The asymptotic value is $2\sigma_q$, reachable through diversification.

The choice of the deterministic method put some restriction on the stochastic method to be used for gaining information regarding the volatility/distribution of the triangle: eg. CHL w/ Bootstrap ok; F-L with CRM ok; CHL w/ CRM NO.

ISVAP REPORT (Quaderno)

[02/11]

Intro for σ_q^2

In order to include the dependence given by the presence of the same q for all the cells, in the simulation process of CRM we have to simulate q and then $K_A \sim P(n_A \cdot q)$, $K_B \sim P(n_B \cdot q)$. After this passage, we have to simulate Z : here, the number of simulation is strictly related to the result of K (extraction from the Poisson).

eg. $K_A = [100; 50; \dots]^T \rightarrow Z_A = \begin{cases} [\dots] \text{ with 100 components} \\ [\dots] \text{ with 50 components} \\ \dots \end{cases}$

With the presence of r_j we have an additional complexity: the cells in the same column are affected by the same value, therefore there is dependency.

If there was not dependence: $K_A = [100; \dots]^T$ and $r_j = [\dots]$ with 100 components $\rightarrow Z_A = \begin{cases} [\dots] \text{ with 100 components} \\ \dots \end{cases}$ with one different value of r for each component (first

value in the vector r_j considered in the first value of the vector Z_A and so on).

In reality we have dependence $\rightarrow$ I cannot apply the previous strategy.

Assume now $K_A = [100; \dots]^T$ and $K_B = [80; \dots]^T$: the value of r_j is the same (if we make the hypothesis that they are in the same column), but the number of simulation is different. How can I solve this problem? r_j simulated a number of times that is given by $\max(K_A, K_B)$.

$r_j = [\dots]$ with 100 components $\rightarrow$ All the values are used for Z_A , while for Z_B we use all the values of r present in Z_A , until the K_B th.

If we sum all the values of the first vector of Z_A , we obtain the value of X (total aggregated amount) in the cell A , for the first simulation.

In the end we have a distribution for each cell of $X_{i,j}$ (PERSO QUALCOSA)

4.5 ONE YEAR VIEW

Aim: quantify the **capital requirements**, instead of the value of the variance of $\sigma^2(\hat{R})$ of the run-off approach.

In this course we will limit to consider **only the risk reserve and only one LoB**: keep in mind the effect of the **aggregation**, both for modules/sub-modules, and for LoB.

As easily comprehensible, we focus on **one year time horizon** for the risk assessment (differently from the run-off approach that considers all the years until the liquidation of all the claims).

We can now assume a triangle for incremental payments (DISEGNINO).

We can interpret $\hat{R}$ as the **amount at disposal** in the **BS** at the **end of the year** (not anymore a random variable): this quantity is used to **cover future claims**.

Which are the **risks involved** in $\hat{R}$?

We have to establish if $\hat{R}$ is sufficient to cover mainly two amounts:

- 1) **Payments of claims already occurred** (before the evaluation of $\hat{R}$) $\rightarrow$ B in the disegnano
- 2) **Reserve at the end of the following year, for claims already occurred** $\rightarrow$ C in the disegnano

In both cases we **include also IBNR** (considered at the time of occurrence).

Both B and C are random variables and they are dependent, because C is evaluated considering also B in the upper triangle.

Until now it may seem that everything is equal to the run-off approach: we can appreciate the difference in the sources of variance considered.

We can now introduce a **new random variable: CDR** (Claims Development Result).

We define it (at the moment) for a single row: $CDR_i = R_i^{D(n)} - X_{i,n-i+1} - R_i^{D(n+1)}$

$R_i^{D(n)}$ is a **deterministic amount**, which represents the claims reserve present in the balance sheet at the end of the year n (the $\hat{R}$ previously considered).

This amount shall cover both incremental payments in a specific cell of the lower triangle (first diagonal of the lower triangle if we consider $D(n)$) and the final claims reserve.

$D(n+1)$ is the triangle in the following year, without considering the row $n+1$ because I am not including the new claims $\rightarrow$ The triangle becomes a **trapezoid**.

Both X and $R_i^{D(n+1)}$ are random variables, therefore CDR_i is a **random variable** too; in addition

X and $R_i^{D(n+1)}$ are **dependent**, because the latter is computed considering $D(n+1)$, which includes the diagonal X .

One year approach **sources of volatility**:

- 1) X : same kind of **volatility** of the **run-off approach**, with both **PV** (given by the fact that the realisation are not in line with the average) and **EV** (volatility related to the estimators). The formula (as we will see) for these two components is exactly equal to the ones given by **Mack**.
- 2) $R_i^{D(n+1)}$: we **don't** consider the **process variance** because the real payments of the following years (after $n+1$) can be observed only 2 years later (starting from n) $\rightarrow$ This risk belongs to the year $n+1$. The only risk present in one year time horizon for the reserve is that the estimation made in n is not equal to what I will estimate in $n+1$ (**updating of the reserve estimation**): this is the **EV**.

Why $R_i^{D(n+1)}$ can be **different** from $R_i^{D(n)}$? Because they use **different information**.

DISEGNINO (IMPO PER CAPIRE)

Total CDR = $\sum_{i=0}^n CDR_i$ and we will have to **include** the **dependency** by row for the estimation variance.

CDR can be also defined as difference of **ultimate costs**: $CDR_i = C_{i,n}^{D(n)} - C_{i,n}^{D(n+1)}$ (simply replacing R with the difference between ultimate cost and cumulative amount on the diagonal).

If we consider the **booked reserve** as reserve for the triangle, then we can make similar reasoning to the one made for **IC-L**: **difference** between two cells represents the **run-off**.

We can apply both **closed formula** and **simulation**: in the former case we are going to refer to the **Merz-Wüthrich formula** (corresponding to the Mack formula but in the one year time horizon); in the latter case we refer to **Re-Reserving** (corresponding to the Bootstrapping and CRM in one year time horizon).

MERZ AND WÜTHRICH

Mack in one year time horizon $\rightarrow$ Same **assumption of Mack**:

- **Mean** $\rightarrow E[C_{i,j+1} | C_{i,j}] = C_{i,j} \cdot \lambda_j$
- **Variance** $\rightarrow Var(C_{i,j+1} | C_{i,j}) = C_{i,j} \cdot \sigma_j^2$
- **Independence of payments**
- **No tail**
- **PC-L** as underlying **deterministic methodology**
- **No discounting** (problem with the regulation for capital requirements, namely SII)

At least, we are interesting in computing $\sigma(CDR)$, even though usually the capital requirements are computed on the quantiles.

We start analysing the expected value (focusing on the single row): $E[CDR_i | D^{(n)}]$ = if we consider the reserve in balance sheet as expectation of future payments (no discounting and no prudence) (CRUCIAL POINT)

$$= E\left[\sum_{j=n-i+1}^n X_{i,j} | D^{(n)}\right] - E[X_{i,n-i+1} | D^{(n)}] - E\left[E\left[\sum_{j=n-i+2}^n X_{i,j} | D^{(n+1)}\right] | D^{(n)}\right] =$$

We can observe that the first two terms can be subtracted, because in $X_{i,n-i+1}$ the subscript is equal to the initial subscript of the first term.

$$= E\left[\sum_{j=n-i+2}^n X_{i,j} | \mathbf{D}^{(n)}\right] - E\left[E\left[\sum_{j=n-i+2}^n X_{i,j} | \mathbf{D}^{(n+1)}\right] | \mathbf{D}^{(n)}\right]$$

The first term is the expectation given the information in the year n ; the second term has, embedded in the expectation given $\mathbf{D}^{(n)}$, the expectation given the information in the year $n+1$.

If the expectation of the year n is exactly equal to the expectation of the year $n+1$, then

$E[\mathbf{CDR}_i | \mathbf{D}^{(n)}] = \mathbf{0}$, meaning that on average CDR has mean null.

Under some assumption this is true (eg. **Classical PC-L** as underlying methodology); if it is not true, then I expect to have a different reserve in future, not including any kind of prudence.

$$\text{Under classical PC-L} \rightarrow \begin{cases} \hat{\lambda}_j^{D(n)} = \frac{\sum_{i=0}^{n-j-1} C_{i,j+1}}{\sum_{i=0}^{n-j-1} C_{i,j}} \text{ for } j = 0, \dots, n-1 \\ \hat{\lambda}_j^{D(n+1)} = \frac{\sum_{i=0}^{n-j} C_{i,j+1}}{\sum_{i=0}^{n-j} C_{i,j}} \text{ for } j = 0, \dots, n-1 \end{cases}$$

To pass from the first to the second we must add at the **numerator** a **random variable** ($\mathbf{C}_{n-j,j+1}$) and at the **denominator** a **deterministic amount** ($\mathbf{C}_{n-j,j}$).

We want to see the if and how the expected value for $\hat{\lambda}_j^{D(n+1)}$ changes **with respect to the one**

$$\text{for } \hat{\lambda}_j^{D(n)}: E[\hat{\lambda}_j^{D(n+1)} | \mathbf{D}^{(n)}] = \frac{\sum_{i=0}^{n-j-1} C_{i,j+1} + E[\mathbf{C}_{n-j,j+1} | \mathbf{D}^{(n)}]}{\sum_{i=0}^{n-j} C_{i,j}}, \text{ because the only element not}$$

deterministic is the value of $\mathbf{C}_{n-j,j+1}$; then if we consider the estimation of the first diagonal given the triangle in n , we can rewrite $\mathbf{C}_{n-j,j+1} = \hat{\lambda}_j^{D(n)} \cdot \mathbf{C}_{n-j,j} \rightarrow$

$$E[\hat{\lambda}_j^{D(n+1)} | \mathbf{D}^{(n)}] = \frac{\hat{\lambda}_j^{D(n)} \sum_{i=0}^{n-j-1} C_{i,j} + \hat{\lambda}_j^{D(n)} \mathbf{C}_{n-j,j}}{\sum_{i=0}^{n-j} C_{i,j}} = \hat{\lambda}_j^{D(n)}$$

This result is in line with the **C-L**, because to pass from $\mathbf{D}^{(n)}$ to $\mathbf{D}^{(n+1)}$ I have to add the next diagonal, but if we consider the expected value (made at time n) of this diagonal, then the development factor does not change, because **on average** you expect that λ **does not change** in time.

As a **consequence** $E[\mathbf{CDR}_i | \mathbf{D}^{(n)}] = E[\mathbf{C}_{i,n}^{D(n)} - \mathbf{C}_{i,n}^{D(n+1)} | \mathbf{D}^{(n)}] = \mathbf{0}$ (If we are considering the classical PC-L): this is demonstrable considering the ultimate cost as the value on the diagonal times a product of lambdas that on average are the same.

This result is **not true** for local **GAAP**, since in this framework it is necessary to introduce some degree of **prudence** for the reserve; therefore it is possible to release in the CDR part of the prudence present in $R_i^{D(n)}$.

In **SII** we have to include the **RM** to the **BEL** (which is exactly the expected value, but with the discounting).

In $\mathbf{CDR}_i = R_i^{D(n)} - X_{i,n-i+1} - R_i^{D(n+1)}$ we have the **discount** for $\mathbf{R}_i^{D(n)}$ in n and for $\mathbf{R}_i^{D(n+1)}$ in $n+1$ (in the lower triangle, therefore $\mathbf{R}_i^{D(n)} < \mathbf{R}_i^{D(n+1)}$ (assuming to have **positive rates**), because I discount for more years.

Also for $\mathbf{X}$, that is not discounted as payments, when we are **inside of the reserve**, the amount to cover $\mathbf{X}$ is considered with **discount for one year**.

So, with **positive rates**, $\mathbf{CDR}_i < \mathbf{0}$ because I discount for one year more: this does not mean that I have a loss on average, because I have to **consider** the **rate of return** (in the year from n to $n+1$) given by the assets invested to cover the the initial BEL.

What I want to have is a **rate of return** $\geq$ **risk free + (possible) volatility adjustment**, in order to cover the 'loss' of the CDR negative. It is crucial to cover liabilities with assets in NL.

If now we include **RM** $\rightarrow (\text{BE} + \text{RM})^{D(n)} - X - (\text{BE} + \text{RM})^{D(n+1)}$, with the **first RM** that is related to the **liability of the present (n)**, while the **second RM** that is related to the same liabilities, with the passage of one year, **without** considering the **new business** $\rightarrow$

RM⁽ⁿ⁾ > RM⁽ⁿ⁺¹⁾, which represents the **releasing** of the RM.

If all goes as **expected** the **RM is a profit**: in fact, in **QIS2** it was possible to use this expected profit as a sort of **safety loading**, in order to reduce the CaR for the reserve risk $\rightarrow \alpha \cdot \text{RM}$, with **α as proportion of claims paid in the following year**.

What is the problem if we include RM in capital requirement computation? **Loop in methodology**: RM depends on SCR_0 , and this one depends on RM if included.

Remember: RM is not affected by the WCS.

Finally: what we proved ($E[CDR] = 0$) is not so helpful in reality, because there are some problems both for local GAAP and for SII.

One year volatility: $PV(X); EV(X); EV(R) \rightarrow$

$msep(\hat{CDR}_i | D^{(n)}) = E[(\hat{CDR}_i - 0)^2 | D^{(n)}] = \dots = \hat{C}_{i,n}^2 [PV(X) + EV(X) + EV(R)]$; minus 0 only if under PC-L.

PV(X)+EV(X): we can take these components from **Mack**.

EV(R): **No Mack**, since we are considering more elements.

[3/11]

$$msep(\hat{CDR}_i | D^{(n)}) = \hat{C}_{i,n}^2 \left[\frac{\sigma_{n-i}^2}{\hat{\lambda}_{n-i}^2 C_{i,n-i}} + \frac{\sigma_{n-i}^2}{\hat{\lambda}_{n-i}^2 \sum_{h=0}^{i-1} C_{h,n-i}} + \sum_{j=n-i+1}^{n-1} \frac{C_{n-j,j}}{\sum_{i=0}^{n-j} C_{i,j}} \left(\frac{\sigma_j^2}{\hat{\lambda}_j^2 \sum_{h=0}^{n-j-1} C_{h,j}} \right) \right]$$

If we go back to slide 27, we can see that both **PV(X)** and **EV(X)** are exactly **equal** to the one of **Mack** formula: in fact in both cases we have the same common factor ($\hat{C}_{i,n}^2$) and a summation with respect to the column, with the number of elements depending on the number of elements to be estimated. Also in this case, there is a lag of one year between the column of the cell to be estimated and the column used for the computation.

(DISEGNINO): **PV** considering the **cumulated amount** is **equal** to the **PV** for the **incremental payments** because the diagonal $n+1$ is composed by the deterministic diagonal n and the incremental payments $\rightarrow$ This is why we can use Mack for one year time horizon.

First term of M-W $\equiv$ **First element of the summation for Mack (PV)**

Second term of M-W $\equiv$ **First element of the summation for Mack (EV)**

For both of them we can have the interpretation of CV^2 .

$$\text{For } i=n \left\{ \begin{array}{l} (PV) \quad \hat{C}_{n,n}^2 \frac{\hat{\sigma}_0^2}{\hat{\lambda}_{n,n}^2 C_{n,0}} \\ (EV) \quad \hat{C}_{n,n}^2 \frac{\hat{\sigma}_0^2}{\hat{\lambda}_0^2 \sum_{h=0}^{n-1} C_{h,0}} \end{array} \right.$$

If we compare the **variability of M-W and Mack**, we can have an idea of the 1 year volatility on the total one $\rightarrow \frac{\sigma(\hat{CDR})}{\sigma(\hat{R})}$: this value depends both on the **size of the triangle** and on the **SS**.

Eg. $i=1 \rightarrow R_i^{D(n+1)} = 0$ since we **don't** have **tail**, meaning that $CDR_1 = R_1^{D(n)} - X_{1,n}$ and

$\frac{\sigma(\hat{CDR}_1)}{\sigma(\hat{R})} = 100\%$. This result is consistent since we are closing all the claims in 1 year.

Now we focus on **EV of the reserve** $\rightarrow \sum_{j=n-i+1}^{n-1} \frac{C_{n-j,j}}{\sum_{i=0}^{n-j} C_{i,j}} \left(\frac{\sigma_j^2}{\hat{\lambda}_{n-i}^2 \sum_{h=0}^{n-j-1} C_{h,j}} \right)$

The element in the '0' is the part of EV based on Mack formula: **Mack applied not for the diagonal n+1** but other cells (**DISEGNINO**)

We have an additional component $\left(\frac{C_{n-j,j}}{\sum_{i=0}^{n-j} C_{i,j}} \right)$: this vector is always (when defined, den $\neq 0$)

between 0 and 1, because we are dividing the numerator for a sum of non negative values that involves also the numerator $\left(\frac{C_{n-j,j}}{C_{0,j} + \dots + C_{n-j,j}} \right)$.

We are reducing the EV with respect the one of Mack: the magnitude of this reduction depends how much the value is near 0.

Eg. $i=n \rightarrow \sum_{j=1}^{n-1} [\cdot]$ ($j=1$, is consistent with lag of one period, in fact we want to estimate the

columns 2); assume to consider only $j=1$: $\frac{C_{n-1,1}}{\sum_{i=0}^{n-1} C_{i,1}}$

(**DISEGNINO CRUCIAL**)

What is the **difference** between $\hat{\lambda}_1^{D(n)}$ and $\hat{\lambda}_1^{D(n+1)}$? The element $C_{n-1,1}$ is considered at the denominator in $\hat{\lambda}_1^{D(n+1)}$ but not in $\hat{\lambda}_1^{D(n)}$ and $C_{n-1,2}$ (which is a random variable) is considered at the numerator in $\hat{\lambda}_1^{D(n+1)}$ but not in $\hat{\lambda}_1^{D(n)}$.

Which is the possibility that the **two individual development factors** are equal? **IFF**

$$\frac{C_{n-1,2}}{C_{n-1,1}} = \hat{\lambda}_1^{D(n)}.$$

Now, assume $C_{n-1,1}^{(1)} = 1\text{€}$ and $C_{n-1,1}^{(2)} = 1 \text{ Mld €}$: in the former case the effect of the realisation of $C_{n-1,2}$ in $\hat{\lambda}_1^{D(n+1)}$ is quite negligible, since we are considering a weighted average, weighted by

$C \rightarrow$ Whatever value for the individual link ratio for the two cells involved, $\hat{\lambda}_1^{D(n)} \approx \hat{\lambda}_1^{D(n+1)}$.

In the latter case, whatever value for the individual link ratio different from $\hat{\lambda}_1^{D(n)}$ makes $\hat{\lambda}_1^{D(n+1)}$ differ very much.

The ratio $\frac{C_{n-1,1}}{\sum_{i=0}^{n-1} C_{i,1}}$ measure the **impact** of the **new information** used in $\hat{\lambda}_1^{D(n+1)}$ on the old

information + the last one: if $C_{n-1,1}$ is large, important difference between λ s and ratio big; otherwise λ s similar and small ratio.

This reasoning must be repeated for all j 's.

The **ratio does not depends on row** $\rightarrow$ We can compute it **one** time for **each column**.

Regarding $\frac{\sigma(\hat{C\hat{D}R_i})}{\sigma(\hat{R_i})}$, for the **diagonal** we have **100%**, whereas for the other cells we are doing

the ratio between the EV of Mack, scaled by $\frac{C_{n-1,1}}{\sum_{i=0}^{n-1} C_{i,1}}$ and the EV of Mack + PV of Mack .

The **distance** from 1 of $\frac{\sigma(\hat{C\hat{D}R_i})}{\sigma(\hat{R_i})}$ **depends** both on the **relevance of the PV** and the

importance of $\frac{C_{n-1,1}}{\sum_{i=0}^{n-1} C_{i,1}}$.

Attention: the sum of one year risk over time is higher than the run-off approach, because we are considering also the EV for R and not only for X.

(83)

Given same triangle and same underlying deterministic methodology (C-L) we can appreciate the differences between the results for Mack and for M-W.

For Mack we can compute the **CV**, while for **M-W** is **not possible**, because the mean is equal to 0.

So when we are dealing with **M-W**, the $CV^{OY} \neq CV(CDR)$,

$$CV^{OY} = CV(X + R^{D(n+1)}) = \frac{\sigma(X + R^{D(n+1)})}{E[X + R]} \text{ with } E[X + R^{D(n+1)}] = R_i^{D(n)}.$$

This is **consistent** because $\sigma(CDR) = \sigma(X + R^{D(n+1)})$.

The dependence between X and R is embedded in the formula for EV(R).

Usually we have **high values** for **OY/TOT Run-off** in **older AY**: this is obvious, because the ratio is composed of few cells.

The ratio usually **decreases**, but **not monotonically**.

Total $\rightarrow CV^{OY}$ can be computed as $\text{Total}(\text{OY/TOT Run-off}) \cdot CV^{MACK}$, because the denominator of the two CVs is the same, the only difference is in the numerator.

CV^{OY} gives us information for the value of $\frac{\text{Cap Req}}{\text{Reserve}}$, without knowing other information, such as skewness.

(84)

The ratio depends a lot on the triangle considered: in this case the **total one year risk** on the **total risk is 90%**, meaning that the **PV** has **not** an **high impact**.

Reference: on traditional triangle OY/Total Run-off between 60% and 95%.

A typical mathematical trick is $Y_i = X_{i,n-i+1} + R_i^{D(n+1)}$, random variable that represents the total amount to be covered in the next year.

In this way: $CDR_i = R_i^{D(n)} - Y_i$.

We have obtained something similar to the premium risk for RT, which means that CDR_i is a **sort of P/L** for the **following year**, just like the second two addendums of $U_1 = U_0 + (1 + \lambda)P - X_1$. For the company as whole, we can consider together premium and reserve risk, with positive correlation (0.5): $U_1 = U_0 + (1 + \lambda)P - X_1 + CDR$

We can decide whether to study Y_i and draw its distribution or study CDR and draw its distribution: for Y_i we have positive skewness (even though this is not important, as seen in Bootstrapping); for CDR we have negative skewness.

$\sigma(CDR)$ depends on $\sigma(Y_i)$.

$$CaR = Y_\epsilon - R_i^{D(n)}$$

[07/11]

As we already said, the value of the ratio $\frac{C_{n-1,1}}{\sum_{i=0}^{n-1} C_{i,1}}$ depends on the column I am considering,

because it is sensitive to the numbers of elements involved in the summation at the denominator: specifically, when we consider higher DY, there are less and less elements in the summation, therefore we expect to observe higher values.

We can now pass from the computation of the CDR for the single row, to the total one:

$$msep(\widehat{CDR} | D^{(n)}) = \sum_{i=1}^n msep(\widehat{CDR}_i | D^{(n)}) + 2 \sum_{i=1}^{n-1} \hat{C}_{i,n} \sum_{h=i+1}^n \hat{C}_{h,n} \left[\frac{\sigma_{n-i}^2}{\hat{\lambda}_{n-i}^2 \sum_{h=0}^{i-1} C_{h,n-i}} + \sum_{j=n-i+1}^{n-1} \frac{C_{n-j,j}}{\sum_{i=0}^{n-j} C_{i,j}} \left(\frac{\sigma_j^2}{\hat{\lambda}_j^2 \sum_{h=0}^{n-j-1} C_{h,j}} \right) \right]$$

(in the second summation $h = i + 1 \rightarrow h > i$)

We can appreciate that the structure of the formula is equal to the Mack's formula

$$(mse(\hat{\mathbf{R}} | \mathbf{D}^{(n)}) = \sum_{i=1}^n mse(\hat{\mathbf{R}}_i | \mathbf{D}^{(n)}) + 2 \sum_{i=1}^{n-1} [\hat{C}_{i,n} (\sum_{h=i+1}^n \hat{C}_{h,n}) \sum_{j=n-i}^{n-1} \frac{\hat{\sigma}_j^2}{\hat{\lambda}_j^2 \sum_{i=0}^{n-j-1} C_{i,j}}]] :$$

the **total prediction variance** is equal to the **sum** of the **singular prediction variances** for the **rows**, plus a **component** that takes into **consideration the correlation effect** (due to dependency), given by the **estimation variance**.

The **main difference** is that for Mack the **dependence** is **between reserves** (or Ultimate Costs alternatively) of different years, whereas for **M-W** it is considered the **dependence** (so the covariance) **between CDR** of different years. In this sense, we know that there are **two random variables involved: incremental payments** (X) and **reserve** ($R^{D(n+1)}$): therefore the covariance that we want to compute is $Cov(\widehat{CDR}_i, \widehat{CDR}_h) \Leftrightarrow CoV(\tilde{Y}_i, \tilde{Y}_h)$.

Fixing both i and h we obtain exactly 2 times $Cov(\widehat{CDR}_i, \widehat{CDR}_h)$, as for Mack; moreover, even for M-W we can notice that the covariance depends on the EV of the AY.

We can finally prove (as already seen with Mack for the reserve instead of the CDR):

$$Cov(\widehat{CDR}_i, \widehat{CDR}_h) = \frac{\hat{C}_{h,n}}{\hat{C}_{i,n}} \cdot EV(\widehat{CDR}_i)$$

Why can I rewrite this? Because the second and third addendums of $msep(\widehat{CDR}_i | \mathbf{D}^{(n)})$ are exactly equal to the part of $msep(\widehat{CDR} | \mathbf{D}^{(n)})$ in the squared brackets.

Excel

With M-W we reduce the variance obtained with Mack proportionally to the column considered: the more the DY is towards 1, the higher is the reduction of the variance, and therefore less similar is the result to Mack.

Fast way to quantify the effect of the covariance:

- Fix i

- Sum for h

$$\text{Compute: } 2 \cdot \hat{C}_{i,n} \sum_{h=i+1}^n \hat{C}_{h,n} \left[\frac{\sigma_{n-i}^2}{\hat{\lambda}_{n-i}^2 \sum_{h=0}^{i-1} C_{h,n-i}} + \sum_{j=n-i+1}^{n-1} \frac{C_{n-j,j}}{\sum_{i=0}^{n-j} C_{i,j}} \left(\frac{\sigma_j^2}{\hat{\lambda}_{n-i}^2 \sum_{h=0}^{n-j-1} C_{h,j}} \right) \right]$$

We can do this only if we are not interested in the covariance of couples of rows; otherwise I fix i and h as seen before.

Even in this example we can see that the total value of $\frac{OY}{Total\ run - off}$ is within the interval

60% – 95 % : in this case we have something similar to 80%.

Why do we care about the number coming from the computation of $\frac{OY}{Total\ run - off}$? From

this value we can have an idea regarding the capital requirement: having the result (CV) of the total run-off methodology we can pass to the one year view; then applying the multiplier (2.58 iff

Normal assumption; 3 SII assumption) times the CV^{OY} we can have the value of

Capital requirements

Reserve

In fact we can pass from CV^{Mack} to CV^{OY} simply multiplying the first value for

$$\frac{OY}{Total\ run - off}; \text{ alternatively it is possible to compute } CV^{OY} \text{ as ratio } \frac{\sqrt{mse(M-W)}}{Reserve}.$$

Main limit of M-W: we **cannot** obtain as result a distribution for the **CDR**, we have only mean and prediction variance.

How can I use the **M-W methodology**?

1) Inside the **standard formula**, mainly as **USP**: in fact it possible to use the SF either as Market Wide (MW) or Undertaking Specific Parameters (USP); the main difference is that in the latter

case the volatility depends on the methodology, whereas in the former one is fixed by the regulation. One of the possible methodology for the reserve risk is exactly M-W: in this case both the mean and the variance are obtained through the M-W formula, while the shape of the distribution is given by the SF.

- 2) In the **internal model**: in this case we have to specify the distribution (for CDR/Y) and the assumption does not come from M-W.

Re-Reserving approach

Contrary to what we have just said for M-W, with this approach I can obtain the shape of the **distribution**.

The re-reserving is a **general methodology**, meaning that it can be applied with many **different total run-off methodologies** (eg. Bootstrap, CRM,...)

We will focus on the **bootstrap** (considering bootstrap + simulation from ODP): in this way we are **fixing** both the **total run-off methodology** and, consequently, the **deterministic approach** (that in this case is the **PC-L**), which strictly depends on the total run-off approach.

As for M-W, the **final aim** is to compute **msep(CDR)**, with the usual components

PV(X), EV(X), EV(R), so we want to build a model that is able to catch these components.

For the payments (X), the reasoning based on **total run-off** or based on **one year contest** is the **same**: **PV(X), EV(X)** must be equal to the one obtained with **bootstrap**.

Given the upper triangle, how can we measure the PV(X) and EV(X)?

We can **replicate** all the **steps** for bootstrap + simulation from ODP: we obtain, in the end, **H** different **triangles** that we call $\ddot{P}^h$. If we decide to focus on one cell of the lower triangle, we can draw the 'empirical' distribution for the incremental payments of that specific cell: in this case we obtain both the PV (from ODP simulations) and the EV (from bootstrap) for X.

Finally, we can say that from the total run-off we already have the distribution for X: of course, in order to consider **X** as the **whole diagonal**, we have to **sum** the values on the **first diagonal** of the **lower triangle** and **draw the distribution for the sum**.

We can now pass to the computation of $R^{D(n+1)}$: given the simulation $\ddot{P}^h$ (with h from 1 to H), I can **associate** to each of them the **original upper triangle** $P_{i,j}$ (**important**: this is equal for all the simulation) and we can add, for each simulation, the first diagonal of the lower triangle $\ddot{P}^h \rightarrow$ We obtain H different triangles $\ddot{P}_{i,j}^{(h)}$ for which the **only difference** is given by the **diagonal n+1**, so it is based on the **distribution of X**.

In this way we are trying to describe which can be possible additional information at disposal at the end of the year n+1, considering **only the existing claims**, because we are considering different values for the r.v payments at each simulation.

The **underlying assumption** is that between the year n and the year n+1 we apply the **same methodology** $\rightarrow$ We apply the **PC-L** to $\ddot{P}_{i,j}^{(h)}$ and we **obtain** $R_i^{D(n+1)(h)}$ for the h-th simulation,

which can be considered in the computation of Y, namely $Y_i^{(h)} = \ddot{P}_{i,n-i+1}^{(h)} + R_i^{D(n+1)(h)}$.

If we **iterate** the process we obtain a vectors $[R_i^{D(n+1)(1)}, \dots, R_i^{D(n+1)(H)}]$ and consequently $[Y_i^{(1)}, \dots, Y_i^{(H)}]$: in each of the component of the latter vector we are considering both **PV(X), EV(X)** (embedded in $\ddot{P}_{i,j}^{(h)}$) and **EV(R)** (present in $R_i^{D(n+1)(h)}$).

Implicitly the methodology takes into account the **dependency between X and R**, because if we change $\ddot{P}_{i,j}^{(h)}$ then also $R_i^{D(n+1)(h)}$ will be different.

As for the other simulation approaches, the goodness of the approach changes with the increase of the number of simulation involved.

Similar to the comparability for the moments between Mack and M-W, we can compare the mean two different means, $\sigma(Y)$ with $\sigma(B + ODP)$ and, differently from the former approaches, the skewness of Re-reserving and the one of B+ODP.

Re-reserving is **not allowed** as **SF**, but it can be applied as **internal model**: from the quantiles of the distribution we can easily define the CaR for the reserve risk.

The distribution of X is not equivalent to the premium risk: in fact in the reserve risk framework we consider only the cash-outflows (payments) for already existing claims, while for the premium risk we include also the expected claims that will occur.

The **main difference** between **closed formula** and **simulation** is in the **discounting** (very important because now we are in SII framework for capital requirements): for **M-W** we cannot directly apply the discounting rates, we can only make the **assumption** that the **CV doesn't change** including the **discounting**; on the other hand, for **Re-Re** we can **directly discount** the reserve for the different discounting rates.

[10/11]

Example (88)

SIFA and AMSES (the second is circa 10 times more than the first one).

In this specific example we consider the **bootstrapping** with the drawing from the **Log Normal** distribution instead of the ODP: even though this is **theoretically not correct**, in terms of **mean** and **standard deviation** the usage of difference distribution has **not any influence**.

In this example we are **comparing** the **one-years approaches** (M-W and Re-Reserving) with the **corresponding run-off approaches** (Mack and Bootstrapping).

We can observe that the **closed formula** and the **simulation approach** yield **different results** in terms of **variability**: this is perfectly in line with the same observed behaviour in the run-off

approach. On the other hand, we can see that the ratio $\frac{\text{OY}}{\text{TotalRO}}$ tends to be **similar** in both of

the cases (M-W and Re-Reserving), with a **better** similitude for the **larger company**, since there is a **better convergence** (given by the size).

This **stability** in terms of proportion is present only if we **consider** the **same triangle** and the **same underlying methodology** (eg. No Re-Reserving with the application of CRM and M-W, because they are based on different deterministic approaches).

The take off of this example is that the change in the methodology is strictly related to the final capital requirements, even if the underlying deterministic methodology is the same: this could be a critical situation from the supervisor point of view, because the company can chose the methodology that yields the lower capital requirement.

In order to solve this problem is quite important the usage of **statistical test methods** to justify the choice of the approach; it is important to state also that there is a **problem** with the **size of the sample** of the data used for these tests, therefore it can be **challenging** to **demonstrate** the goodness of the **choice**.

We had already encountered this problematic for the computation of the BE, but in the framework of capital requirements it is more important.

The **level of approval** needed for the computation of the capital requirements depends on how the company decides to face this calculation, namely we have **three different approaches**:

Internal model (partial/total), **Standard Formula with USP** (partial usage of the data of the company), **Standard Formula MW** (Market Wide, with no usage of internal data).

If the company uses the **SF MW**, then there is **no need of further approval** and justification of the approach, because it is fixed in SII regulation; already with **SF USP** is important to **justify** the methodology used but with the **internal model** the **level of approval** needed is certainly **higher**.

4.6 One year approach and Capital requirement

We can now compute the SCR for reserve risk, reminding that we are considering only one LoB and only reserve risk.

For now, let's consider Partial (because we are considering only one risk) Internal Model.

$\tilde{Y} = \sum_{i=1}^n X_{i,n-i+1} + R^{D(n+1)}$, it represents the **obligation (paid and reserve)** that I need to **face**

in the next year with the **reserve** $R^{D(n)}$. If we compare this random variable with

$C\tilde{D}R = R^{D(n)} - \sum_{i=0}^n X_{i,n-i+1} - R^{D(n+1)}$ (that represents the **profit/loss of the reserve**) then:

- (Under some specific assumptions) the expected value is different $\rightarrow E[C\tilde{D}R] = 0$ and therefore $E[\tilde{Y}] = R^{D(n)}$
- The **standard deviation** is the same $\rightarrow \sigma(C\tilde{D}R) = \sigma(\tilde{Y})$
- The **skewness** is the **same in magnitude**, but with the **opposite sign** $\rightarrow \gamma(C\tilde{D}R) = -\gamma(\tilde{Y})$, similarly to the RT $\gamma(\tilde{U}) = -\gamma(\tilde{X})$

We can **rewrite** the profit/loss of the reserve as follows: $C\tilde{D}R = R^{D(n)} - \tilde{Y}$

So I can compute the capital required either on CDR or on Y and I obtain the same result, just like it happens with the computation either on U or on X.

If we don't consider the initial capital, then the CDR is equal to the risk reserve $\rightarrow$ **Risk reserve:**

$\tilde{U}_t = \tilde{U}_{t-1} + C\tilde{D}R_t$ or, more accurately, $\tilde{U}_t = \tilde{U}_{t-1}(1+j) + C\tilde{D}R_t(1+j)^{\frac{1}{2}}$, with the usual assumption of homogeneity within the year.

CDR: the **worse events** are on the **LHS**, because it represents the **value of the profit** (q_α with $\alpha = 0.05\%$).

Y: the **worse events** are on the **RHS**, because it represents the **value of the obligation/losses** (q_α with $\alpha = 99.5\%$).

$SCR_\alpha^{Reserve} = q_\alpha(Y) - R^{D(n)}$ with different values of α depending on the application (eg. $\alpha = 99.5\%$ (SII); $\alpha = 99.97\%$ (Rating)). This formula is equivalent to $X_\epsilon - P$ of RT, without considering the safety loading.

(DISEGNINO)

How to estimate $q_\alpha(Y)$? It depends on the method used by the company/fixed by the regulation: eg. With **Re-Reserving** is possible to **directly** compute the **quantile**, while with **M-W** we have to **fix the distribution of Y**, like the **Log-Normal** $\rightarrow$ In this case, given $E[Y]$ and $\sigma(Y)$, we must calibrate the parameters with the MM and then we can compute the quantile.

How can we take into consideration the discounting?

- **Re-Reserving:** it is possible to consider directly $R^{D(n+1)}$ discounted. In this case, we don't have anymore $E[CDR] = 0$, because (as we have already said) $R^{D(n+1)}$ is discounted for one year less than $R^{D(n)}$. Part of the CDR can be covered by financial return of the first year, but this is not easy to be introduced in the computation: in fact, we can consider it, introducing a part linked to the market risk in the reserve risk calculation, or we can neglect it and compute the new mean.
- **C-W:** the usual **assumption** is that the **CV** is **not affected** by the introduction of discounting.

RM is excluded by the capital requirement computation (loop problem, already treated), even if it may seem reasonable to consider it to decrease the capital, as presented in QIS2.

This problematic is analogous to the absence of safety loading in the premium risk calculation and it is present also for the internal model: in fact, the supervisor authority is not likely to accept its presence in the internal model, mainly for prudence reasons (even if it is possible also to insert a negative RM/Safety loading).

Now, let's consider Standard Formula

NL premium-reserve capital requirement: $SCR_{99.5\%}^{Res,SF} = 3\sigma V$ both for **MW** and for **USP** (the difference stands in the estimation of σ).

We can observe the presence of **three components in this formula:**

- **Multiplier** $\rightarrow 3$

- **Volume measure** (Net BE) $\rightarrow V$

- **Relative volatility measure** (it catches the volatility of the ratio $\frac{CDR}{BE^{D(n)}} \rightarrow \sigma$

Other way in which we can see the SCR:

$$SCR = q_\alpha(Y)E[Y] \approx k \cdot \frac{\sigma(Y)}{E[Y]} \cdot E[Y] = k \cdot \frac{\sigma(Y)}{V} \cdot V$$

There is the **approximation** sign because, in case of **non-Normal distribution** (such as the Log-Normal), **k** doesn't assume only one value by a **different value** for each **CV** of the distribution (we can write k_γ because it depends on $\gamma \rightarrow$ If CV increases, both γ and k_γ increase).

Example:

- Normal distribution $\rightarrow \begin{cases} CV(Y) = 10\% \rightarrow k \cdot \frac{\sigma(Y)}{E[Y]} = 25.8\% \\ CV(Y) = 20\% \rightarrow k \cdot \frac{\sigma(Y)}{E[Y]} = 51.6\% \end{cases}$ we can observe that with a

doubled CV also the factor $k \cdot \frac{\sigma(Y)}{E[Y]}$ is doubled

- Log-Normal $\rightarrow \begin{cases} CV(Y) = 10\% \rightarrow k \cdot \frac{\sigma(Y)}{E[Y]} \approx 10\% \cdot 2.7\% \\ CV(Y) = 20\% \rightarrow k \cdot \frac{\sigma(Y)}{E[Y]} \approx 20\% \cdot 3.2\% \end{cases}$ in this case with a doubled CV,

the factor $k \cdot \frac{\sigma(Y)}{E[Y]}$ growth more than the double, because there is an higher skewness.

The **multiplier** equals to **3** was a **compromise** that **EIOPA** had to make, in order to have a fixed value in the SF: it is computed with the underlying **assumption** of **Log-Normal with CV $\approx 15\%$** .

In all the QISs, **k** was considered as variable: $\rho(\sigma) \cdot V = K_\gamma \cdot \sigma \cdot V$; in the last version $K_\gamma = 3$.

If the **Log-Normal** is a **good assumption** for the company, then the **fixed multiplier** leads either to an **overestimation** (when **CV < 15%**) or to an **underestimation** (**CV > 15%**).

This behaviour can be problematic, because usually the smaller companies have higher CV, and they are exactly the most likely to apply the SF $\rightarrow$ **Systematic underestimation of the risk**.

σ is computed for **each risk** (premium and reserve) and for **each LoB**, then we **aggregate** the different result, considering a **linear correlation** $\rightarrow \sigma$ final is the aggregation of risks and LoBs. The same situation for **V**: this value includes the total volume of premium and reserve, for all the LoBs.

For **reserve**, the **volume measure** is the **BE** of the claims reserve, while for the **premium** it will be based on the **amount of premiums**.

Reserve risk $\rightarrow \sigma^{\text{LoB}} = \frac{\sigma(Y)}{\text{BE}}$ for each LoB but we can have **different** possibilities for the

computation, according to the methodology used:

- **MW** $\rightarrow \sigma^{\text{Res risk, LoB}}$ is given and it is based on the market wide calibration, meaning that it comes from different types of companies and markets.
- **USP** $\rightarrow \sigma^{\text{Res risk, LoB}} = C_{\text{LoB}} \sigma_{\text{USP}} + (1 - C_{\text{LoB}}) \sigma_{\text{MW}}$, so the formula is partially based on the real data of the company; there are **two possible methodologies** for σ_{USP} : **Merz-Wüthrich** or

Standard run-off ratio. The aim of both of them is to compute $\sigma\left(\frac{CDR}{RD(n)}\right)$, **without**

discounting both for numerator and for denominator, but the **approach** is **different**: with **M-W** we consider the **triangle** (as usual), while in the **other approach** we consider the empirical **time series for the CDR**.

C_{LoB} is a **weight** that attributes **importance** to **USP** based on the length of data: higher time

horizon, higher weight for USP. Practically: $C_{MTPL} = C_{GTPL} = C_{Credit} = C_{Suretyship} = 1$ with at **least 15 years**; $C_{Other\ LoB} = 1$ with at **least 10 years** of time horizon.

For M-W application, C_{LoB} depends on the size of the triangle (since there is no time series): the usual 12x12 is not sufficient, therefore the insurance companies that uses USP tend to enlarge the triangle (reaching 15x15) also for balance sheet purposes.

Drivers of difference (for a single LoB):

- **Volatility:** σ_{MW} and σ_{USP} for MW and USP; the two previous values and $\frac{\sigma(Y)}{R^{D(n)}}$ to pass to Internal model. In the latter case, the difference can come both from the numerator and from the denominator (for example, using Re-Reserving instead of MW).
- **Shape (skewness):** no difference between MW and USP; difference depending on value of $k = \frac{q(Y) - E[Y]}{\sigma(Y)}$ (which depends on the distribution) to pass to Internal Model.

If we consider more LoB, there is also the effect of the aggregation, which depends on the volume of the LoB, meaning that if the difference comes from LoBs with limited volume, then also the difference is limited.

For **IM** we have to decide also the method for the **aggregation**: in place of **linear correlation** (present in **SF**), it is possible to consider different possibilities, also to include the **tail dependency** in the aggregation —> If one extreme event occurs, it is likely to be observed in all the LoBs that have tail dependency.

In this optic, it is important to remember that **un-correlation doesn't imply independency** (**outside of the Gaussian example**, 'Uncorrelated and jointly gaussian implies independent').

COPULA FUNCTIONS

Topic: how to aggregate different risks and LoBs.

Inside the SF the aggregation is linear: main problem —> Tail dependence: extreme events that affect jointly the tail of different distributions or different LoBs.

How to deal with tail dependence? Copula function.

The framework for the Copula function is the **Sklar's Theorem**: 'Any multivariate joint distribution can be written in terms of univariate marginal distribution functions and a Copula which describes the dependence structure between the variables' —> Y_{MTPL}, Y_{GTPL} r.vs ; F=CDF; C=function,

therefore $F(Y_1, Y_2) = C(F_1(Y_1), F_2(Y_2))$ with Y_1, Y_2 respectively values of MTPL and GTPL.

There are some important assumptions to be satisfied to apply this theorem, such as the fact that the marginal distributions must be continuous.

In this way we can split the components of the marginals and the conditional distribution.

Changing the C we can have different $F(Y_1, Y_2)$ for the same $F(Y_1), F(Y_2)$: to describe the linear dependence there is a specific Copula.

How to define C?

In general $C : [0,1]^2 \rightarrow [0,1]$; $C[u_1, u_2] = P[U_1 \leq u_1; U_2 \leq u_2] = F(u_1, u_2)$ in which U_1, U_2 are uniformly distributed.

For different C, the only difference is the dependence of U_1, U_2 while the uniform marginal distributions remain the same for both the variables.

If we simulate numbers from a Copula, the output is a matrix with one dimension for the numbers of simulations and one column for each random variable involved.

From C we obtain the characteristics of dependence; from F we obtain the characteristics of marginals —> C(F) we have both the effects.

Each CDF can be interpreted as a random variable: $Z = F_{Y_1}(y_1)$ and it can be interpreted as a uniform rv, thanks to the **Probability integral transformation**.

In this way we can use Z in place of the CDFs: we can pass from $C(u_1, u_2)$ to $F(Y_1, Y_2)$.

$C[u_1, u_2] = P[F_{Y_1}(y_1) \leq u_1; F_{Y_2}(y_2) \leq u_2]$ the advantage is that we know u_1, u_2 —>

$P[Y_1 \leq F_{Y_1}^{-1}(u_1); Y_2 \leq F_{Y_2}^{-1}(u_2)]$

F can associate to the matrix of u_1, u_2 a matrix of y_1, y_2 (DISEGNO)

In this way I have both the dependence (from C) and the marginal functions (from F_{y_1}, F_{y_2}).

Suppose $\rho(u_1, u_2) = 0.5$, then, after a sufficient numbers of simulations, also $\rho(Y_1, Y_2) = 0.5$ because we are using the inversion of the dependent random variables.

Now we can compute for the whole vector $Y_1 + Y_2 = [* , \dots ,]^T$ with

$$\sigma^2(*) = \sigma^2(Y_1) + \sigma^2(Y_2) + 2 \cdot 0.5\sigma(Y_1)\sigma(Y_2).$$

Types of Copulas:

- **Elliptical** —> The **Gaussian** is the most used one for representing the linear correlation: given the fact that it is symmetric, then it doesn't consider the tail dependence; if $X_1 \sim N$; $X_2 \sim N$ and there is linear correlation, so $(X_1, X_2) \sim N(\underline{\mu}, \underline{\sigma}^2)$. In Gaussian the parameters needed is the correlation matrix: with high numbers of simulations the correlation is in line with the one of the matrix.
Another one is the **T-student**.
- **Archimedean** —> Not linear correlation but tail dependence (eg. **Gumbel**). Problem: dependence only on one parameter, so it is difficult to compute the dependence when there are more than 2 rvs.

With non linear correlation which is the coefficient that captures the dependence?

$$\tau_{Kendall} = \frac{\sum_{i < j} \text{sign}(x_i - x_j) \text{sign}(y_i - y_j)}{\binom{n}{2}} \text{ assuming } X, Y \text{ rvs} \rightarrow \textbf{Measure of concordance}$$

EXAMPLE QUADERNO

The value inside the summation is +1 when x_i, x_j are in the same order of y_i, y_j , disregarding the magnitude the difference; the same value is -1 when the order for x and y is different.

The maximum value for the numerator is $\binom{n}{2} \rightarrow \tau = 1$ (eg. $Y = X^2, \rho = 0, \tau = 1$)

The minimum value for the numerator is $-\binom{n}{2} \rightarrow \tau = -1$

Finally, $-1 \leq \tau \leq 1$, like $-1 \leq \rho \leq 1$ but considering the order instead of the correlation coefficient.

The correlation coefficient of ranking is equal to τ .

τ is important for calibration of parameters of the Copula for the Archimedean, because I don't have linear relationship.

For the insurance sector we can have problem with the usage of the Copula, because of the size of the sample for reserving, while for financial sector there are not problem.

